

# KIT SAFETY DATA SHEET



Kit Product Name CWD SAP Combo Kit

Kit Catalog Number(s) 3600032, 3600033

Revision date 14-May-2025

## Kit Contents

| Catalog Number(s) | Product Name                     |
|-------------------|----------------------------------|
| 7360A             | R2 - Washing Solution            |
| 5145B             | R3 - Negative Control            |
| 5145C             | R4 - Positive Control            |
| 5145D             | R6 - Sample Diluent              |
| 5145E             | R7- Concentrated (10X) Conjugate |
| 7435S             | R8 - Peroxidase Substrate Buffer |
| 7436J             | R9 - Chromogen: TMB Solution     |
| 7438L             | R10 - Stopping Solution          |
| 5144A             | Grinding Tube                    |
| 5144B             | Reagent A - Denaturing Solution  |
| 1641              | Reagent B - Clarifying Solution  |
| 5144D             | Reagent C1 - Resolving Solution  |
| 3551103C1         | R1 - Solid Phase Antibody        |



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
US OSHA Hazard Communication Standard (29 CFR 1910.1200)

Revision date 14-May-2025

Revision Number 1.1

## 1. Identification

### Product identifier

**Product Name** R2 - Washing Solution

### Other means of identification

**Catalog Number(s)** 7360A

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Life Science  
2000 Alfred Nobel Drive  
Hercules, California 94547

#### **Technical Service**

1-800-424-6723  
support@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification of the substance or mixture

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200)

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

#### Hazard statements

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200)

### **Hazards classified under paragraph (d)(1)(ii) of 1910.1200**

No information available.

### Other information

No information available.

### 3. Composition/information on ingredients

#### Substance

Not applicable.

#### Mixture

| Chemical name   | CAS No.   | Weight-% | Trade secret |
|-----------------|-----------|----------|--------------|
| Sodium chloride | 7647-14-5 | 10 - 20  | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

### 4. First-aid measures

#### Description of first aid measures

|              |                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                    |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a physician. |
| Skin contact | Wash skin with soap and water.                                                                                          |
| Ingestion    | Rinse mouth.                                                                                                            |

#### Most important symptoms and effects, both acute and delayed

|                     |                           |
|---------------------|---------------------------|
| Symptoms            | No information available. |
| Effects of Exposure | No information available. |

#### Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

### 5. Fire-fighting measures

|                                                                |                                                                                                                                       |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                               |
| Unsuitable extinguishing media                                 | Do not scatter spilled material with high pressure water streams.                                                                     |
| Specific hazards arising from the chemical                     | No information available.                                                                                                             |
| Explosion data                                                 |                                                                                                                                       |
| Sensitivity to mechanical impact                               | None.                                                                                                                                 |
| Sensitivity to static discharge                                | None.                                                                                                                                 |
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.<br>Use personal protection equipment. |

### 6. Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Ensure adequate ventilation.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

**7. Handling and storage****Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store according to product and label instructions.

**8. Exposure controls/personal protection****Working area parameters, subject to mandatory control (MAC or TSEL)**

**Exposure Limits** This product, as supplied, contains materials that do not have reportable occupational exposure limits or are not subject to the reporting requirements of the local jurisdiction.

**Biological occupational exposure limits** This product, as supplied, contains materials that do not have reportable biological exposure limits or are not subject to the reporting requirements of the local jurisdiction.

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** Appropriate eye/face protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Hand protection** Appropriate hand protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Skin and body protection** Appropriate skin and body protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Respiratory protection** Appropriate respiratory protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                                |                  |
|--------------------------------|------------------|
| Appearance                     | aqueous solution |
| Physical state                 | Liquid           |
| Color                          | colorless        |
| Odor (includes odor threshold) | Odorless         |

| Property                                | Values            | Remarks • Method |
|-----------------------------------------|-------------------|------------------|
| pH                                      | 7.4               |                  |
| Melting point / freezing point          | < 0 °C / 32 °F    |                  |
| Initial boiling point and boiling range | > 100 °C / 212 °F |                  |
| Flash point                             | No data available | None known       |
| Evaporation rate                        | No data available | None known       |
| Flammability                            | No data available | None known       |
| Flammability Limit in Air               |                   | None known       |
| Upper flammability or explosive limits  | No data available |                  |
| Lower flammability or explosive limits  | No data available |                  |
| Vapor pressure                          | No data available | None known       |
| Relative vapor density                  | No data available | None known       |
| Relative density                        | No data available | None known       |
| Water solubility                        | Miscible in water | None known       |
| Solubility(ies)                         | No data available | None known       |
| Partition coefficient                   | No data available | None known       |
| Autoignition temperature                | No data available | None known       |
| Decomposition temperature               | No data available | None known       |
| Kinematic viscosity                     | No data available | None known       |
| Dynamic viscosity                       | No data available | None known       |

### Other information

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

### Information on likely routes of exposure

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

**Symptoms related to the physical, chemical and toxicological characteristics**

**Symptoms** No information available.

**Acute toxicity** No information available.

**Numerical measures of toxicity**

The following ATE values have been calculated for the mixture

ATEmix (oral) 19,231.30 mg/kg  
ATEmix (dermal) 493,096.60 mg/kg

**Component Information**

| Chemical name                | Oral LD50            | Dermal LD50              | Inhalation LC50       |
|------------------------------|----------------------|--------------------------|-----------------------|
| Sodium chloride<br>7647-14-5 | = 3550 mg/kg ( Rat ) | > 10000 mg/kg ( Rabbit ) | > 42 mg/L ( Rat ) 1 h |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** No information available.

**Serious eye damage/eye irritation** No information available.

**Respiratory or skin sensitization** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

**STOT - repeated exposure** No information available.

**Aspiration hazard** No information available.

**Other adverse effects** No information available.

**Interactive effects** No information available.

**12. Ecological information****Ecotoxicity**

| Chemical name                | Algae/aquatic plants | Fish                                                                    | Toxicity to microorganisms | Crustacea                                                             |
|------------------------------|----------------------|-------------------------------------------------------------------------|----------------------------|-----------------------------------------------------------------------|
| Sodium chloride<br>7647-14-5 | -                    | LC50: 5560 - 6080mg/L<br>(96h, Lepomis macrochirus)<br>LC50: =12946mg/L | -                          | EC50: =1000mg/L (48h, Daphnia magna)<br>EC50: 340.7 - 469.2mg/L (48h, |

|  |  |                                                                                                                                                                                                                                               |  |                |
|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------|
|  |  | (96h, Lepomis macrochirus)<br>LC50: 6020 - 7070mg/L<br>(96h, Pimephales promelas)<br>LC50: =7050mg/L (96h, Pimephales promelas)<br>LC50: 6420 - 6700mg/L<br>(96h, Pimephales promelas)<br>LC50: 4747 - 7824mg/L<br>(96h, Oncorhynchus mykiss) |  | Daphnia magna) |
|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------|

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

**Other adverse effects** No information available.

### 13. Disposal considerations

#### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

### 14. Transport information

**DOT** Not regulated

**TDG** Not regulated

**MEX** Not regulated

**ICAO (air)** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

### 15. Regulatory information

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories** Contact supplier for inventory compliance status

**US Federal Regulations****TSCA**

All chemical components are listed on the TSCA Inventory.

**SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

**CAA (Clean Air Act)**

This product does not contain any substances regulated as pollutants pursuant to Clean Air Act (CAA).

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**US State Regulations****California Proposition 65**

This product does not contain any Proposition 65 chemicals

**U.S. State Right-to-Know Regulations**

| Chemical name      | New Jersey | Massachusetts | Pennsylvania |
|--------------------|------------|---------------|--------------|
| Water<br>7732-18-5 | -          | -             | X            |

**U.S. EPA Label Information**

**EPA Pesticide Registration Number** Not applicable

**16. Other information**

|                    |                         |                       |                           |                              |
|--------------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b><u>NFPA</u></b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b><u>HMIS</u></b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> - |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend**

|             |                                                                                                 |
|-------------|-------------------------------------------------------------------------------------------------|
| <b>ADN</b>  | Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways (Europe) |
| <b>ADR</b>  | Agreement concerning the International Carriage of Dangerous Goods by Road (Europe)             |
| <b>ATE</b>  | Acute Toxicity Estimate                                                                         |
| <b>ASTM</b> | American Society for the Testing of Materials                                                   |
| <b>BEL</b>  | Biological exposure limits                                                                      |
| <b>bw</b>   | Body weight                                                                                     |

|                |                                                                                     |
|----------------|-------------------------------------------------------------------------------------|
| <b>Ceiling</b> | Maximum limit value                                                                 |
| <b>CMR</b>     | Carcinogen, Mutagen or Reproductive Toxicant                                        |
| <b>DOT</b>     | Department of Transportation (United States)                                        |
| <b>EmS</b>     | Emergency Schedule                                                                  |
| <b>EPA</b>     | Environmental Protection Agency                                                     |
| <b>GHS</b>     | Globally Harmonized System                                                          |
| <b>HMIS</b>    | Hazardous Materials Identification System                                           |
| <b>IATA</b>    | International Air Transport Association                                             |
| <b>ICAO</b>    | International Civil Aviation Organization                                           |
| <b>IMDG</b>    | International Maritime Dangerous Goods                                              |
| <b>IMO</b>     | International Maritime Organization                                                 |
| <b>ISO</b>     | International Organization for Standardization                                      |
| <b>LC50</b>    | Lethal Concentration to 50% of a test population                                    |
| <b>LD50</b>    | Lethal Dose to 50% of a test population (Median Lethal Dose)                        |
| <b>MARPOL</b>  | International Convention for the Prevention of Pollution from Ships                 |
| <b>NFPA</b>    | National Fire Protection Association                                                |
| <b>NIOSH</b>   | National Institute for Occupational Safety and Health                               |
| <b>n.o.s.</b>  | Not Otherwise Specified                                                             |
| <b>NOAEL</b>   | No Observed Adverse Effect Level                                                    |
| <b>OECD</b>    | Organization for Economic Cooperation and Development                               |
| <b>OEL</b>     | Occupational exposure limits                                                        |
| <b>OSHA</b>    | Occupational Safety and Health Administration of the US Department of Labor         |
| <b>PBT</b>     | Persistent, Bioaccumulative and Toxic substance                                     |
| <b>PMT</b>     | Persistent, Mobile and Toxic                                                        |
| <b>PPE</b>     | Personal protective equipment                                                       |
| <b>RID</b>     | Agreement concerning the International Carriage of Dangerous Goods by Rail (Europe) |
| <b>SARA</b>    | Superfund Amendments and Reauthorization Act                                        |
| <b>SDS</b>     | Safety Data Sheet                                                                   |
| <b>STEL</b>    | Short Term Exposure Limit                                                           |
| <b>STOT RE</b> | Specific target organ toxicity - Repeated exposure                                  |
| <b>STOT SE</b> | Specific target organ toxicity - Single exposure                                    |
| <b>TDG</b>     | Transport of Dangerous Goods (Canada)                                               |
| <b>TSCA</b>    | Toxic Substances Control Act (United States)                                        |
| <b>TWA</b>     | Time-Weighted Average                                                               |
| <b>UN</b>      | United Nations                                                                      |
| <b>VOC</b>     | Volatile organic compounds                                                          |
| <b>vPvB</b>    | Very Persistent and Very Bioaccumulative                                            |
| <b>vPvM</b>    | Very Persistent and Very Mobile                                                     |
| <b>Sk</b>      | Skin notation                                                                       |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AELG(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.  
**Revision date** 14-May-2025  
**Revision Note** Reviewed existing information and made minor updates.  
**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the

date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
US OSHA Hazard Communication Standard (29 CFR 1910.1200)

Revision date 14-May-2025

Revision Number 3.1

## 1. Identification

### Product identifier

**Product Name** R3 - Negative Control

### Other means of identification

**Catalog Number(s)** 5145B

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Life Science  
2000 Alfred Nobel Drive  
Hercules, California 94547

**Technical Service** 1-800-424-6723  
support@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification of the substance or mixture

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200)

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

#### Hazard statements

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200)

### Hazards classified under paragraph (d)(1)(ii) of 1910.1200

No information available.

### Other information

No information available.

### 3. Composition/information on ingredients

#### Substance

Not applicable.

#### Mixture

The product contains no substances which at their given concentration, are considered to be hazardous to health

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

### 4. First-aid measures

#### Description of first aid measures

|              |                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                    |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a physician. |
| Skin contact | Wash skin with soap and water.                                                                                          |
| Ingestion    | Rinse mouth.                                                                                                            |

#### Most important symptoms and effects, both acute and delayed

|                     |                           |
|---------------------|---------------------------|
| Symptoms            | No information available. |
| Effects of Exposure | No information available. |

#### Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

### 5. Fire-fighting measures

|                                                                |                                                                                                                                       |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                               |
| Unsuitable extinguishing media                                 | Do not scatter spilled material with high pressure water streams.                                                                     |
| Specific hazards arising from the chemical                     | No information available.                                                                                                             |
| Explosion data                                                 |                                                                                                                                       |
| Sensitivity to mechanical impact                               | None.                                                                                                                                 |
| Sensitivity to static discharge                                | None.                                                                                                                                 |
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.<br>Use personal protection equipment. |

### 6. Accidental release measures

#### Personal precautions, protective equipment and emergency procedures

**Personal precautions** Ensure adequate ventilation.

#### **Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

### **7. Handling and storage**

#### **Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

#### **Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store according to product and label instructions.

### **8. Exposure controls/personal protection**

#### **Working area parameters, subject to mandatory control (MAC or TSEL)**

**Exposure Limits** This product, as supplied, contains materials that do not have reportable occupational exposure limits or are not subject to the reporting requirements of the local jurisdiction.

**Biological occupational exposure limits** This product, as supplied, contains materials that do not have reportable biological exposure limits or are not subject to the reporting requirements of the local jurisdiction.

#### **Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

#### **Individual protection measures, such as personal protective equipment**

**Eye/face protection** Appropriate eye/face protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Hand protection** Appropriate hand protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Skin and body protection** Appropriate skin and body protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Respiratory protection** Appropriate respiratory protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                                |                  |
|--------------------------------|------------------|
| Appearance                     | aqueous solution |
| Physical state                 | Liquid           |
| Color                          | white            |
| Odor (includes odor threshold) | Odorless         |

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      | 7.4               |                         |
| Melting point / freezing point          | 0 °C / 32 °F      |                         |
| Initial boiling point and boiling range | 100 °C / 212 °F   |                         |
| Flash point                             | No data available | None known              |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapor pressure                          | No data available | None known              |
| Relative vapor density                  | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Miscible in water |                         |
| Solubility(ies)                         | No data available | None known              |
| Partition coefficient                   | No data available | None known              |
| Autoignition temperature                | No data available | None known              |
| Decomposition temperature               | No data available | None known              |
| Kinematic viscosity                     | No data available | None known              |
| Dynamic viscosity                       | No data available | None known              |

### Other information

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

### Information on likely routes of exposure

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

**Symptoms related to the physical, chemical and toxicological characteristics**

Symptoms No information available.

Acute toxicity No information available.

Numerical measures of toxicity

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

Skin corrosion/irritation No information available.

Serious eye damage/eye irritation No information available.

Respiratory or skin sensitization No information available.

Germ cell mutagenicity No information available.

Carcinogenicity No information available.

Reproductive toxicity No information available.

STOT - single exposure No information available.

STOT - repeated exposure No information available.

Aspiration hazard No information available.

Other adverse effects No information available.

Interactive effects No information available.

**12. Ecological information**

Ecotoxicity

Persistence and degradability No information available.

Bioaccumulation There is no data for this product.

Other adverse effects No information available.

### 13. Disposal considerations

#### Disposal methods

|                                            |                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| <b>Contaminated packaging</b>              | Do not reuse empty containers.                                                                                  |

### 14. Transport information

|                          |               |
|--------------------------|---------------|
| <b><u>DOT</u></b>        | Not regulated |
| <b><u>TDG</u></b>        | Not regulated |
| <b><u>MEX</u></b>        | Not regulated |
| <b><u>ICAO (air)</u></b> | Not regulated |
| <b><u>IATA</u></b>       | Not regulated |
| <b><u>IMDG</u></b>       | Not regulated |

### 15. Regulatory information

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories** Contact supplier for inventory compliance status

#### US Federal Regulations

##### **TSCA**

All chemical components are listed on the TSCA Inventory.

##### **SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

##### **SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

##### **CWA (Clean Water Act)**

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

##### **CAA (Clean Air Act)**

This product does not contain any substances regulated as pollutants pursuant to Clean Air Act (CAA).

##### **CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive

Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

### US State Regulations

#### **California Proposition 65**

This product does not contain any Proposition 65 chemicals

### **U.S. State Right-to-Know Regulations**

| Chemical name      | New Jersey | Massachusetts | Pennsylvania |
|--------------------|------------|---------------|--------------|
| Water<br>7732-18-5 | -          | -             | X            |

### U.S. EPA Label Information

**EPA Pesticide Registration Number** Not applicable

## **16. Other information**

|             |                         |                       |                           |                              |
|-------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> - |

### **Key or legend to abbreviations and acronyms used in the safety data sheet**

#### **Legend**

|                |                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------|
| <b>ADN</b>     | Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways (Europe) |
| <b>ADR</b>     | Agreement concerning the International Carriage of Dangerous Goods by Road (Europe)             |
| <b>ATE</b>     | Acute Toxicity Estimate                                                                         |
| <b>ASTM</b>    | American Society for the Testing of Materials                                                   |
| <b>BEL</b>     | Biological exposure limits                                                                      |
| <b>bw</b>      | Body weight                                                                                     |
| <b>Ceiling</b> | Maximum limit value                                                                             |
| <b>CMR</b>     | Carcinogen, Mutagen or Reproductive Toxicant                                                    |
| <b>DOT</b>     | Department of Transportation (United States)                                                    |
| <b>EmS</b>     | Emergency Schedule                                                                              |
| <b>EPA</b>     | Environmental Protection Agency                                                                 |
| <b>GHS</b>     | Globally Harmonized System                                                                      |
| <b>HMIS</b>    | Hazardous Materials Identification System                                                       |
| <b>IATA</b>    | International Air Transport Association                                                         |
| <b>ICAO</b>    | International Civil Aviation Organization                                                       |
| <b>IMDG</b>    | International Maritime Dangerous Goods                                                          |
| <b>IMO</b>     | International Maritime Organization                                                             |
| <b>ISO</b>     | International Organization for Standardization                                                  |
| <b>LC50</b>    | Lethal Concentration to 50% of a test population                                                |
| <b>LD50</b>    | Lethal Dose to 50% of a test population (Median Lethal Dose)                                    |
| <b>MARPOL</b>  | International Convention for the Prevention of Pollution from Ships                             |
| <b>NFPA</b>    | National Fire Protection Association                                                            |
| <b>NIOSH</b>   | National Institute for Occupational Safety and Health                                           |
| <b>n.o.s.</b>  | Not Otherwise Specified                                                                         |
| <b>NOAEL</b>   | No Observed Adverse Effect Level                                                                |
| <b>OECD</b>    | Organization for Economic Cooperation and Development                                           |
| <b>OEL</b>     | Occupational exposure limits                                                                    |
| <b>OSHA</b>    | Occupational Safety and Health Administration of the US Department of Labor                     |
| <b>PBT</b>     | Persistent, Bioaccumulative and Toxic substance                                                 |
| <b>PMT</b>     | Persistent, Mobile and Toxic                                                                    |
| <b>PPE</b>     | Personal protective equipment                                                                   |
| <b>RID</b>     | Agreement concerning the International Carriage of Dangerous Goods by Rail (Europe)             |
| <b>SARA</b>    | Superfund Amendments and Reauthorization Act                                                    |

|                |                                                    |
|----------------|----------------------------------------------------|
| <b>SDS</b>     | Safety Data Sheet                                  |
| <b>STEL</b>    | Short Term Exposure Limit                          |
| <b>STOT RE</b> | Specific target organ toxicity - Repeated exposure |
| <b>STOT SE</b> | Specific target organ toxicity - Single exposure   |
| <b>TDG</b>     | Transport of Dangerous Goods (Canada)              |
| <b>TSCA</b>    | Toxic Substances Control Act (United States)       |
| <b>TWA</b>     | Time-Weighted Average                              |
| <b>UN</b>      | United Nations                                     |
| <b>VOC</b>     | Volatile organic compounds                         |
| <b>vPvB</b>    | Very Persistent and Very Bioaccumulative           |
| <b>vPvM</b>    | Very Persistent and Very Mobile                    |
| <b>Sk</b>      | Skin notation                                      |

### Key literature references and sources for data used to compile the SDS

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.  
**Revision date** 14-May-2025  
**Revision Note** Reviewed existing information and made minor updates.

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
US OSHA Hazard Communication Standard (29 CFR 1910.1200)

Revision date 14-May-2025

Revision Number 2.1

## 1. Identification

### Product identifier

Product Name R4 - Positive Control

### Other means of identification

Catalog Number(s) 5145C

### Recommended use of the chemical and restrictions on use

Recommended use Laboratory chemicals

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Life Science  
2000 Alfred Nobel Drive  
Hercules, California 94547

#### Technical Service

1-800-424-6723  
support@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification of the substance or mixture

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200)

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

#### Hazard statements

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200)

### Hazards classified under paragraph (d)(1)(ii) of 1910.1200

No information available.

### Other information

No information available.

### 3. Composition/information on ingredients

#### Substance

Not applicable.

#### Mixture

| Chemical name   | CAS No.   | Weight-% | Trade secret |
|-----------------|-----------|----------|--------------|
| Sodium chloride | 7647-14-5 | 10 - 20  | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

### 4. First-aid measures

#### Description of first aid measures

|              |                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                    |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a physician. |
| Skin contact | Wash skin with soap and water.                                                                                          |
| Ingestion    | Rinse mouth.                                                                                                            |

#### Most important symptoms and effects, both acute and delayed

|                     |                           |
|---------------------|---------------------------|
| Symptoms            | No information available. |
| Effects of Exposure | No information available. |

#### Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

### 5. Fire-fighting measures

|                                                                |                                                                                                                                       |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                               |
| Unsuitable extinguishing media                                 | Do not scatter spilled material with high pressure water streams.                                                                     |
| Specific hazards arising from the chemical                     | No information available.                                                                                                             |
| Explosion data                                                 |                                                                                                                                       |
| Sensitivity to mechanical impact                               | None.                                                                                                                                 |
| Sensitivity to static discharge                                | None.                                                                                                                                 |
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.<br>Use personal protection equipment. |

### 6. Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Ensure adequate ventilation.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

**7. Handling and storage****Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store according to product and label instructions.

**8. Exposure controls/personal protection****Working area parameters, subject to mandatory control (MAC or TSEL)**

**Exposure Limits** This product, as supplied, contains materials that do not have reportable occupational exposure limits or are not subject to the reporting requirements of the local jurisdiction.

**Biological occupational exposure limits** This product, as supplied, contains materials that do not have reportable biological exposure limits or are not subject to the reporting requirements of the local jurisdiction.

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** Appropriate eye/face protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Hand protection** Appropriate hand protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Skin and body protection** Appropriate skin and body protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Respiratory protection** Appropriate respiratory protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                                |                             |
|--------------------------------|-----------------------------|
| Appearance                     | powder or cake, lyophilized |
| Physical state                 | Solid                       |
| Color                          | white                       |
| Odor (includes odor threshold) | Odorless                    |

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      | 7.4               |                         |
| Melting point / freezing point          | No data available | None known              |
| Initial boiling point and boiling range | No data available | None known              |
| Flash point                             | No data available | None known              |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapor pressure                          | No data available | None known              |
| Relative vapor density                  | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Soluble in water  | None known              |
| Solubility(ies)                         | No data available | None known              |
| Partition coefficient                   | No data available | None known              |
| Autoignition temperature                | No data available | None known              |
| Decomposition temperature               | No data available | None known              |
| Kinematic viscosity                     | No data available | None known              |
| Dynamic viscosity                       | No data available | None known              |

### Other information

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

### Information on likely routes of exposure

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

**Symptoms related to the physical, chemical and toxicological characteristics**

**Symptoms** No information available.

**Acute toxicity** No information available.

**Numerical measures of toxicity**

The following ATE values have been calculated for the mixture

ATEmix (oral) 8,878.40 mg/kg

ATEmix (inhalation-dust/mist) 7.97 mg/l

**Component Information**

| Chemical name                | Oral LD50            | Dermal LD50              | Inhalation LC50       |
|------------------------------|----------------------|--------------------------|-----------------------|
| Sodium chloride<br>7647-14-5 | = 3550 mg/kg ( Rat ) | > 10000 mg/kg ( Rabbit ) | > 42 mg/L ( Rat ) 1 h |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** No information available.

**Serious eye damage/eye irritation** No information available.

**Respiratory or skin sensitization** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

**STOT - repeated exposure** No information available.

**Aspiration hazard** No information available.

**Other adverse effects** No information available.

**Interactive effects** No information available.

**12. Ecological information****Ecotoxicity**

| Chemical name                | Algae/aquatic plants | Fish                                                                    | Toxicity to microorganisms | Crustacea                                                             |
|------------------------------|----------------------|-------------------------------------------------------------------------|----------------------------|-----------------------------------------------------------------------|
| Sodium chloride<br>7647-14-5 | -                    | LC50: 5560 - 6080mg/L<br>(96h, Lepomis macrochirus)<br>LC50: =12946mg/L | -                          | EC50: =1000mg/L (48h, Daphnia magna)<br>EC50: 340.7 - 469.2mg/L (48h, |

|  |  |                                                                                                                                                                                                                                               |  |                |
|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------|
|  |  | (96h, Lepomis macrochirus)<br>LC50: 6020 - 7070mg/L<br>(96h, Pimephales promelas)<br>LC50: =7050mg/L (96h, Pimephales promelas)<br>LC50: 6420 - 6700mg/L<br>(96h, Pimephales promelas)<br>LC50: 4747 - 7824mg/L<br>(96h, Oncorhynchus mykiss) |  | Daphnia magna) |
|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------|

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

**Other adverse effects** No information available.

### 13. Disposal considerations

#### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

### 14. Transport information

DOT Not regulated

TDG Not regulated

MEX Not regulated

ICAO (air) Not regulated

IATA Not regulated

IMDG Not regulated

### 15. Regulatory information

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

International Inventories Contact supplier for inventory compliance status

**US Federal Regulations****TSCA**

All chemical components are listed on the TSCA Inventory.

**SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

**CAA (Clean Air Act)**

This product does not contain any substances regulated as pollutants pursuant to Clean Air Act (CAA).

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**US State Regulations****California Proposition 65**

This product does not contain any Proposition 65 chemicals

**U.S. State Right-to-Know Regulations**

This product does not contain any substances regulated under applicable state right-to-know regulations

**U.S. EPA Label Information**

**EPA Pesticide Registration Number** Not applicable

**16. Other information**

|             |                         |                       |                           |                              |
|-------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> - |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend**

|                |                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------|
| <b>ADN</b>     | Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways (Europe) |
| <b>ADR</b>     | Agreement concerning the International Carriage of Dangerous Goods by Road (Europe)             |
| <b>ATE</b>     | Acute Toxicity Estimate                                                                         |
| <b>ASTM</b>    | American Society for the Testing of Materials                                                   |
| <b>BEL</b>     | Biological exposure limits                                                                      |
| <b>bw</b>      | Body weight                                                                                     |
| <b>Ceiling</b> | Maximum limit value                                                                             |
| <b>CMR</b>     | Carcinogen, Mutagen or Reproductive Toxicant                                                    |
| <b>DOT</b>     | Department of Transportation (United States)                                                    |

|                |                                                                                     |
|----------------|-------------------------------------------------------------------------------------|
| <b>EmS</b>     | Emergency Schedule                                                                  |
| <b>EPA</b>     | Environmental Protection Agency                                                     |
| <b>GHS</b>     | Globally Harmonized System                                                          |
| <b>HMIS</b>    | Hazardous Materials Identification System                                           |
| <b>IATA</b>    | International Air Transport Association                                             |
| <b>ICAO</b>    | International Civil Aviation Organization                                           |
| <b>IMDG</b>    | International Maritime Dangerous Goods                                              |
| <b>IMO</b>     | International Maritime Organization                                                 |
| <b>ISO</b>     | International Organization for Standardization                                      |
| <b>LC50</b>    | Lethal Concentration to 50% of a test population                                    |
| <b>LD50</b>    | Lethal Dose to 50% of a test population (Median Lethal Dose)                        |
| <b>MARPOL</b>  | International Convention for the Prevention of Pollution from Ships                 |
| <b>NFPA</b>    | National Fire Protection Association                                                |
| <b>NIOSH</b>   | National Institute for Occupational Safety and Health                               |
| <b>n.o.s.</b>  | Not Otherwise Specified                                                             |
| <b>NOAEL</b>   | No Observed Adverse Effect Level                                                    |
| <b>OECD</b>    | Organization for Economic Cooperation and Development                               |
| <b>OEL</b>     | Occupational exposure limits                                                        |
| <b>OSHA</b>    | Occupational Safety and Health Administration of the US Department of Labor         |
| <b>PBT</b>     | Persistent, Bioaccumulative and Toxic substance                                     |
| <b>PMT</b>     | Persistent, Mobile and Toxic                                                        |
| <b>PPE</b>     | Personal protective equipment                                                       |
| <b>RID</b>     | Agreement concerning the International Carriage of Dangerous Goods by Rail (Europe) |
| <b>SARA</b>    | Superfund Amendments and Reauthorization Act                                        |
| <b>SDS</b>     | Safety Data Sheet                                                                   |
| <b>STEL</b>    | Short Term Exposure Limit                                                           |
| <b>STOT RE</b> | Specific target organ toxicity - Repeated exposure                                  |
| <b>STOT SE</b> | Specific target organ toxicity - Single exposure                                    |
| <b>TDG</b>     | Transport of Dangerous Goods (Canada)                                               |
| <b>TSCA</b>    | Toxic Substances Control Act (United States)                                        |
| <b>TWA</b>     | Time-Weighted Average                                                               |
| <b>UN</b>      | United Nations                                                                      |
| <b>VOC</b>     | Volatile organic compounds                                                          |
| <b>vPvB</b>    | Very Persistent and Very Bioaccumulative                                            |
| <b>vPvM</b>    | Very Persistent and Very Mobile                                                     |
| <b>Sk</b>      | Skin notation                                                                       |

#### Key literature references and sources for data used to compile the SDS

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.  
**Revision date** 14-May-2025  
**Revision Note** Reviewed existing information and made minor updates.  
**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other

materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
US OSHA Hazard Communication Standard (29 CFR 1910.1200)

Revision date 14-May-2025

Revision Number 2.1

## 1. Identification

### Product identifier

Product Name R6 - Sample Diluent

### Other means of identification

Catalog Number(s) 5145D

### Recommended use of the chemical and restrictions on use

Recommended use Laboratory chemicals

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Life Science  
2000 Alfred Nobel Drive  
Hercules, California 94547

Technical Service 1-800-424-6723  
support@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification of the substance or mixture

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200)

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

#### Hazard statements

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200)

### Hazards classified under paragraph (d)(1)(ii) of 1910.1200

No information available.

### Other information

No information available.

### 3. Composition/information on ingredients

#### Substance

Not applicable.

#### Mixture

The product contains no substances which at their given concentration, are considered to be hazardous to health

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

### 4. First-aid measures

#### Description of first aid measures

|              |                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                    |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a physician. |
| Skin contact | Wash skin with soap and water.                                                                                          |
| Ingestion    | Rinse mouth.                                                                                                            |

#### Most important symptoms and effects, both acute and delayed

|                     |                           |
|---------------------|---------------------------|
| Symptoms            | No information available. |
| Effects of Exposure | No information available. |

#### Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

### 5. Fire-fighting measures

|                                                                |                                                                                                                                       |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                               |
| Unsuitable extinguishing media                                 | Do not scatter spilled material with high pressure water streams.                                                                     |
| Specific hazards arising from the chemical                     | No information available.                                                                                                             |
| Explosion data                                                 |                                                                                                                                       |
| Sensitivity to mechanical impact                               | None.                                                                                                                                 |
| Sensitivity to static discharge                                | None.                                                                                                                                 |
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.<br>Use personal protection equipment. |

### 6. Accidental release measures

#### Personal precautions, protective equipment and emergency procedures

**Personal precautions** Ensure adequate ventilation.

#### **Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

### **7. Handling and storage**

#### **Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

#### **Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store according to product and label instructions.

### **8. Exposure controls/personal protection**

#### **Working area parameters, subject to mandatory control (MAC or TSEL)**

**Exposure Limits** This product, as supplied, contains materials that do not have reportable occupational exposure limits or are not subject to the reporting requirements of the local jurisdiction.

**Biological occupational exposure limits** This product, as supplied, contains materials that do not have reportable biological exposure limits or are not subject to the reporting requirements of the local jurisdiction.

#### **Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

#### **Individual protection measures, such as personal protective equipment**

**Eye/face protection** Appropriate eye/face protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Hand protection** Appropriate hand protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Skin and body protection** Appropriate skin and body protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Respiratory protection** Appropriate respiratory protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                                |                  |
|--------------------------------|------------------|
| Appearance                     | aqueous solution |
| Physical state                 | Liquid           |
| Color                          | colorless        |
| Odor (includes odor threshold) | Odorless         |

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      | 7.4               |                         |
| Melting point / freezing point          | 0 °C / 32 °F      |                         |
| Initial boiling point and boiling range | 100 °C / 212 °F   |                         |
| Flash point                             | No data available | None known              |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapor pressure                          | No data available | None known              |
| Relative vapor density                  | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Miscible in water | None known              |
| Solubility(ies)                         | No data available | None known              |
| Partition coefficient                   | No data available | None known              |
| Autoignition temperature                | No data available | None known              |
| Decomposition temperature               | No data available | None known              |
| Kinematic viscosity                     | No data available | None known              |
| Dynamic viscosity                       | No data available | None known              |

### Other information

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

### Information on likely routes of exposure

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

**Symptoms related to the physical, chemical and toxicological characteristics**

Symptoms No information available.

Acute toxicity No information available.

Numerical measures of toxicity

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

Skin corrosion/irritation No information available.

Serious eye damage/eye irritation No information available.

Respiratory or skin sensitization No information available.

Germ cell mutagenicity No information available.

Carcinogenicity No information available.

Reproductive toxicity No information available.

STOT - single exposure No information available.

STOT - repeated exposure No information available.

Aspiration hazard No information available.

Other adverse effects No information available.

Interactive effects No information available.

**12. Ecological information**

Ecotoxicity

Persistence and degradability No information available.

Bioaccumulation There is no data for this product.

Other adverse effects No information available.

### 13. Disposal considerations

#### Disposal methods

|                                            |                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| <b>Contaminated packaging</b>              | Do not reuse empty containers.                                                                                  |

### 14. Transport information

|                          |               |
|--------------------------|---------------|
| <b><u>DOT</u></b>        | Not regulated |
| <b><u>TDG</u></b>        | Not regulated |
| <b><u>MEX</u></b>        | Not regulated |
| <b><u>ICAO (air)</u></b> | Not regulated |
| <b><u>IATA</u></b>       | Not regulated |
| <b><u>IMDG</u></b>       | Not regulated |

### 15. Regulatory information

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories** Contact supplier for inventory compliance status

#### US Federal Regulations

##### **TSCA**

All chemical components are listed on the TSCA Inventory.

##### **SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

##### **SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

##### **CWA (Clean Water Act)**

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

##### **CAA (Clean Air Act)**

This product does not contain any substances regulated as pollutants pursuant to Clean Air Act (CAA).

##### **CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive

Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

### US State Regulations

#### **California Proposition 65**

This product does not contain any Proposition 65 chemicals

### **U.S. State Right-to-Know Regulations**

| Chemical name      | New Jersey | Massachusetts | Pennsylvania |
|--------------------|------------|---------------|--------------|
| Water<br>7732-18-5 | -          | -             | X            |

### U.S. EPA Label Information

**EPA Pesticide Registration Number** Not applicable

## **16. Other information**

|             |                         |                       |                           |                              |
|-------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> - |

### **Key or legend to abbreviations and acronyms used in the safety data sheet**

#### **Legend**

|                |                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------|
| <b>ADN</b>     | Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways (Europe) |
| <b>ADR</b>     | Agreement concerning the International Carriage of Dangerous Goods by Road (Europe)             |
| <b>ATE</b>     | Acute Toxicity Estimate                                                                         |
| <b>ASTM</b>    | American Society for the Testing of Materials                                                   |
| <b>BEL</b>     | Biological exposure limits                                                                      |
| <b>bw</b>      | Body weight                                                                                     |
| <b>Ceiling</b> | Maximum limit value                                                                             |
| <b>CMR</b>     | Carcinogen, Mutagen or Reproductive Toxicant                                                    |
| <b>DOT</b>     | Department of Transportation (United States)                                                    |
| <b>EmS</b>     | Emergency Schedule                                                                              |
| <b>EPA</b>     | Environmental Protection Agency                                                                 |
| <b>GHS</b>     | Globally Harmonized System                                                                      |
| <b>HMIS</b>    | Hazardous Materials Identification System                                                       |
| <b>IATA</b>    | International Air Transport Association                                                         |
| <b>ICAO</b>    | International Civil Aviation Organization                                                       |
| <b>IMDG</b>    | International Maritime Dangerous Goods                                                          |
| <b>IMO</b>     | International Maritime Organization                                                             |
| <b>ISO</b>     | International Organization for Standardization                                                  |
| <b>LC50</b>    | Lethal Concentration to 50% of a test population                                                |
| <b>LD50</b>    | Lethal Dose to 50% of a test population (Median Lethal Dose)                                    |
| <b>MARPOL</b>  | International Convention for the Prevention of Pollution from Ships                             |
| <b>NFPA</b>    | National Fire Protection Association                                                            |
| <b>NIOSH</b>   | National Institute for Occupational Safety and Health                                           |
| <b>n.o.s.</b>  | Not Otherwise Specified                                                                         |
| <b>NOAEL</b>   | No Observed Adverse Effect Level                                                                |
| <b>OECD</b>    | Organization for Economic Cooperation and Development                                           |
| <b>OEL</b>     | Occupational exposure limits                                                                    |
| <b>OSHA</b>    | Occupational Safety and Health Administration of the US Department of Labor                     |
| <b>PBT</b>     | Persistent, Bioaccumulative and Toxic substance                                                 |
| <b>PMT</b>     | Persistent, Mobile and Toxic                                                                    |
| <b>PPE</b>     | Personal protective equipment                                                                   |
| <b>RID</b>     | Agreement concerning the International Carriage of Dangerous Goods by Rail (Europe)             |
| <b>SARA</b>    | Superfund Amendments and Reauthorization Act                                                    |

|                |                                                    |
|----------------|----------------------------------------------------|
| <b>SDS</b>     | Safety Data Sheet                                  |
| <b>STEL</b>    | Short Term Exposure Limit                          |
| <b>STOT RE</b> | Specific target organ toxicity - Repeated exposure |
| <b>STOT SE</b> | Specific target organ toxicity - Single exposure   |
| <b>TDG</b>     | Transport of Dangerous Goods (Canada)              |
| <b>TSCA</b>    | Toxic Substances Control Act (United States)       |
| <b>TWA</b>     | Time-Weighted Average                              |
| <b>UN</b>      | United Nations                                     |
| <b>VOC</b>     | Volatile organic compounds                         |
| <b>vPvB</b>    | Very Persistent and Very Bioaccumulative           |
| <b>vPvM</b>    | Very Persistent and Very Mobile                    |
| <b>Sk</b>      | Skin notation                                      |

#### Key literature references and sources for data used to compile the SDS

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.  
**Revision date** 14-May-2025  
**Revision Note** Reviewed existing information and made minor updates.

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
US OSHA Hazard Communication Standard (29 CFR 1910.1200)

Revision date 14-May-2025

Revision Number 2.1

## 1. Identification

### Product identifier

**Product Name** R7- Concentrated (10X) Conjugate

### Other means of identification

**Catalog Number(s)** 5145E

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Life Science  
2000 Alfred Nobel Drive  
Hercules, California 94547

#### **Technical Service**

1-800-424-6723  
support@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification of the substance or mixture

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200)

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

#### Hazard statements

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200)

### **Hazards classified under paragraph (d)(1)(ii) of 1910.1200**

No information available.

### Other information

Contains animal source material. (Mouse). (Cattle).

### 3. Composition/information on ingredients

#### Substance

Not applicable.

#### Mixture

| Chemical name      | CAS No. | Weight-% | Trade secret |
|--------------------|---------|----------|--------------|
| 1,2,3-Propanetriol | 56-81-5 | 10 - 20  | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

### 4. First-aid measures

#### Description of first aid measures

|              |                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                    |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a physician. |
| Skin contact | Wash skin with soap and water.                                                                                          |
| Ingestion    | Rinse mouth.                                                                                                            |

#### Most important symptoms and effects, both acute and delayed

|                     |                           |
|---------------------|---------------------------|
| Symptoms            | No information available. |
| Effects of Exposure | No information available. |

#### Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

### 5. Fire-fighting measures

|                                                                |                                                                                                                                       |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                               |
| Unsuitable extinguishing media                                 | Do not scatter spilled material with high pressure water streams.                                                                     |
| Specific hazards arising from the chemical                     | No information available.                                                                                                             |
| Explosion data                                                 |                                                                                                                                       |
| Sensitivity to mechanical impact                               | None.                                                                                                                                 |
| Sensitivity to static discharge                                | None.                                                                                                                                 |
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.<br>Use personal protection equipment. |

### 6. Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Ensure adequate ventilation.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

**7. Handling and storage****Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store according to product and label instructions.

**8. Exposure controls/personal protection****Working area parameters, subject to mandatory control (MAC or TSEL)****Exposure Limits**

| Chemical name                 | ACGIH TLV | OSHA PEL                                                                                                                                                                                                                               | NIOSH |
|-------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1,2,3-Propanetriol<br>56-81-5 | -         | TWA: 15 mg/m <sup>3</sup> mist, total particulate<br>TWA: 5 mg/m <sup>3</sup> mist, respirable fraction<br>(vacated) TWA: 10 mg/m <sup>3</sup> mist, total particulate<br>(vacated) TWA: 5 mg/m <sup>3</sup> mist, respirable fraction | -     |

**Biological occupational exposure limits** This product, as supplied, contains materials that do not have reportable biological exposure limits or are not subject to the reporting requirements of the local jurisdiction.

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** Appropriate eye/face protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Hand protection** Appropriate hand protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

|                                 |                                                                                                                                                                                                                                                                                               |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b> | Appropriate skin and body protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.                                                                                                         |
| <b>Respiratory protection</b>   | Appropriate respiratory protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                                       |                  |
|---------------------------------------|------------------|
| <b>Appearance</b>                     | aqueous solution |
| <b>Physical state</b>                 | Liquid           |
| <b>Color</b>                          | colorless        |
| <b>Odor (includes odor threshold)</b> | Odorless         |

| <u>Property</u>                                | <u>Values</u>     | <u>Remarks • Method</u> |
|------------------------------------------------|-------------------|-------------------------|
| <b>pH</b>                                      | 7.4               |                         |
| <b>Melting point / freezing point</b>          | No data available | None known              |
| <b>Initial boiling point and boiling range</b> | > 100 °C / 212 °F |                         |
| <b>Flash point</b>                             | No data available | None known              |
| <b>Evaporation rate</b>                        | No data available | None known              |
| <b>Flammability</b>                            | No data available | None known              |
| <b>Flammability Limit in Air</b>               |                   | None known              |
| <b>Upper flammability or explosive limits</b>  | No data available |                         |
| <b>Lower flammability or explosive limits</b>  | No data available |                         |
| <b>Vapor pressure</b>                          | No data available | None known              |
| <b>Relative vapor density</b>                  | No data available | None known              |
| <b>Relative density</b>                        | No data available | None known              |
| <b>Water solubility</b>                        | Miscible in water | None known              |
| <b>Solubility(ies)</b>                         | No data available | None known              |
| <b>Partition coefficient</b>                   | No data available | None known              |
| <b>Autoignition temperature</b>                | No data available | None known              |
| <b>Decomposition temperature</b>               | No data available | None known              |
| <b>Kinematic viscosity</b>                     | No data available | None known              |
| <b>Dynamic viscosity</b>                       | No data available | None known              |

### Other information

## 10. Stability and reactivity

|                                           |                                           |
|-------------------------------------------|-------------------------------------------|
| <b>Reactivity</b>                         | No information available.                 |
| <b>Chemical stability</b>                 | Stable under normal conditions.           |
| <b>Possibility of hazardous reactions</b> | None under normal processing.             |
| <b>Conditions to avoid</b>                | None known based on information supplied. |
| <b>Incompatible materials</b>             | None known based on information supplied. |
| <b>Hazardous decomposition products</b>   | None known based on information supplied. |

### Information on likely routes of exposure

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** Specific test data for the substance or mixture is not available.

**Ingestion** Specific test data for the substance or mixture is not available.

#### **Symptoms related to the physical, chemical and toxicological characteristics**

**Symptoms** No information available.

**Acute toxicity** No information available.

**Numerical measures of toxicity**

#### **Component Information**

| Chemical name                 | Oral LD50             | Dermal LD50          | Inhalation LC50         |
|-------------------------------|-----------------------|----------------------|-------------------------|
| 1,2,3-Propanetriol<br>56-81-5 | = 27200 mg/kg ( Rat ) | > 10 g/kg ( Rabbit ) | > 5.85 mg/L ( Rat ) 4 h |

#### **Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** No information available.

**Serious eye damage/eye irritation** No information available.

**Respiratory or skin sensitization** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

**STOT - repeated exposure** No information available.

**Target organ effects** Kidney, Respiratory system, Eyes, Skin.

**Aspiration hazard** No information available.

**Other adverse effects** No information available.

**Interactive effects** No information available.

## **12. Ecological information**

**Ecotoxicity**

| Chemical name                 | Algae/aquatic plants | Fish                                            | Toxicity to microorganisms | Crustacea |
|-------------------------------|----------------------|-------------------------------------------------|----------------------------|-----------|
| 1,2,3-Propanetriol<br>56-81-5 | -                    | LC50: 51 - 57mL/L (96h,<br>Oncorhynchus mykiss) | -                          | -         |

**Persistence and degradability** No information available.

#### Bioaccumulation

#### Component Information

| Chemical name                 | Partition coefficient |
|-------------------------------|-----------------------|
| 1,2,3-Propanetriol<br>56-81-5 | -1.75                 |

**Other adverse effects** No information available.

### 13. Disposal considerations

#### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

### 14. Transport information

**DOT** Not regulated

**TDG** Not regulated

**MEX** Not regulated

**ICAO (air)** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

### 15. Regulatory information

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories** Contact supplier for inventory compliance status

#### US Federal Regulations

**TSCA**

All chemical components are listed on the TSCA Inventory.

**SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

**CAA (Clean Air Act)**

This product does not contain any substances regulated as pollutants pursuant to Clean Air Act (CAA).

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**US State Regulations****California Proposition 65**

This product does not contain any Proposition 65 chemicals

**U.S. State Right-to-Know Regulations**

| Chemical name                 | New Jersey | Massachusetts | Pennsylvania |
|-------------------------------|------------|---------------|--------------|
| Water<br>7732-18-5            | -          | -             | X            |
| 1,2,3-Propanetriol<br>56-81-5 | X          | X             | X            |

**U.S. EPA Label Information**

**EPA Pesticide Registration Number** Not applicable

**16. Other information**

|             |                         |                       |                           |                              |
|-------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> - |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend**

|                |                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------|
| <b>ADN</b>     | Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways (Europe) |
| <b>ADR</b>     | Agreement concerning the International Carriage of Dangerous Goods by Road (Europe)             |
| <b>ATE</b>     | Acute Toxicity Estimate                                                                         |
| <b>ASTM</b>    | American Society for the Testing of Materials                                                   |
| <b>BEL</b>     | Biological exposure limits                                                                      |
| <b>bw</b>      | Body weight                                                                                     |
| <b>Ceiling</b> | Maximum limit value                                                                             |

|                |                                                                                     |
|----------------|-------------------------------------------------------------------------------------|
| <b>CMR</b>     | Carcinogen, Mutagen or Reproductive Toxicant                                        |
| <b>DOT</b>     | Department of Transportation (United States)                                        |
| <b>EmS</b>     | Emergency Schedule                                                                  |
| <b>EPA</b>     | Environmental Protection Agency                                                     |
| <b>GHS</b>     | Globally Harmonized System                                                          |
| <b>HMIS</b>    | Hazardous Materials Identification System                                           |
| <b>IATA</b>    | International Air Transport Association                                             |
| <b>ICAO</b>    | International Civil Aviation Organization                                           |
| <b>IMDG</b>    | International Maritime Dangerous Goods                                              |
| <b>IMO</b>     | International Maritime Organization                                                 |
| <b>ISO</b>     | International Organization for Standardization                                      |
| <b>LC50</b>    | Lethal Concentration to 50% of a test population                                    |
| <b>LD50</b>    | Lethal Dose to 50% of a test population (Median Lethal Dose)                        |
| <b>MARPOL</b>  | International Convention for the Prevention of Pollution from Ships                 |
| <b>NFPA</b>    | National Fire Protection Association                                                |
| <b>NIOSH</b>   | National Institute for Occupational Safety and Health                               |
| <b>n.o.s.</b>  | Not Otherwise Specified                                                             |
| <b>NOAEL</b>   | No Observed Adverse Effect Level                                                    |
| <b>OECD</b>    | Organization for Economic Cooperation and Development                               |
| <b>OEL</b>     | Occupational exposure limits                                                        |
| <b>OSHA</b>    | Occupational Safety and Health Administration of the US Department of Labor         |
| <b>PBT</b>     | Persistent, Bioaccumulative and Toxic substance                                     |
| <b>PMT</b>     | Persistent, Mobile and Toxic                                                        |
| <b>PPE</b>     | Personal protective equipment                                                       |
| <b>RID</b>     | Agreement concerning the International Carriage of Dangerous Goods by Rail (Europe) |
| <b>SARA</b>    | Superfund Amendments and Reauthorization Act                                        |
| <b>SDS</b>     | Safety Data Sheet                                                                   |
| <b>STEL</b>    | Short Term Exposure Limit                                                           |
| <b>STOT RE</b> | Specific target organ toxicity - Repeated exposure                                  |
| <b>STOT SE</b> | Specific target organ toxicity - Single exposure                                    |
| <b>TDG</b>     | Transport of Dangerous Goods (Canada)                                               |
| <b>TSCA</b>    | Toxic Substances Control Act (United States)                                        |
| <b>TWA</b>     | Time-Weighted Average                                                               |
| <b>UN</b>      | United Nations                                                                      |
| <b>VOC</b>     | Volatile organic compounds                                                          |
| <b>vPvB</b>    | Very Persistent and Very Bioaccumulative                                            |
| <b>vPvM</b>    | Very Persistent and Very Mobile                                                     |
| <b>Sk</b>      | Skin notation                                                                       |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.  
**Revision date** 14-May-2025  
**Revision Note** Reviewed existing information and made minor updates.  
**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage,

transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
US OSHA Hazard Communication Standard (29 CFR 1910.1200)

Revision date 14-May-2025

Revision Number 1.1

## 1. Identification

### Product identifier

**Product Name** R8 - Peroxidase Substrate Buffer

### Other means of identification

**Catalog Number(s)** 7435S

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Life Science  
2000 Alfred Nobel Drive  
Hercules, California 94547

#### **Technical Service**

1-800-424-6723  
support@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification of the substance or mixture

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200).

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

#### Hazard statements

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200).

### **Hazards classified under paragraph (d)(1)(ii) of 1910.1200**

No information available.

### Other information

No information available.

### 3. Composition/information on ingredients

#### Substance

Not applicable.

#### Mixture

| Chemical name  | CAS No.  | Weight-% | Trade secret |
|----------------|----------|----------|--------------|
| Citric acid    | 77-92-9  | 1 - 2.5  | *            |
| Sodium acetate | 127-09-3 | 1 - 2.5  | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

### 4. First-aid measures

#### Description of first aid measures

|              |                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                    |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a physician. |
| Skin contact | Wash skin with soap and water.                                                                                          |
| Ingestion    | Rinse mouth.                                                                                                            |

#### Most important symptoms and effects, both acute and delayed

Symptoms No information available.

Effects of Exposure No information available.

#### Indication of any immediate medical attention and special treatment needed

Note to physicians Treat symptomatically.

### 5. Fire-fighting measures

|                                                                |                                                                                                                                       |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                               |
| Unsuitable extinguishing media                                 | Do not scatter spilled material with high pressure water streams.                                                                     |
| Specific hazards arising from the chemical                     | No information available.                                                                                                             |
| Explosion data                                                 |                                                                                                                                       |
| Sensitivity to mechanical impact                               | None.                                                                                                                                 |
| Sensitivity to static discharge                                | None.                                                                                                                                 |
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.<br>Use personal protection equipment. |

### 6. Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Ensure adequate ventilation.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

**7. Handling and storage****Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store according to product and label instructions.

**8. Exposure controls/personal protection****Working area parameters, subject to mandatory control (MAC or TSEL)**

**Exposure Limits** The following ingredients are the only ingredients of the product above the cut-off level (or level that contributes to the hazard classification of the mixture) which have an exposure limit applicable in the region for which this safety data sheet is intended or other recommended limit. At this time, the other relevant constituents have no known exposure limits from the sources listed here.

**Biological occupational exposure limits** This product, as supplied, contains materials that do not have reportable biological exposure limits or are not subject to the reporting requirements of the local jurisdiction.

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** Appropriate eye/face protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Hand protection** Appropriate hand protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Skin and body protection** Appropriate skin and body protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Respiratory protection** Appropriate respiratory protection should be selected and used according to the chemical

nature, hazards and use of this product and safety requirements of the local jurisdiction. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                                |                  |
|--------------------------------|------------------|
| Appearance                     | aqueous solution |
| Physical state                 | Liquid           |
| Color                          | colorless        |
| Odor (includes odor threshold) | Odorless         |

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      | 4                 |                         |
| Melting point / freezing point          | No data available | None known              |
| Initial boiling point and boiling range | 100 °C / 212 °F   |                         |
| Flash point                             | No data available | None known              |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapor pressure                          | No data available | None known              |
| Relative vapor density                  | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Miscible in water | None known              |
| Solubility(ies)                         | No data available | None known              |
| Partition coefficient                   | No data available | None known              |
| Autoignition temperature                | No data available | None known              |
| Decomposition temperature               | No data available | None known              |
| Kinematic viscosity                     | No data available | None known              |
| Dynamic viscosity                       | No data available | None known              |

### Other information

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

### Information on likely routes of exposure

|             |                                                                   |
|-------------|-------------------------------------------------------------------|
| Inhalation  | Specific test data for the substance or mixture is not available. |
| Eye contact | Specific test data for the substance or mixture is not available. |

**Skin contact** Specific test data for the substance or mixture is not available.

**Ingestion** Specific test data for the substance or mixture is not available.

#### Symptoms related to the physical, chemical and toxicological characteristics

**Symptoms** No information available.

**Acute toxicity** No information available.

#### **Numerical measures of toxicity**

The following ATE values have been calculated for the mixture

|                               |                  |
|-------------------------------|------------------|
| ATEmix (oral)                 | 86,720.30 mg/kg  |
| ATEmix (dermal)               | 118,666.20 mg/kg |
| ATEmix (inhalation-dust/mist) | 424.20 mg/l      |

#### **Component Information**

| Chemical name              | Oral LD50            | Dermal LD50          | Inhalation LC50        |
|----------------------------|----------------------|----------------------|------------------------|
| Citric acid<br>77-92-9     | = 3 g/kg ( Rat )     | > 2000 mg/kg ( Rat ) | -                      |
| Sodium acetate<br>127-09-3 | = 3530 mg/kg ( Rat ) | > 10 g/kg ( Rabbit ) | > 5.6 mg/L ( Rat ) 4 h |

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Skin corrosion/irritation** No information available.

**Serious eye damage/eye irritation** No information available.

**Respiratory or skin sensitization** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

**STOT - repeated exposure** No information available.

**Aspiration hazard** No information available.

**Other adverse effects** No information available.

**Interactive effects** No information available.

## **12. Ecological information**

**Ecotoxicity** The environmental impact of this product has not been fully investigated.

| Chemical name              | Algae/aquatic plants | Fish                                               | Toxicity to microorganisms | Crustacea                                    |
|----------------------------|----------------------|----------------------------------------------------|----------------------------|----------------------------------------------|
| Citric acid<br>77-92-9     | -                    | LC50: =1516mg/L (96h, <i>Lepomis macrochirus</i> ) | -                          | -                                            |
| Sodium acetate<br>127-09-3 | -                    | LC50: >100mg/L (96h, <i>Danio rerio</i> )          | -                          | EC50: >1000mg/L (48h, <i>Daphnia magna</i> ) |

**Persistence and degradability** No information available.

#### Bioaccumulation

#### Component Information

| Chemical name          | Partition coefficient |
|------------------------|-----------------------|
| Citric acid<br>77-92-9 | -1.72                 |

**Other adverse effects** No information available.

### 13. Disposal considerations

#### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

### 14. Transport information

**DOT** Not regulated

**TDG** Not regulated

**MEX** Not regulated

**ICAO (air)** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

### 15. Regulatory information

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories** Contact supplier for inventory compliance status

**US Federal Regulations****TSCA**

All chemical components are listed on the TSCA Inventory.

**SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

This product contains the following substances which are regulated pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

**CAA (Clean Air Act)**

This product does not contain any substances regulated as pollutants pursuant to Clean Air Act (CAA).

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**US State Regulations****California Proposition 65**

This product does not contain any Proposition 65 chemicals

**U.S. State Right-to-Know Regulations**

| Chemical name                              | New Jersey | Massachusetts | Pennsylvania |
|--------------------------------------------|------------|---------------|--------------|
| Water<br>7732-18-5                         | -          | -             | X            |
| Dimethyl sulfoxide<br>67-68-5              | X          | -             | -            |
| Ethylenediaminetetraacetic acid<br>60-00-4 | X          | X             | X            |
| Hydrogen peroxide<br>7722-84-1             | X          | X             | X            |

**U.S. EPA Label Information**

**EPA Pesticide Registration Number** Not applicable

**16. Other information**

|             |                         |                       |                           |                              |
|-------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> - |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend**

**ADN** Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways (Europe)

|                |                                                                                     |
|----------------|-------------------------------------------------------------------------------------|
| <b>ADR</b>     | Agreement concerning the International Carriage of Dangerous Goods by Road (Europe) |
| <b>ATE</b>     | Acute Toxicity Estimate                                                             |
| <b>ASTM</b>    | American Society for the Testing of Materials                                       |
| <b>BEL</b>     | Biological exposure limits                                                          |
| <b>bw</b>      | Body weight                                                                         |
| <b>Ceiling</b> | Maximum limit value                                                                 |
| <b>CMR</b>     | Carcinogen, Mutagen or Reproductive Toxicant                                        |
| <b>DOT</b>     | Department of Transportation (United States)                                        |
| <b>EmS</b>     | Emergency Schedule                                                                  |
| <b>EPA</b>     | Environmental Protection Agency                                                     |
| <b>GHS</b>     | Globally Harmonized System                                                          |
| <b>HMIS</b>    | Hazardous Materials Identification System                                           |
| <b>IATA</b>    | International Air Transport Association                                             |
| <b>ICAO</b>    | International Civil Aviation Organization                                           |
| <b>IMDG</b>    | International Maritime Dangerous Goods                                              |
| <b>IMO</b>     | International Maritime Organization                                                 |
| <b>ISO</b>     | International Organization for Standardization                                      |
| <b>LC50</b>    | Lethal Concentration to 50% of a test population                                    |
| <b>LD50</b>    | Lethal Dose to 50% of a test population (Median Lethal Dose)                        |
| <b>MARPOL</b>  | International Convention for the Prevention of Pollution from Ships                 |
| <b>NFPA</b>    | National Fire Protection Association                                                |
| <b>NIOSH</b>   | National Institute for Occupational Safety and Health                               |
| <b>n.o.s.</b>  | Not Otherwise Specified                                                             |
| <b>NOAEL</b>   | No Observed Adverse Effect Level                                                    |
| <b>OECD</b>    | Organization for Economic Cooperation and Development                               |
| <b>OEL</b>     | Occupational exposure limits                                                        |
| <b>OSHA</b>    | Occupational Safety and Health Administration of the US Department of Labor         |
| <b>PBT</b>     | Persistent, Bioaccumulative and Toxic substance                                     |
| <b>PMT</b>     | Persistent, Mobile and Toxic                                                        |
| <b>PPE</b>     | Personal protective equipment                                                       |
| <b>RID</b>     | Agreement concerning the International Carriage of Dangerous Goods by Rail (Europe) |
| <b>SARA</b>    | Superfund Amendments and Reauthorization Act                                        |
| <b>SDS</b>     | Safety Data Sheet                                                                   |
| <b>STEL</b>    | Short Term Exposure Limit                                                           |
| <b>STOT RE</b> | Specific target organ toxicity - Repeated exposure                                  |
| <b>STOT SE</b> | Specific target organ toxicity - Single exposure                                    |
| <b>TDG</b>     | Transport of Dangerous Goods (Canada)                                               |
| <b>TSCA</b>    | Toxic Substances Control Act (United States)                                        |
| <b>TWA</b>     | Time-Weighted Average                                                               |
| <b>UN</b>      | United Nations                                                                      |
| <b>VOC</b>     | Volatile organic compounds                                                          |
| <b>vPvB</b>    | Very Persistent and Very Bioaccumulative                                            |
| <b>vPvM</b>    | Very Persistent and Very Mobile                                                     |
| <b>Sk</b>      | Skin notation                                                                       |

### Key literature references and sources for data used to compile the SDS

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Prepared By**

Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date**

14-May-2025

**Revision Note**

Reviewed existing information and made minor updates.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
US OSHA Hazard Communication Standard (29 CFR 1910.1200)

Revision date 14-May-2025

Revision Number 1.1

## 1. Identification

### Product identifier

**Product Name** R9 - Chromogen: TMB Solution

### Other means of identification

**Catalog Number(s)** 7436J

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Life Science  
2000 Alfred Nobel Drive  
Hercules, California 94547

#### **Technical Service**

1-800-424-6723  
support@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification of the substance or mixture

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200).

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

#### Hazard statements

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200).

### **Hazards classified under paragraph (d)(1)(ii) of 1910.1200**

No information available.

### Other information

Causes mild skin irritation.

### 3. Composition/information on ingredients

#### Substance

Not applicable.

#### Mixture

| Chemical name     | CAS No.   | Weight-%   | Trade secret |
|-------------------|-----------|------------|--------------|
| Hydrochloric acid | 7647-01-0 | 0.3 - 0.99 | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

### 4. First-aid measures

#### Description of first aid measures

|              |                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                    |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a physician. |
| Skin contact | Wash skin with soap and water.                                                                                          |
| Ingestion    | Rinse mouth.                                                                                                            |

#### Most important symptoms and effects, both acute and delayed

|                     |                                                     |
|---------------------|-----------------------------------------------------|
| Symptoms            | Prolonged contact may cause redness and irritation. |
| Effects of Exposure | No information available.                           |

#### Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

### 5. Fire-fighting measures

|                                                                |                                                                                                                                       |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                               |
| Unsuitable extinguishing media                                 | Do not scatter spilled material with high pressure water streams.                                                                     |
| Specific hazards arising from the chemical                     | No information available.                                                                                                             |
| Explosion data                                                 |                                                                                                                                       |
| Sensitivity to mechanical impact                               | None.                                                                                                                                 |
| Sensitivity to static discharge                                | None.                                                                                                                                 |
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.<br>Use personal protection equipment. |

### 6. Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Ensure adequate ventilation.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

**7. Handling and storage****Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store according to product and label instructions.

**8. Exposure controls/personal protection****Working area parameters, subject to mandatory control (MAC or TSEL)****Exposure Limits**

| Chemical name                  | ACGIH TLV      | OSHA PEL                                                                                                             | NIOSH                                                          |
|--------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Hydrochloric acid<br>7647-01-0 | Ceiling: 2 ppm | (vacated) Ceiling: 5 ppm<br>(vacated) Ceiling: 7 mg/m <sup>3</sup><br>Ceiling: 5 ppm<br>Ceiling: 7 mg/m <sup>3</sup> | IDLH: 50 ppm<br>Ceiling: 5 ppm<br>Ceiling: 7 mg/m <sup>3</sup> |

**Biological occupational exposure limits** This product, as supplied, contains materials that do not have reportable biological exposure limits or are not subject to the reporting requirements of the local jurisdiction.

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Hand protection** Wear suitable gloves.

**Skin and body protection** Wear suitable protective clothing.

**Respiratory protection** Appropriate respiratory protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                                |                  |
|--------------------------------|------------------|
| Appearance                     | aqueous solution |
| Physical state                 | Liquid           |
| Color                          | colorless        |
| Odor (includes odor threshold) | Odorless         |

| Property                                | Values            | Remarks • Method |
|-----------------------------------------|-------------------|------------------|
| pH                                      | 4                 |                  |
| Melting point / freezing point          | No data available | None known       |
| Initial boiling point and boiling range | > 100 °C / 212 °F |                  |
| Flash point                             | No data available | None known       |
| Evaporation rate                        | No data available | None known       |
| Flammability                            | No data available | None known       |
| Flammability Limit in Air               |                   | None known       |
| Upper flammability or explosive limits  | No data available |                  |
| Lower flammability or explosive limits  | No data available |                  |
| Vapor pressure                          | No data available | None known       |
| Relative vapor density                  | No data available | None known       |
| Relative density                        | No data available | None known       |
| Water solubility                        | Miscible in water |                  |
| Solubility(ies)                         | No data available | None known       |
| Partition coefficient                   | No data available | None known       |
| Autoignition temperature                | No data available | None known       |
| Decomposition temperature               | No data available | None known       |
| Kinematic viscosity                     | No data available | None known       |
| Dynamic viscosity                       | No data available | None known       |

### Other information

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

### Information on likely routes of exposure

|              |                                                                                                |
|--------------|------------------------------------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available.                              |
| Eye contact  | Specific test data for the substance or mixture is not available.                              |
| Skin contact | Specific test data for the substance or mixture is not available. Causes mild skin irritation. |

**Ingestion**

Specific test data for the substance or mixture is not available.

**Symptoms related to the physical, chemical and toxicological characteristics****Symptoms**

Prolonged contact may cause redness and irritation.

**Acute toxicity**

No information available.

**Numerical measures of toxicity****Component Information**

| Chemical name                  | Oral LD50               | Dermal LD50             | Inhalation LC50         |
|--------------------------------|-------------------------|-------------------------|-------------------------|
| Hydrochloric acid<br>7647-01-0 | 238 - 277 mg/kg ( Rat ) | > 5010 mg/kg ( Rabbit ) | = 1.68 mg/L ( Rat ) 1 h |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure****Skin corrosion/irritation**

Classification based on data available for ingredients. Causes mild skin irritation.

**Serious eye damage/eye irritation**

No information available.

**Respiratory or skin sensitization**

No information available.

**Germ cell mutagenicity**

No information available.

**Carcinogenicity**

No information available.

The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Chemical name                  | ACGIH                                          | IARC    | NTP | OSHA |
|--------------------------------|------------------------------------------------|---------|-----|------|
| Hydrochloric acid<br>7647-01-0 | A4 - Not Classifiable as<br>a Human Carcinogen | Group 3 | -   | X    |

**Legend****ACGIH (American Conference of Governmental Industrial Hygienists)**

A4 - Not Classifiable as a Human Carcinogen

**IARC (International Agency for Research on Cancer)**

Group 3 - Not Classifiable as to Carcinogenicity in Humans

**Occupational Safety and Health Administration of the US Department of Labor**

X - Present

**Reproductive toxicity**

No information available.

**STOT - single exposure**

No information available.

**STOT - repeated exposure**

No information available.

**Aspiration hazard**

No information available.

**Other adverse effects**

No information available.

**Interactive effects**

No information available.

**12. Ecological information**

|                                      |                                                                           |
|--------------------------------------|---------------------------------------------------------------------------|
| <b>Ecotoxicity</b>                   | The environmental impact of this product has not been fully investigated. |
| <b>Persistence and degradability</b> | No information available.                                                 |
| <b>Bioaccumulation</b>               | There is no data for this product.                                        |
| <b>Other adverse effects</b>         | No information available.                                                 |

### 13. Disposal considerations

#### Disposal methods

|                                            |                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| <b>Contaminated packaging</b>              | Do not reuse empty containers.                                                                                  |

### 14. Transport information

|                          |               |
|--------------------------|---------------|
| <b><u>DOT</u></b>        | Not regulated |
| <b><u>TDG</u></b>        | Not regulated |
| <b><u>MEX</u></b>        | Not regulated |
| <b><u>ICAO (air)</u></b> | Not regulated |
| <b><u>IATA</u></b>       | Not regulated |
| <b><u>IMDG</u></b>       | Not regulated |

### 15. Regulatory information

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories** Contact supplier for inventory compliance status

#### US Federal Regulations

##### **TSCA**

All chemical components are listed on the TSCA Inventory.

##### **SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical

or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

This product contains the following substances which are regulated pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

| Chemical name                  | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants | CWA - Hazardous Substances |
|--------------------------------|-----------------------------|------------------------|---------------------------|----------------------------|
| Hydrochloric acid<br>7647-01-0 | 5000 lb                     | -                      | -                         | X                          |

**CAA (Clean Air Act)**

This product does not contain any substances regulated as pollutants pursuant to Clean Air Act (CAA).

**CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302).

| Chemical name                  | Hazardous Substances RQs | Extremely Hazardous Substances RQs | Reportable Quantity (RQ)                   |
|--------------------------------|--------------------------|------------------------------------|--------------------------------------------|
| Hydrochloric acid<br>7647-01-0 | 5000 lb                  | 5000 lb                            | RQ 5000 lb final RQ<br>RQ 2270 kg final RQ |

**US State Regulations****California Proposition 65**

This product does not contain any Proposition 65 chemicals

**U.S. State Right-to-Know Regulations**

| Chemical name                  | New Jersey | Massachusetts | Pennsylvania |
|--------------------------------|------------|---------------|--------------|
| Water<br>7732-18-5             | -          | -             | X            |
| Hydrochloric acid<br>7647-01-0 | X          | X             | X            |

**U.S. EPA Label Information**

**EPA Pesticide Registration Number** Not applicable

**16. Other information**

|             |                         |                       |                           |                              |
|-------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> - |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend**

|             |                                                                                                 |
|-------------|-------------------------------------------------------------------------------------------------|
| <b>ADN</b>  | Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways (Europe) |
| <b>ADR</b>  | Agreement concerning the International Carriage of Dangerous Goods by Road (Europe)             |
| <b>ATE</b>  | Acute Toxicity Estimate                                                                         |
| <b>ASTM</b> | American Society for the Testing of Materials                                                   |
| <b>BEL</b>  | Biological exposure limits                                                                      |
| <b>bw</b>   | Body weight                                                                                     |

|                |                                                                                     |
|----------------|-------------------------------------------------------------------------------------|
| <b>Ceiling</b> | Maximum limit value                                                                 |
| <b>CMR</b>     | Carcinogen, Mutagen or Reproductive Toxicant                                        |
| <b>DOT</b>     | Department of Transportation (United States)                                        |
| <b>EmS</b>     | Emergency Schedule                                                                  |
| <b>EPA</b>     | Environmental Protection Agency                                                     |
| <b>GHS</b>     | Globally Harmonized System                                                          |
| <b>HMIS</b>    | Hazardous Materials Identification System                                           |
| <b>IATA</b>    | International Air Transport Association                                             |
| <b>ICAO</b>    | International Civil Aviation Organization                                           |
| <b>IMDG</b>    | International Maritime Dangerous Goods                                              |
| <b>IMO</b>     | International Maritime Organization                                                 |
| <b>ISO</b>     | International Organization for Standardization                                      |
| <b>LC50</b>    | Lethal Concentration to 50% of a test population                                    |
| <b>LD50</b>    | Lethal Dose to 50% of a test population (Median Lethal Dose)                        |
| <b>MARPOL</b>  | International Convention for the Prevention of Pollution from Ships                 |
| <b>NFPA</b>    | National Fire Protection Association                                                |
| <b>NIOSH</b>   | National Institute for Occupational Safety and Health                               |
| <b>n.o.s.</b>  | Not Otherwise Specified                                                             |
| <b>NOAEL</b>   | No Observed Adverse Effect Level                                                    |
| <b>OECD</b>    | Organization for Economic Cooperation and Development                               |
| <b>OEL</b>     | Occupational exposure limits                                                        |
| <b>OSHA</b>    | Occupational Safety and Health Administration of the US Department of Labor         |
| <b>PBT</b>     | Persistent, Bioaccumulative and Toxic substance                                     |
| <b>PMT</b>     | Persistent, Mobile and Toxic                                                        |
| <b>PPE</b>     | Personal protective equipment                                                       |
| <b>RID</b>     | Agreement concerning the International Carriage of Dangerous Goods by Rail (Europe) |
| <b>SARA</b>    | Superfund Amendments and Reauthorization Act                                        |
| <b>SDS</b>     | Safety Data Sheet                                                                   |
| <b>STEL</b>    | Short Term Exposure Limit                                                           |
| <b>STOT RE</b> | Specific target organ toxicity - Repeated exposure                                  |
| <b>STOT SE</b> | Specific target organ toxicity - Single exposure                                    |
| <b>TDG</b>     | Transport of Dangerous Goods (Canada)                                               |
| <b>TSCA</b>    | Toxic Substances Control Act (United States)                                        |
| <b>TWA</b>     | Time-Weighted Average                                                               |
| <b>UN</b>      | United Nations                                                                      |
| <b>VOC</b>     | Volatile organic compounds                                                          |
| <b>vPvB</b>    | Very Persistent and Very Bioaccumulative                                            |
| <b>vPvM</b>    | Very Persistent and Very Mobile                                                     |
| <b>Sk</b>      | Skin notation                                                                       |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AELG(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.  
**Revision date** 14-May-2025  
**Revision Note** Reviewed existing information and made minor updates.  
**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the

date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
US OSHA Hazard Communication Standard (29 CFR 1910.1200)

Revision date 24-Feb-2021

Revision Number 1

## 1. Identification

### Product identifier

**Product Name** Discontinued - R9 - Chromogen: TMB Solution - Platelia Rabies

### Other means of identification

**Catalog Number(s)** 1719

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Life Science  
2000 Alfred Nobel Drive  
Hercules, California 94547

**Technical Service** 1-800-424-6723  
support@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification of the substance or mixture

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200).

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

#### Hazard statements

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200).

### Hazards classified under paragraph (d)(1)(ii) of 1910.1200

No information available.

### Other information

Causes mild skin irritation.

### 3. Composition/information on ingredients

#### Substance

Not applicable.

#### Mixture

| Chemical name     | CAS No.   | Weight-%   | Trade secret |
|-------------------|-----------|------------|--------------|
| Hydrochloric acid | 7647-01-0 | 0.3 - 0.99 | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

### 4. First-aid measures

#### Description of first aid measures

|              |                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                    |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a physician. |
| Skin contact | Wash skin with soap and water.                                                                                          |
| Ingestion    | Rinse mouth.                                                                                                            |

#### Most important symptoms and effects, both acute and delayed

**Symptoms** Prolonged contact may cause redness and irritation.

**Effects of Exposure** No information available.

#### Indication of any immediate medical attention and special treatment needed

**Note to physicians** Treat symptomatically.

### 5. Fire-fighting measures

**Suitable Extinguishing Media** Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Unsuitable extinguishing media** Do not scatter spilled material with high pressure water streams.

**Specific hazards arising from the chemical** No information available.

#### **Explosion data**

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Special protective equipment and precautions for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.  
Use personal protection equipment.

### 6. Accidental release measures

### Personal precautions, protective equipment and emergency procedures

**Personal precautions** Ensure adequate ventilation.

### Methods and material for containment and cleaning up

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## 7. Handling and storage

### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

### Conditions for safe storage, including any incompatibilities

**Storage Conditions** Store according to product and label instructions.

## 8. Exposure controls/personal protection

### Working area parameters, subject to mandatory control (MAC or TSEL)

#### Exposure Limits

| Chemical name                  | ACGIH TLV      | OSHA PEL                                                                                                             | NIOSH                                                          |
|--------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Hydrochloric acid<br>7647-01-0 | Ceiling: 2 ppm | (vacated) Ceiling: 5 ppm<br>(vacated) Ceiling: 7 mg/m <sup>3</sup><br>Ceiling: 5 ppm<br>Ceiling: 7 mg/m <sup>3</sup> | IDLH: 50 ppm<br>Ceiling: 5 ppm<br>Ceiling: 7 mg/m <sup>3</sup> |

**Biological occupational exposure limits** This product, as supplied, contains materials that do not have reportable biological exposure limits or are not subject to the reporting requirements of the local jurisdiction.

### Appropriate engineering controls

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Hand protection** Wear suitable gloves.

**Skin and body protection** Wear suitable protective clothing.

**Respiratory protection** Appropriate respiratory protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction. If

exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                                |                  |
|--------------------------------|------------------|
| Appearance                     | aqueous solution |
| Physical state                 | Liquid           |
| Color                          | colorless        |
| Odor (includes odor threshold) | Odorless         |

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      | 4                 |                         |
| Melting point / freezing point          | No data available | None known              |
| Initial boiling point and boiling range | > 100 °C / 212 °F |                         |
| Flash point                             | No data available | None known              |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapor pressure                          | No data available | None known              |
| Relative vapor density                  | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Miscible in water |                         |
| Solubility(ies)                         | No data available | None known              |
| Partition coefficient                   | No data available | None known              |
| Autoignition temperature                | No data available | None known              |
| Decomposition temperature               | No data available | None known              |
| Kinematic viscosity                     | No data available | None known              |
| Dynamic viscosity                       | No data available | None known              |

### Other information

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

### Information on likely routes of exposure

|             |                                                                   |
|-------------|-------------------------------------------------------------------|
| Inhalation  | Specific test data for the substance or mixture is not available. |
| Eye contact | Specific test data for the substance or mixture is not available. |

**Skin contact** Specific test data for the substance or mixture is not available. Causes mild skin irritation.

**Ingestion** Specific test data for the substance or mixture is not available.

**Symptoms related to the physical, chemical and toxicological characteristics**

**Symptoms** Prolonged contact may cause redness and irritation.

**Acute toxicity** No information available.

**Numerical measures of toxicity**

**Component Information**

| Chemical name                  | Oral LD50               | Dermal LD50             | Inhalation LC50         |
|--------------------------------|-------------------------|-------------------------|-------------------------|
| Hydrochloric acid<br>7647-01-0 | 238 - 277 mg/kg ( Rat ) | > 5010 mg/kg ( Rabbit ) | = 1.68 mg/L ( Rat ) 1 h |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** Classification based on data available for ingredients. Causes mild skin irritation.

**Serious eye damage/eye irritation** No information available.

**Respiratory or skin sensitization** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Chemical name                  | ACGIH                                          | IARC    | NTP | OSHA |
|--------------------------------|------------------------------------------------|---------|-----|------|
| Hydrochloric acid<br>7647-01-0 | A4 - Not Classifiable as<br>a Human Carcinogen | Group 3 | -   | X    |

**Legend**

**ACGIH (American Conference of Governmental Industrial Hygienists)**

A4 - Not Classifiable as a Human Carcinogen

**IARC (International Agency for Research on Cancer)**

Group 3 - Not Classifiable as to Carcinogenicity in Humans

**Occupational Safety and Health Administration of the US Department of Labor**

X - Present

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

**STOT - repeated exposure** No information available.

**Aspiration hazard** No information available.

**Other adverse effects** No information available.

Interactive effects No information available.

## 12. Ecological information

Ecotoxicity The environmental impact of this product has not been fully investigated.

Persistence and degradability No information available.

Bioaccumulation There is no data for this product.

Other adverse effects No information available.

## 13. Disposal considerations

### Disposal methods

Waste from residues/unused products Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

Contaminated packaging Do not reuse empty containers.

## 14. Transport information

DOT Not regulated

TDG Not regulated

MEX Not regulated

ICAO (air) Not regulated

IATA Not regulated

IMDG Not regulated

## 15. Regulatory information

### International Regulations

The Montreal Protocol on Substances that Deplete the Ozone Layer Not applicable

The Stockholm Convention on Persistent Organic Pollutants Not applicable

The Rotterdam Convention Not applicable

International Inventories Contact supplier for inventory compliance status

US Federal Regulations

**TSCA**

All chemical components are listed on the TSCA Inventory.

**SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

This product contains the following substances which are regulated pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

| Chemical name                  | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants | CWA - Hazardous Substances |
|--------------------------------|-----------------------------|------------------------|---------------------------|----------------------------|
| Hydrochloric acid<br>7647-01-0 | 5000 lb                     | -                      | -                         | X                          |

**CAA (Clean Air Act)**

This product does not contain any substances regulated as pollutants pursuant to Clean Air Act (CAA).

**CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302).

| Chemical name                  | Hazardous Substances RQs | Extremely Hazardous Substances RQs | Reportable Quantity (RQ)                   |
|--------------------------------|--------------------------|------------------------------------|--------------------------------------------|
| Hydrochloric acid<br>7647-01-0 | 5000 lb                  | 5000 lb                            | RQ 5000 lb final RQ<br>RQ 2270 kg final RQ |

**US State Regulations**

**California Proposition 65**

This product does not contain any Proposition 65 chemicals

**U.S. State Right-to-Know Regulations**

| Chemical name                  | New Jersey | Massachusetts | Pennsylvania |
|--------------------------------|------------|---------------|--------------|
| Water<br>7732-18-5             | -          | -             | X            |
| Hydrochloric acid<br>7647-01-0 | X          | X             | X            |

**U.S. EPA Label Information**

**EPA Pesticide Registration Number** Not applicable

**16. Other information**

|             |                  |                |                    |                       |
|-------------|------------------|----------------|--------------------|-----------------------|
| <b>NFPA</b> | Health hazards 0 | Flammability 0 | Instability 0      | Special hazards -     |
| <b>HMIS</b> | Health hazards 0 | Flammability 0 | Physical hazards 0 | Personal protection - |

**Key or legend to abbreviations and acronyms used in the safety data sheet**

**Legend**

|                |                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------|
| <b>ADN</b>     | Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways (Europe) |
| <b>ADR</b>     | Agreement concerning the International Carriage of Dangerous Goods by Road (Europe)             |
| <b>ATE</b>     | Acute Toxicity Estimate                                                                         |
| <b>ASTM</b>    | American Society for the Testing of Materials                                                   |
| <b>BEL</b>     | Biological exposure limits                                                                      |
| <b>bw</b>      | Body weight                                                                                     |
| <b>Ceiling</b> | Maximum limit value                                                                             |
| <b>CMR</b>     | Carcinogen, Mutagen or Reproductive Toxicant                                                    |
| <b>DOT</b>     | Department of Transportation (United States)                                                    |
| <b>EmS</b>     | Emergency Schedule                                                                              |
| <b>EPA</b>     | Environmental Protection Agency                                                                 |
| <b>GHS</b>     | Globally Harmonized System                                                                      |
| <b>HMIS</b>    | Hazardous Materials Identification System                                                       |
| <b>IATA</b>    | International Air Transport Association                                                         |
| <b>ICAO</b>    | International Civil Aviation Organization                                                       |
| <b>IMDG</b>    | International Maritime Dangerous Goods                                                          |
| <b>IMO</b>     | International Maritime Organization                                                             |
| <b>ISO</b>     | International Organization for Standardization                                                  |
| <b>LC50</b>    | Lethal Concentration to 50% of a test population                                                |
| <b>LD50</b>    | Lethal Dose to 50% of a test population (Median Lethal Dose)                                    |
| <b>MARPOL</b>  | International Convention for the Prevention of Pollution from Ships                             |
| <b>NFPA</b>    | National Fire Protection Association                                                            |
| <b>NIOSH</b>   | National Institute for Occupational Safety and Health                                           |
| <b>n.o.s.</b>  | Not Otherwise Specified                                                                         |
| <b>NOAEL</b>   | No Observed Adverse Effect Level                                                                |
| <b>OECD</b>    | Organization for Economic Cooperation and Development                                           |
| <b>OEL</b>     | Occupational exposure limits                                                                    |
| <b>OSHA</b>    | Occupational Safety and Health Administration of the US Department of Labor                     |
| <b>PBT</b>     | Persistent, Bioaccumulative and Toxic substance                                                 |
| <b>PMT</b>     | Persistent, Mobile and Toxic                                                                    |
| <b>PPE</b>     | Personal protective equipment                                                                   |
| <b>RID</b>     | Agreement concerning the International Carriage of Dangerous Goods by Rail (Europe)             |
| <b>SARA</b>    | Superfund Amendments and Reauthorization Act                                                    |
| <b>SDS</b>     | Safety Data Sheet                                                                               |
| <b>STEL</b>    | Short Term Exposure Limit                                                                       |
| <b>STOT RE</b> | Specific target organ toxicity - Repeated exposure                                              |
| <b>STOT SE</b> | Specific target organ toxicity - Single exposure                                                |
| <b>TDG</b>     | Transport of Dangerous Goods (Canada)                                                           |
| <b>TSCA</b>    | Toxic Substances Control Act (United States)                                                    |
| <b>TWA</b>     | Time-Weighted Average                                                                           |
| <b>UN</b>      | United Nations                                                                                  |
| <b>VOC</b>     | Volatile organic compounds                                                                      |
| <b>vPvB</b>    | Very Persistent and Very Bioaccumulative                                                        |
| <b>vPvM</b>    | Very Persistent and Very Mobile                                                                 |
| <b>Sk</b>      | Skin notation                                                                                   |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
Environmental Protection Agency  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
National Institute of Technology and Evaluation (NITE)  
Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program

Organization for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

**Prepared By**

Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date**

24-Feb-2021

**Revision Note**

This product has been discontinued and the Safety Data Sheet will not be updated.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
US OSHA Hazard Communication Standard (29 CFR 1910.1200)

Revision date 14-May-2025

Revision Number 2.1

## 1. Identification

### Product identifier

Product Name R10 - Stopping Solution

### Other means of identification

Catalog Number(s) 7438L

### Recommended use of the chemical and restrictions on use

Recommended use Laboratory chemicals

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Life Science  
2000 Alfred Nobel Drive  
Hercules, California 94547

Technical Service 1-800-424-6723  
support@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification of the substance or mixture

|                                   |            |
|-----------------------------------|------------|
| Skin corrosion/irritation         | Category 2 |
| Serious eye damage/eye irritation | Category 2 |

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements



Warning

### Hazard statements

Causes skin irritation.  
Causes serious eye irritation.

**Precautionary Statements - Prevention**

Wash face, hands and any exposed skin thoroughly after handling.

Wear protective gloves, protective clothing, eye protection and face protection.

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

If eye irritation persists: Get medical advice/attention.

IF ON SKIN: Wash with plenty of water and soap.

If skin irritation occurs: Get medical advice/attention.

**Hazards classified under paragraph (d)(1)(ii) of 1910.1200**

No information available.

**Other information**

No information available.

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

| Chemical name | CAS No.   | Weight-% | Trade secret |
|---------------|-----------|----------|--------------|
| Sulfuric acid | 7664-93-9 | 2.5 - 5  | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

**4. First-aid measures****Description of first aid measures**

|                                           |                                                                                                                                                                                                                                                                                     |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b>                     | Show this safety data sheet to the doctor in attendance.                                                                                                                                                                                                                            |
| <b>Inhalation</b>                         | Remove to fresh air. Get medical attention immediately if symptoms occur.                                                                                                                                                                                                           |
| <b>Eye contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Keep eye wide open while rinsing. Do not rub affected area. Get medical attention if irritation develops and persists. |
| <b>Skin contact</b>                       | Wash off immediately with soap and plenty of water for at least 15 minutes. Get medical attention if irritation develops and persists.                                                                                                                                              |
| <b>Ingestion</b>                          | Rinse mouth. Never give anything by mouth to an unconscious person. Do NOT induce vomiting. Call a physician.                                                                                                                                                                       |
| <b>Self-protection of the first aider</b> | Avoid contact with skin, eyes or clothing. Wear personal protective clothing (see section 8).                                                                                                                                                                                       |

**Most important symptoms and effects, both acute and delayed**

**Symptoms** May cause redness and tearing of the eyes. Burning sensation.

**Effects of Exposure** No information available.

**Indication of any immediate medical attention and special treatment needed**

**Note to physicians** Treat symptomatically.

## 5. Fire-fighting measures

|                                                                       |                                                                                                                                    |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>                                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| <b>Unsuitable extinguishing media</b>                                 | Do not scatter spilled material with high pressure water streams.                                                                  |
| <b>Specific hazards arising from the chemical</b>                     | No information available.                                                                                                          |
| <b>Explosion data</b>                                                 |                                                                                                                                    |
| <b>Sensitivity to mechanical impact</b>                               | None.                                                                                                                              |
| <b>Sensitivity to static discharge</b>                                | None.                                                                                                                              |
| <b>Special protective equipment and precautions for fire-fighters</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

## 6. Accidental release measures

### Personal precautions, protective equipment and emergency procedures

|                             |                                                                                                                        |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Personal precautions</b> | Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. |
| <b>Other information</b>    | Refer to protective measures listed in Sections 7 and 8.                                                               |

### Methods and material for containment and cleaning up

|                                        |                                                                                      |
|----------------------------------------|--------------------------------------------------------------------------------------|
| <b>Methods for containment</b>         | Prevent further leakage or spillage if safe to do so.                                |
| <b>Methods for cleaning up</b>         | Pick up and transfer to properly labeled containers.                                 |
| <b>Prevention of secondary hazards</b> | Clean contaminated objects and areas thoroughly observing environmental regulations. |

## 7. Handling and storage

### Precautions for safe handling

|                                       |                                                                                                                                                                                                                             |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Advice on safe handling</b>        | Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse. |
| <b>General hygiene considerations</b> | Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product. Avoid contact with skin, eyes or clothing.                                                                                |

### Conditions for safe storage, including any incompatibilities

|                           |                                                                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Storage Conditions</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Store according to product and label instructions. |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------|

## 8. Exposure controls/personal protection

### Working area parameters, subject to mandatory control (MAC or TSEL)

#### Exposure Limits

| Chemical name              | ACGIH TLV                                                 | OSHA PEL                                                       | NIOSH                                                  |
|----------------------------|-----------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------|
| Sulfuric acid<br>7664-93-9 | TWA: 0.2 mg/m <sup>3</sup> thoracic<br>particulate matter | TWA: 1 mg/m <sup>3</sup><br>(vacated) TWA: 1 mg/m <sup>3</sup> | IDLH: 15 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> |

**Biological occupational exposure limits** This product, as supplied, contains materials that do not have reportable biological exposure limits or are not subject to the reporting requirements of the local jurisdiction.

### Appropriate engineering controls

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

**Eye/face protection** If splashes are likely to occur, wear safety glasses with side-shields.

**Hand protection** Wear suitable gloves. Impervious gloves.

**Skin and body protection** Wear suitable protective clothing. Long sleeved clothing.

**Respiratory protection** Appropriate respiratory protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

## **9. Physical and chemical properties**

### Information on basic physical and chemical properties

**Appearance** aqueous solution  
**Physical state** Liquid  
**Color** colorless  
**Odor (includes odor threshold)** Odorless

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      | 2.6               |                         |
| Melting point / freezing point          | 0 °C / 32 °F      |                         |
| Initial boiling point and boiling range | 100 °C / 212 °F   |                         |
| Flash point                             | No data available | None known              |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapor pressure                          | No data available | None known              |
| Relative vapor density                  | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Miscible in water | None known              |
| Solubility(ies)                         | No data available | None known              |
| Partition coefficient                   | No data available | None known              |
| Autoignition temperature                | No data available | None known              |
| Decomposition temperature               | No data available | None known              |
| Kinematic viscosity                     | No data available | None known              |
| Dynamic viscosity                       | No data available | None known              |

### Other information

## 10. Stability and reactivity

|                                           |                                                      |
|-------------------------------------------|------------------------------------------------------|
| <b>Reactivity</b>                         | No information available.                            |
| <b>Chemical stability</b>                 | Stable under normal conditions.                      |
| <b>Possibility of hazardous reactions</b> | None under normal processing.                        |
| <b>Conditions to avoid</b>                | None known based on information supplied.            |
| <b>Incompatible materials</b>             | Strong acids. Strong bases. Strong oxidizing agents. |
| <b>Hazardous decomposition products</b>   | None known based on information supplied.            |

### Information on likely routes of exposure

#### Product Information

|                     |                                                                                                                                                              |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available. May cause irritation of respiratory tract.                                                 |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available. Causes serious eye irritation (based on components). May cause redness, itching, and pain. |
| <b>Skin contact</b> | Specific test data for the substance or mixture is not available. Causes skin irritation (based on components).                                              |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea.            |

### Symptoms related to the physical, chemical and toxicological characteristics

|                       |                                                     |
|-----------------------|-----------------------------------------------------|
| <b>Symptoms</b>       | Redness. May cause redness and tearing of the eyes. |
| <b>Acute toxicity</b> | No information available.                           |

#### Numerical measures of toxicity

#### Component Information

| Chemical name              | Oral LD50            | Dermal LD50 | Inhalation LC50          |
|----------------------------|----------------------|-------------|--------------------------|
| Sulfuric acid<br>7664-93-9 | = 2140 mg/kg ( Rat ) | -           | = 0.375 mg/L ( Rat ) 4 h |

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                          |                                                                                        |
|------------------------------------------|----------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Classification based on data available for ingredients. Causes skin irritation.        |
| <b>Serious eye damage/eye irritation</b> | Classification based on data available for ingredients. Causes serious eye irritation. |
| <b>Respiratory or skin sensitization</b> | No information available.                                                              |
| <b>Germ cell mutagenicity</b>            | No information available.                                                              |
| <b>Carcinogenicity</b>                   | No information available.                                                              |

The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Chemical name              | ACGIH                                                                                     | IARC    | NTP   | OSHA |
|----------------------------|-------------------------------------------------------------------------------------------|---------|-------|------|
| Sulfuric acid<br>7664-93-9 | A2<br>A2 - Suspected Human<br>Carcinogen (contained<br>in strong inorganic acid<br>mists) | Group 1 | Known | X    |

**Legend****ACGIH (American Conference of Governmental Industrial Hygienists)**

A2 - Suspected Human Carcinogen

**IARC (International Agency for Research on Cancer)**

Group 1 - Carcinogenic to Humans

**NTP (National Toxicology Program)**

Known - Known Carcinogen

**Occupational Safety and Health Administration of the US Department of Labor**

X - Present

**Reproductive toxicity** No information available.**STOT - single exposure** No information available.**STOT - repeated exposure** No information available.**Target organ effects** Respiratory system, Eyes, Skin, Teeth.**Aspiration hazard** No information available.**Other adverse effects** No information available.**Interactive effects** No information available.**12. Ecological information****Ecotoxicity**

| Chemical name              | Algae/aquatic plants | Fish                                       | Toxicity to microorganisms | Crustacea |
|----------------------------|----------------------|--------------------------------------------|----------------------------|-----------|
| Sulfuric acid<br>7664-93-9 | -                    | LC50: >500mg/L (96h,<br>Brachydanio rerio) | -                          | -         |

**Persistence and degradability** No information available.**Bioaccumulation** There is no data for this product.**Other adverse effects** No information available.**13. Disposal considerations****Disposal methods**

|                                            |                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| <b>Contaminated packaging</b>              | Do not reuse empty containers.                                                                                  |
| <b>California Hazardous Waste Status</b>   | This product contains one or more substances that are listed with the State of California as a hazardous waste. |

## 14. Transport information

|                   |               |
|-------------------|---------------|
| <b>DOT</b>        | Not regulated |
| <b>TDG</b>        | Not regulated |
| <b>MEX</b>        | Not regulated |
| <b>ICAO (air)</b> | Not regulated |
| <b>IATA</b>       | Not regulated |
| <b>IMDG</b>       | Not regulated |

## 15. Regulatory information

### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories** Contact supplier for inventory compliance status

### US Federal Regulations

#### **TSCA**

All chemical components are listed on the TSCA Inventory.

#### **SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

#### **SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

#### **CWA (Clean Water Act)**

This product contains the following substances which are regulated pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

| Chemical name              | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants | CWA - Hazardous Substances |
|----------------------------|-----------------------------|------------------------|---------------------------|----------------------------|
| Sulfuric acid<br>7664-93-9 | 1000 lb                     | -                      | -                         | X                          |

#### **CAA (Clean Air Act)**

This product does not contain any substances regulated as pollutants pursuant to Clean Air Act (CAA).

**CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302).

| Chemical name              | Hazardous Substances RQs | Extremely Hazardous Substances RQs | Reportable Quantity (RQ)                  |
|----------------------------|--------------------------|------------------------------------|-------------------------------------------|
| Sulfuric acid<br>7664-93-9 | 1000 lb                  | 1000 lb                            | RQ 1000 lb final RQ<br>RQ 454 kg final RQ |

**US State Regulations****California Proposition 65**

This product contains the following Proposition 65 chemicals:

| Chemical name             | California Proposition 65 |
|---------------------------|---------------------------|
| Sulfuric acid - 7664-93-9 | Carcinogen                |

**U.S. State Right-to-Know Regulations**

| Chemical name              | New Jersey | Massachusetts | Pennsylvania |
|----------------------------|------------|---------------|--------------|
| Water<br>7732-18-5         | -          | -             | X            |
| Sulfuric acid<br>7664-93-9 | X          | X             | X            |

**U.S. EPA Label Information**

**EPA Pesticide Registration Number** Not applicable

**16. Other information**

|             |                         |                       |                           |                              |
|-------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> - |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend**

|                |                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------|
| <b>ADN</b>     | Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways (Europe) |
| <b>ADR</b>     | Agreement concerning the International Carriage of Dangerous Goods by Road (Europe)             |
| <b>ATE</b>     | Acute Toxicity Estimate                                                                         |
| <b>ASTM</b>    | American Society for the Testing of Materials                                                   |
| <b>BEL</b>     | Biological exposure limits                                                                      |
| <b>bw</b>      | Body weight                                                                                     |
| <b>Ceiling</b> | Maximum limit value                                                                             |
| <b>CMR</b>     | Carcinogen, Mutagen or Reproductive Toxicant                                                    |
| <b>DOT</b>     | Department of Transportation (United States)                                                    |
| <b>EmS</b>     | Emergency Schedule                                                                              |
| <b>EPA</b>     | Environmental Protection Agency                                                                 |
| <b>GHS</b>     | Globally Harmonized System                                                                      |
| <b>HMIS</b>    | Hazardous Materials Identification System                                                       |
| <b>IATA</b>    | International Air Transport Association                                                         |
| <b>ICAO</b>    | International Civil Aviation Organization                                                       |
| <b>IMDG</b>    | International Maritime Dangerous Goods                                                          |
| <b>IMO</b>     | International Maritime Organization                                                             |
| <b>ISO</b>     | International Organization for Standardization                                                  |
| <b>LC50</b>    | Lethal Concentration to 50% of a test population                                                |
| <b>LD50</b>    | Lethal Dose to 50% of a test population (Median Lethal Dose)                                    |
| <b>MARPOL</b>  | International Convention for the Prevention of Pollution from Ships                             |
| <b>NFPA</b>    | National Fire Protection Association                                                            |
| <b>NIOSH</b>   | National Institute for Occupational Safety and Health                                           |
| <b>n.o.s.</b>  | Not Otherwise Specified                                                                         |
| <b>NOAEL</b>   | No Observed Adverse Effect Level                                                                |

|                |                                                                                     |
|----------------|-------------------------------------------------------------------------------------|
| <b>OECD</b>    | Organization for Economic Cooperation and Development                               |
| <b>OEL</b>     | Occupational exposure limits                                                        |
| <b>OSHA</b>    | Occupational Safety and Health Administration of the US Department of Labor         |
| <b>PBT</b>     | Persistent, Bioaccumulative and Toxic substance                                     |
| <b>PMT</b>     | Persistent, Mobile and Toxic                                                        |
| <b>PPE</b>     | Personal protective equipment                                                       |
| <b>RID</b>     | Agreement concerning the International Carriage of Dangerous Goods by Rail (Europe) |
| <b>SARA</b>    | Superfund Amendments and Reauthorization Act                                        |
| <b>SDS</b>     | Safety Data Sheet                                                                   |
| <b>STEL</b>    | Short Term Exposure Limit                                                           |
| <b>STOT RE</b> | Specific target organ toxicity - Repeated exposure                                  |
| <b>STOT SE</b> | Specific target organ toxicity - Single exposure                                    |
| <b>TDG</b>     | Transport of Dangerous Goods (Canada)                                               |
| <b>TSCA</b>    | Toxic Substances Control Act (United States)                                        |
| <b>TWA</b>     | Time-Weighted Average                                                               |
| <b>UN</b>      | United Nations                                                                      |
| <b>VOC</b>     | Volatile organic compounds                                                          |
| <b>vPvB</b>    | Very Persistent and Very Bioaccumulative                                            |
| <b>vPvM</b>    | Very Persistent and Very Mobile                                                     |
| <b>Sk</b>      | Skin notation                                                                       |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 14-May-2025

**Revision Note** Reviewed existing information and made minor updates.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
US OSHA Hazard Communication Standard (29 CFR 1910.1200)

Revision date 14-May-2025

Revision Number 1.1

## 1. Identification

### Product identifier

Product Name Grinding Tube

### Other means of identification

Catalog Number(s) 5144A

### Recommended use of the chemical and restrictions on use

Recommended use Laboratory chemicals

Restrictions on use No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Life Science  
2000 Alfred Nobel Drive  
Hercules, California 94547

#### Technical Service

1-800-424-6723  
support@bio-rad.com

### Emergency telephone number

24 Hour Emergency Phone Number CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification of the substance or mixture

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200).

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

#### Hazard statements

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200).

### Hazards classified under paragraph (d)(1)(ii) of 1910.1200

No information available.

### Other information

No information available.

### 3. Composition/information on ingredients

#### Substance

Not applicable.

#### Mixture

The product contains no substances which at their given concentration, are considered to be hazardous to health

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

### 4. First-aid measures

#### Description of first aid measures

|              |                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                    |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a physician. |
| Skin contact | Wash skin with soap and water.                                                                                          |
| Ingestion    | Rinse mouth.                                                                                                            |

#### Most important symptoms and effects, both acute and delayed

|                     |                           |
|---------------------|---------------------------|
| Symptoms            | No information available. |
| Effects of Exposure | No information available. |

#### Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

### 5. Fire-fighting measures

|                                                                |                                                                                                                                       |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                               |
| Unsuitable extinguishing media                                 | Do not scatter spilled material with high pressure water streams.                                                                     |
| Specific hazards arising from the chemical                     | No information available.                                                                                                             |
| Explosion data                                                 |                                                                                                                                       |
| Sensitivity to mechanical impact                               | None.                                                                                                                                 |
| Sensitivity to static discharge                                | None.                                                                                                                                 |
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.<br>Use personal protection equipment. |

### 6. Accidental release measures

#### Personal precautions, protective equipment and emergency procedures

**Personal precautions** Ensure adequate ventilation.

#### **Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

### **7. Handling and storage**

#### **Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

#### **Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store according to product and label instructions.

### **8. Exposure controls/personal protection**

#### **Working area parameters, subject to mandatory control (MAC or TSEL)**

**Exposure Limits** This product, as supplied, contains materials that do not have reportable occupational exposure limits or are not subject to the reporting requirements of the local jurisdiction.

**Biological occupational exposure limits** This product, as supplied, contains materials that do not have reportable biological exposure limits or are not subject to the reporting requirements of the local jurisdiction.

#### **Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

#### **Individual protection measures, such as personal protective equipment**

**Eye/face protection** Appropriate eye/face protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Hand protection** Appropriate hand protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Skin and body protection** Appropriate skin and body protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Respiratory protection** Appropriate respiratory protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                                |                  |
|--------------------------------|------------------|
| Appearance                     | aqueous solution |
| Physical state                 | Liquid           |
| Color                          | colorless        |
| Odor (includes odor threshold) | Odorless         |

| Property                                | Values            | Remarks • Method |
|-----------------------------------------|-------------------|------------------|
| pH                                      | 7                 |                  |
| Melting point / freezing point          | 0 °C / 32 °F      |                  |
| Initial boiling point and boiling range | > 100 °C / 212 °F |                  |
| Flash point                             | No data available | None known       |
| Evaporation rate                        | No data available | None known       |
| Flammability                            | No data available | None known       |
| Flammability Limit in Air               |                   | None known       |
| Upper flammability or explosive limits  | No data available |                  |
| Lower flammability or explosive limits  | No data available |                  |
| Vapor pressure                          | No data available | None known       |
| Relative vapor density                  | No data available | None known       |
| Relative density                        | No data available | None known       |
| Water solubility                        | Miscible in water |                  |
| Solubility(ies)                         | No data available | None known       |
| Partition coefficient                   | No data available | None known       |
| Autoignition temperature                | No data available | None known       |
| Decomposition temperature               | No data available | None known       |
| Kinematic viscosity                     | No data available | None known       |
| Dynamic viscosity                       | No data available | None known       |

### Other information

## 10. Stability and reactivity

|                                    |                                                                                                                                                                                           |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reactivity                         | No information available.                                                                                                                                                                 |
| Chemical stability                 | Stable under normal conditions.                                                                                                                                                           |
| Possibility of hazardous reactions | Avoid contact with metals. This product contains sodium azide. Sodium azide can react with copper, brass, lead, and solder in piping systems to form explosive compounds and toxic gases. |
| Conditions to avoid                | None known based on information supplied.                                                                                                                                                 |
| Incompatible materials             | Metals.                                                                                                                                                                                   |
| Hazardous decomposition products   | None known based on information supplied.                                                                                                                                                 |

### Information on likely routes of exposure

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |

**Ingestion**

Specific test data for the substance or mixture is not available.

**Symptoms related to the physical, chemical and toxicological characteristics****Symptoms**

No information available.

**Acute toxicity**

No information available.

**Numerical measures of toxicity****Delayed and immediate effects as well as chronic effects from short and long-term exposure****Skin corrosion/irritation**

No information available.

**Serious eye damage/eye irritation**

No information available.

**Respiratory or skin sensitization**

No information available.

**Germ cell mutagenicity**

No information available.

**Carcinogenicity**

No information available.

**Reproductive toxicity**

No information available.

**STOT - single exposure**

No information available.

**STOT - repeated exposure**

No information available.

**Aspiration hazard**

No information available.

**Other adverse effects**

No information available.

**Interactive effects**

No information available.

**12. Ecological information****Ecotoxicity**

The environmental impact of this product has not been fully investigated.

**Persistence and degradability**

No information available.

**Bioaccumulation**

There is no data for this product.

**Other adverse effects**

No information available.

### 13. Disposal considerations

#### Disposal methods

**Waste from residues/unused products**

Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. Flush pipes with water frequently if discarding solutions containing sodium azide into metal piping systems.

**Contaminated packaging**

Do not reuse empty containers.

### 14. Transport information

**DOT**

Not regulated

**TDG**

Not regulated

**MEX**

Not regulated

**ICAO (air)**

Not regulated

**IATA**

Not regulated

**IMDG**

Not regulated

### 15. Regulatory information

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories**

Contact supplier for inventory compliance status

**US Federal Regulations****TSCA**

All chemical components are listed on the TSCA Inventory.

**SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

**CAA (Clean Air Act)**

This product does not contain any substances regulated as pollutants pursuant to Clean Air Act (CAA).

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**US State Regulations****California Proposition 65**

This product does not contain any Proposition 65 chemicals

**U.S. State Right-to-Know Regulations**

| Chemical name              | New Jersey | Massachusetts | Pennsylvania |
|----------------------------|------------|---------------|--------------|
| Water<br>7732-18-5         | -          | -             | X            |
| Sodium azide<br>26628-22-8 | X          | X             | X            |

**U.S. EPA Label Information**

**EPA Pesticide Registration Number** Not applicable

**16. Other information**

|             |                         |                       |                           |                              |
|-------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> - |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend**

|                |                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------|
| <b>ADN</b>     | Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways (Europe) |
| <b>ADR</b>     | Agreement concerning the International Carriage of Dangerous Goods by Road (Europe)             |
| <b>ATE</b>     | Acute Toxicity Estimate                                                                         |
| <b>ASTM</b>    | American Society for the Testing of Materials                                                   |
| <b>BEL</b>     | Biological exposure limits                                                                      |
| <b>bw</b>      | Body weight                                                                                     |
| <b>Ceiling</b> | Maximum limit value                                                                             |
| <b>CMR</b>     | Carcinogen, Mutagen or Reproductive Toxicant                                                    |
| <b>DOT</b>     | Department of Transportation (United States)                                                    |
| <b>EmS</b>     | Emergency Schedule                                                                              |
| <b>EPA</b>     | Environmental Protection Agency                                                                 |
| <b>GHS</b>     | Globally Harmonized System                                                                      |
| <b>HMIS</b>    | Hazardous Materials Identification System                                                       |
| <b>IATA</b>    | International Air Transport Association                                                         |
| <b>ICAO</b>    | International Civil Aviation Organization                                                       |
| <b>IMDG</b>    | International Maritime Dangerous Goods                                                          |
| <b>IMO</b>     | International Maritime Organization                                                             |
| <b>ISO</b>     | International Organization for Standardization                                                  |
| <b>LC50</b>    | Lethal Concentration to 50% of a test population                                                |
| <b>LD50</b>    | Lethal Dose to 50% of a test population (Median Lethal Dose)                                    |
| <b>MARPOL</b>  | International Convention for the Prevention of Pollution from Ships                             |
| <b>NFPA</b>    | National Fire Protection Association                                                            |
| <b>NIOSH</b>   | National Institute for Occupational Safety and Health                                           |
| <b>n.o.s.</b>  | Not Otherwise Specified                                                                         |
| <b>NOAEL</b>   | No Observed Adverse Effect Level                                                                |
| <b>OECD</b>    | Organization for Economic Cooperation and Development                                           |
| <b>OEL</b>     | Occupational exposure limits                                                                    |
| <b>OSHA</b>    | Occupational Safety and Health Administration of the US Department of Labor                     |
| <b>PBT</b>     | Persistent, Bioaccumulative and Toxic substance                                                 |

|                |                                                                                     |
|----------------|-------------------------------------------------------------------------------------|
| <b>PMT</b>     | Persistent, Mobile and Toxic                                                        |
| <b>PPE</b>     | Personal protective equipment                                                       |
| <b>RID</b>     | Agreement concerning the International Carriage of Dangerous Goods by Rail (Europe) |
| <b>SARA</b>    | Superfund Amendments and Reauthorization Act                                        |
| <b>SDS</b>     | Safety Data Sheet                                                                   |
| <b>STEL</b>    | Short Term Exposure Limit                                                           |
| <b>STOT RE</b> | Specific target organ toxicity - Repeated exposure                                  |
| <b>STOT SE</b> | Specific target organ toxicity - Single exposure                                    |
| <b>TDG</b>     | Transport of Dangerous Goods (Canada)                                               |
| <b>TSCA</b>    | Toxic Substances Control Act (United States)                                        |
| <b>TWA</b>     | Time-Weighted Average                                                               |
| <b>UN</b>      | United Nations                                                                      |
| <b>VOC</b>     | Volatile organic compounds                                                          |
| <b>vPvB</b>    | Very Persistent and Very Bioaccumulative                                            |
| <b>vPvM</b>    | Very Persistent and Very Mobile                                                     |
| <b>Sk</b>      | Skin notation                                                                       |

#### Key literature references and sources for data used to compile the SDS

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.  
**Revision date** 14-May-2025  
**Revision Note** Reviewed existing information and made minor updates.  
**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
US OSHA Hazard Communication Standard (29 CFR 1910.1200)

Revision date 14-May-2025

Revision Number 1.1

## 1. Identification

### Product identifier

**Product Name** Reagent A - Denaturing Solution

### Other means of identification

**Catalog Number(s)** 5144B

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Life Science  
2000 Alfred Nobel Drive  
Hercules, California 94547

**Technical Service** 1-800-424-6723  
support@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification of the substance or mixture

|                                           |            |
|-------------------------------------------|------------|
| Acute toxicity - Inhalation (Dusts/Mists) | Category 3 |
| Skin corrosion/irritation                 | Category 2 |
| Serious eye damage/eye irritation         | Category 1 |

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements



Danger

### Hazard statements

Toxic if inhaled.

Causes skin irritation.

|Causes serious eye damage.

**Precautionary Statements - Prevention**

Use only outdoors or in a well-ventilated area.

Wash face, hands and any exposed skin thoroughly after handling.

Wear protective gloves, protective clothing, eye protection and face protection.

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

Immediately call a POISON CENTER or doctor.

IF ON SKIN: Wash with plenty of water and soap.

If skin irritation occurs: Get medical advice/attention.

IF INHALED: Remove person to fresh air and keep comfortable for breathing.

Call a POISON CENTER or doctor.

**Precautionary Statements - Storage**

Store in a well-ventilated place. Keep container tightly closed.

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable.

**Unknown acute toxicity**

18.49 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (dust/mist).

**Hazards classified under paragraph (d)(1)(ii) of 1910.1200**

No information available.

**Other information**

Toxic to aquatic life with long lasting effects. Harmful to aquatic life.

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

| Chemical name                                                                                  | CAS No.   | Weight-% | Trade secret |
|------------------------------------------------------------------------------------------------|-----------|----------|--------------|
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-.omega.<br>-hydroxy- | 9002-93-1 | 5 - 10   | *            |
| Glycine, N-methyl-N-(1-oxododecyl)-, sodium salt                                               | 137-16-6  | 5 - 10   | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

**4. First-aid measures****Description of first aid measures****General advice**

Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.

**Inhalation**

Remove to fresh air. Get medical attention immediately if symptoms occur. If breathing has stopped, give artificial respiration. Get medical attention immediately. Immediate medical attention is required. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. If breathing is difficult, (trained personnel should) give oxygen.

**Eye contact**

Get immediate medical attention. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue

rinsing. Keep eye wide open while rinsing. Do not rub affected area.

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin contact</b>                       | Wash off immediately with soap and plenty of water for at least 15 minutes. Get medical attention if irritation develops and persists.                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Ingestion</b>                          | Rinse mouth. Never give anything by mouth to an unconscious person. Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Self-protection of the first aider</b> | Avoid contact with skin, eyes or clothing. Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Do not breathe vapor or mist. Use personal protective equipment as required. See section 8 for more information. |

#### **Most important symptoms and effects, both acute and delayed**

|                            |                                                                        |
|----------------------------|------------------------------------------------------------------------|
| <b>Symptoms</b>            | Burning sensation. Coughing and/ or wheezing. Difficulty in breathing. |
| <b>Effects of Exposure</b> | No information available.                                              |

#### **Indication of any immediate medical attention and special treatment needed**

|                           |                        |
|---------------------------|------------------------|
| <b>Note to physicians</b> | Treat symptomatically. |
|---------------------------|------------------------|

### **5. Fire-fighting measures**

|                                                                       |                                                                                                                                    |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>                                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| <b>Unsuitable extinguishing media</b>                                 | Do not scatter spilled material with high pressure water streams.                                                                  |
| <b>Specific hazards arising from the chemical</b>                     | No information available.                                                                                                          |
| <b>Explosion data</b>                                                 |                                                                                                                                    |
| <b>Sensitivity to mechanical impact</b>                               | None.                                                                                                                              |
| <b>Sensitivity to static discharge</b>                                | None.                                                                                                                              |
| <b>Special protective equipment and precautions for fire-fighters</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

### **6. Accidental release measures**

#### **Personal precautions, protective equipment and emergency procedures**

|                             |                                                                                                                                                                                                                                        |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal precautions</b> | Avoid contact with skin, eyes or clothing. Use personal protective equipment as required. Ensure adequate ventilation. Do not breathe vapor or mist. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. |
| <b>Other information</b>    | Refer to protective measures listed in Sections 7 and 8.                                                                                                                                                                               |

#### **Methods and material for containment and cleaning up**

|                                        |                                                                                      |
|----------------------------------------|--------------------------------------------------------------------------------------|
| <b>Methods for containment</b>         | Prevent further leakage or spillage if safe to do so.                                |
| <b>Methods for cleaning up</b>         | Pick up and transfer to properly labeled containers.                                 |
| <b>Prevention of secondary hazards</b> | Clean contaminated objects and areas thoroughly observing environmental regulations. |

## 7. Handling and storage

### Precautions for safe handling

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Advice on safe handling</b>        | Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse. Do not breathe vapor or mist. In case of insufficient ventilation, wear suitable respiratory equipment. Handle product only in closed system or provide appropriate exhaust ventilation.                                                              |
| <b>General hygiene considerations</b> | Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product. Do not breathe vapor or mist. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Contaminated work clothing must not be allowed out of the workplace. Regular cleaning of equipment, work area and clothing is recommended. Wash hands before breaks and immediately after handling the product. |

### Conditions for safe storage, including any incompatibilities

|                           |                                                                                                                                                                                 |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Storage Conditions</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Store locked up. Keep out of the reach of children. Store according to product and label instructions. |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 8. Exposure controls/personal protection

### Working area parameters, subject to mandatory control (MAC or TSEL)

|                                                |                                                                                                                                                                                    |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Limits</b>                         | This product, as supplied, contains materials that do not have reportable occupational exposure limits or are not subject to the reporting requirements of the local jurisdiction. |
| <b>Biological occupational exposure limits</b> | This product, as supplied, contains materials that do not have reportable biological exposure limits or are not subject to the reporting requirements of the local jurisdiction.   |

### Appropriate engineering controls

|                             |                                                     |
|-----------------------------|-----------------------------------------------------|
| <b>Engineering controls</b> | Showers<br>Eyewash stations<br>Ventilation systems. |
|-----------------------------|-----------------------------------------------------|

### Individual protection measures, such as personal protective equipment

|                                 |                                                                                                                                                                                                                                                                                               |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye/face protection</b>      | Tight sealing safety goggles.                                                                                                                                                                                                                                                                 |
| <b>Hand protection</b>          | Wear suitable gloves. Impervious gloves.                                                                                                                                                                                                                                                      |
| <b>Skin and body protection</b> | Wear suitable protective clothing. Long sleeved clothing.                                                                                                                                                                                                                                     |
| <b>Respiratory protection</b>   | Appropriate respiratory protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                       |                  |
|-----------------------|------------------|
| <b>Appearance</b>     | aqueous solution |
| <b>Physical state</b> | Liquid           |
| <b>Color</b>          | colorless        |

Odor (includes odor threshold) Odorless

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      | 7                 |                         |
| Melting point / freezing point          | No data available | None known              |
| Initial boiling point and boiling range | No data available | None known              |
| Flash point                             | No data available | None known              |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapor pressure                          | No data available | None known              |
| Relative vapor density                  | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Miscible in water |                         |
| Solubility(ies)                         | No data available | None known              |
| Partition coefficient                   | No data available | None known              |
| Autoignition temperature                | No data available | None known              |
| Decomposition temperature               | No data available | None known              |
| Kinematic viscosity                     | No data available | None known              |
| Dynamic viscosity                       | No data available | None known              |

Other information

## 10. Stability and reactivity

|                                    |                                                      |
|------------------------------------|------------------------------------------------------|
| Reactivity                         | No information available.                            |
| Chemical stability                 | Stable under normal conditions.                      |
| Possibility of hazardous reactions | None under normal processing.                        |
| Conditions to avoid                | Excessive heat.                                      |
| Incompatible materials             | Strong acids. Strong bases. Strong oxidizing agents. |
| Hazardous decomposition products   | None known based on information supplied.            |

Information on likely routes of exposure

**Product Information**

|              |                                                                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. May cause irritation of respiratory tract. Toxic by inhalation (based on components). |
| Eye contact  | Specific test data for the substance or mixture is not available. Causes serious eye damage. May cause irreversible damage to eyes.                     |
| Skin contact | Specific test data for the substance or mixture is not available. Causes skin irritation (based on components).                                         |
| Ingestion    | Specific test data for the substance or mixture is not available. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea.       |

Symptoms related to the physical, chemical and toxicological characteristics

|          |                                                                                   |
|----------|-----------------------------------------------------------------------------------|
| Symptoms | Redness. Burning. May cause blindness. May cause redness and tearing of the eyes. |
|----------|-----------------------------------------------------------------------------------|

Coughing and/ or wheezing. Difficulty in breathing.

**Acute toxicity** Harmful by inhalation.

#### Numerical measures of toxicity

The following ATE values have been calculated for the mixture

ATEmix (oral) 13,128.60 mg/kg  
ATEmix (inhalation-dust/mist) 0.5014 mg/l

#### Unknown acute toxicity

18.49 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (dust/mist)

#### Component Information

| Chemical name                                                                                               | Oral LD50            | Dermal LD50 | Inhalation LC50             |
|-------------------------------------------------------------------------------------------------------------|----------------------|-------------|-----------------------------|
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)ph<br>enyl]-.omega.-hydroxy-<br>9002-93-1 | = 1800 mg/kg ( Rat ) | -           | -                           |
| Glycine, N-methyl-N-(1-oxododecyl)-,<br>sodium salt<br>137-16-6                                             | > 5000 mg/kg ( Rat ) | -           | 0.05 - 0.5 mg/L ( Rat ) 4 h |

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                          |                                                                                                  |
|------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Classification based on data available for ingredients. Causes skin irritation.                  |
| <b>Serious eye damage/eye irritation</b> | Classification based on data available for ingredients. Causes burns. Causes serious eye damage. |
| <b>Respiratory or skin sensitization</b> | No information available.                                                                        |
| <b>Germ cell mutagenicity</b>            | No information available.                                                                        |
| <b>Carcinogenicity</b>                   | No information available.                                                                        |
| <b>Reproductive toxicity</b>             | No information available.                                                                        |
| <b>STOT - single exposure</b>            | No information available.                                                                        |
| <b>STOT - repeated exposure</b>          | No information available.                                                                        |
| <b>Aspiration hazard</b>                 | No information available.                                                                        |
| <b>Other adverse effects</b>             | No information available.                                                                        |
| <b>Interactive effects</b>               | No information available.                                                                        |

## 12. Ecological information

**Ecotoxicity** Toxic to aquatic life with long lasting effects. Harmful to aquatic life.

| Chemical name | Algae/aquatic plants | Fish | Toxicity to | Crustacea |
|---------------|----------------------|------|-------------|-----------|
|---------------|----------------------|------|-------------|-----------|

|                                                                    |   |                                      | microorganisms |   |
|--------------------------------------------------------------------|---|--------------------------------------|----------------|---|
| Glycine,<br>N-methyl-N-(1-oxododecyl)-,<br>sodium salt<br>137-16-6 | - | LC50: =107mg/L (96h,<br>Danio rerio) | -              | - |

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

**Other adverse effects** No information available.

| Chemical name                                                                                  | EU - REACH (1907/2006) - Article 59(1)<br>- Candidate List of Substances of Very<br>High Concern (SVHC) for Authorisation | EU - REACH (1907/2006) - Endocrine<br>Disruptor Assessment List of<br>Substances |
|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-.ome<br>ga.-hydroxy- | Endocrine disrupting properties                                                                                           | -                                                                                |

### 13. Disposal considerations

#### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

### 14. Transport information

DOT Not regulated

TDG Not regulated

MEX Not regulated

ICAO (air) Not regulated

IATA Not regulated

IMDG Not regulated

### 15. Regulatory information

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories** Contact supplier for inventory compliance status

**US Federal Regulations****TSCA**

All chemical components are listed on the TSCA Inventory.

**SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

**CAA (Clean Air Act)**

This product does not contain any substances regulated as pollutants pursuant to Clean Air Act (CAA).

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**US State Regulations****California Proposition 65**

This product does not contain any Proposition 65 chemicals

**U.S. State Right-to-Know Regulations**

| Chemical name      | New Jersey | Massachusetts | Pennsylvania |
|--------------------|------------|---------------|--------------|
| Water<br>7732-18-5 | -          | -             | X            |

**U.S. EPA Label Information**

**EPA Pesticide Registration Number** Not applicable

**16. Other information**

|             |                         |                       |                           |                              |
|-------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> - |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend**

|                |                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------|
| <b>ADN</b>     | Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways (Europe) |
| <b>ADR</b>     | Agreement concerning the International Carriage of Dangerous Goods by Road (Europe)             |
| <b>ATE</b>     | Acute Toxicity Estimate                                                                         |
| <b>ASTM</b>    | American Society for the Testing of Materials                                                   |
| <b>BEL</b>     | Biological exposure limits                                                                      |
| <b>bw</b>      | Body weight                                                                                     |
| <b>Ceiling</b> | Maximum limit value                                                                             |

|                |                                                                                     |
|----------------|-------------------------------------------------------------------------------------|
| <b>CMR</b>     | Carcinogen, Mutagen or Reproductive Toxicant                                        |
| <b>DOT</b>     | Department of Transportation (United States)                                        |
| <b>EmS</b>     | Emergency Schedule                                                                  |
| <b>EPA</b>     | Environmental Protection Agency                                                     |
| <b>GHS</b>     | Globally Harmonized System                                                          |
| <b>HMIS</b>    | Hazardous Materials Identification System                                           |
| <b>IATA</b>    | International Air Transport Association                                             |
| <b>ICAO</b>    | International Civil Aviation Organization                                           |
| <b>IMDG</b>    | International Maritime Dangerous Goods                                              |
| <b>IMO</b>     | International Maritime Organization                                                 |
| <b>ISO</b>     | International Organization for Standardization                                      |
| <b>LC50</b>    | Lethal Concentration to 50% of a test population                                    |
| <b>LD50</b>    | Lethal Dose to 50% of a test population (Median Lethal Dose)                        |
| <b>MARPOL</b>  | International Convention for the Prevention of Pollution from Ships                 |
| <b>NFPA</b>    | National Fire Protection Association                                                |
| <b>NIOSH</b>   | National Institute for Occupational Safety and Health                               |
| <b>n.o.s.</b>  | Not Otherwise Specified                                                             |
| <b>NOAEL</b>   | No Observed Adverse Effect Level                                                    |
| <b>OECD</b>    | Organization for Economic Cooperation and Development                               |
| <b>OEL</b>     | Occupational exposure limits                                                        |
| <b>OSHA</b>    | Occupational Safety and Health Administration of the US Department of Labor         |
| <b>PBT</b>     | Persistent, Bioaccumulative and Toxic substance                                     |
| <b>PMT</b>     | Persistent, Mobile and Toxic                                                        |
| <b>PPE</b>     | Personal protective equipment                                                       |
| <b>RID</b>     | Agreement concerning the International Carriage of Dangerous Goods by Rail (Europe) |
| <b>SARA</b>    | Superfund Amendments and Reauthorization Act                                        |
| <b>SDS</b>     | Safety Data Sheet                                                                   |
| <b>STEL</b>    | Short Term Exposure Limit                                                           |
| <b>STOT RE</b> | Specific target organ toxicity - Repeated exposure                                  |
| <b>STOT SE</b> | Specific target organ toxicity - Single exposure                                    |
| <b>TDG</b>     | Transport of Dangerous Goods (Canada)                                               |
| <b>TSCA</b>    | Toxic Substances Control Act (United States)                                        |
| <b>TWA</b>     | Time-Weighted Average                                                               |
| <b>UN</b>      | United Nations                                                                      |
| <b>VOC</b>     | Volatile organic compounds                                                          |
| <b>vPvB</b>    | Very Persistent and Very Bioaccumulative                                            |
| <b>vPvM</b>    | Very Persistent and Very Mobile                                                     |
| <b>Sk</b>      | Skin notation                                                                       |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 14-May-2025

**Revision Note** Reformatted and updated existing information.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage,

transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
US OSHA Hazard Communication Standard (29 CFR 1910.1200)

Revision date 25-Jan-2023

Revision Number 2

## 1. Identification

### Product identifier

**Product Name** Discontinued - Reagent 1 - Denaturing Solution

### Other means of identification

**Catalog Number(s)** 1769

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Life Science  
2000 Alfred Nobel Drive  
Hercules, California 94547

**Technical Service** 1-800-424-6723  
support@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification of the substance or mixture

|                                           |            |
|-------------------------------------------|------------|
| Acute toxicity - Inhalation (Dusts/Mists) | Category 3 |
| Skin corrosion/irritation                 | Category 2 |
| Serious eye damage/eye irritation         | Category 1 |

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements



Danger

### Hazard statements

Toxic if inhaled.

Causes skin irritation.

Causes serious eye damage.

**Precautionary Statements - Prevention**

Use only outdoors or in a well-ventilated area.

Wash face, hands and any exposed skin thoroughly after handling.

Wear protective gloves, protective clothing, eye protection and face protection.

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

Immediately call a POISON CENTER or doctor.

IF ON SKIN: Wash with plenty of water and soap.

If skin irritation occurs: Get medical advice/attention.

IF INHALED: Remove person to fresh air and keep comfortable for breathing.

Call a POISON CENTER or doctor.

**Precautionary Statements - Storage**

Store in a well-ventilated place. Keep container tightly closed.

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable.

**Unknown acute toxicity**

18.49 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (dust/mist).

**Hazards classified under paragraph (d)(1)(ii) of 1910.1200**

No information available.

**Other information**

Toxic to aquatic life with long lasting effects. Harmful to aquatic life.

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

| Chemical name                                                                                  | CAS No.   | Weight-% | Trade secret |
|------------------------------------------------------------------------------------------------|-----------|----------|--------------|
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-.omega.<br>-hydroxy- | 9002-93-1 | 5 - 10   | *            |
| Glycine, N-methyl-N-(1-oxododecyl)-, sodium salt                                               | 137-16-6  | 5 - 10   | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

**4. First-aid measures****Description of first aid measures****General advice**

Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.

**Inhalation**

Remove to fresh air. Get medical attention immediately if symptoms occur. If breathing has stopped, give artificial respiration. Get medical attention immediately. Immediate medical attention is required. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. If breathing is difficult, (trained personnel should) give oxygen.

**Eye contact**

Get immediate medical attention. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue

rinsing. Keep eye wide open while rinsing. Do not rub affected area.

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin contact</b>                       | Wash off immediately with soap and plenty of water for at least 15 minutes. Get medical attention if irritation develops and persists.                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Ingestion</b>                          | Rinse mouth. Never give anything by mouth to an unconscious person. Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Self-protection of the first aider</b> | Avoid contact with skin, eyes or clothing. Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Do not breathe vapor or mist. Use personal protective equipment as required. See section 8 for more information. |

#### **Most important symptoms and effects, both acute and delayed**

|                            |                                                                        |
|----------------------------|------------------------------------------------------------------------|
| <b>Symptoms</b>            | Burning sensation. Coughing and/ or wheezing. Difficulty in breathing. |
| <b>Effects of Exposure</b> | No information available.                                              |

#### **Indication of any immediate medical attention and special treatment needed**

|                           |                        |
|---------------------------|------------------------|
| <b>Note to physicians</b> | Treat symptomatically. |
|---------------------------|------------------------|

### **5. Fire-fighting measures**

|                                                                       |                                                                                                                                    |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>                                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| <b>Unsuitable extinguishing media</b>                                 | Do not scatter spilled material with high pressure water streams.                                                                  |
| <b>Specific hazards arising from the chemical</b>                     | No information available.                                                                                                          |
| <b>Explosion data</b>                                                 |                                                                                                                                    |
| <b>Sensitivity to mechanical impact</b>                               | None.                                                                                                                              |
| <b>Sensitivity to static discharge</b>                                | None.                                                                                                                              |
| <b>Special protective equipment and precautions for fire-fighters</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

### **6. Accidental release measures**

#### **Personal precautions, protective equipment and emergency procedures**

|                             |                                                                                                                                                                                                                                        |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal precautions</b> | Avoid contact with skin, eyes or clothing. Use personal protective equipment as required. Ensure adequate ventilation. Do not breathe vapor or mist. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. |
| <b>Other information</b>    | Refer to protective measures listed in Sections 7 and 8.                                                                                                                                                                               |

#### **Methods and material for containment and cleaning up**

|                                        |                                                                                      |
|----------------------------------------|--------------------------------------------------------------------------------------|
| <b>Methods for containment</b>         | Prevent further leakage or spillage if safe to do so.                                |
| <b>Methods for cleaning up</b>         | Pick up and transfer to properly labeled containers.                                 |
| <b>Prevention of secondary hazards</b> | Clean contaminated objects and areas thoroughly observing environmental regulations. |

## 7. Handling and storage

### Precautions for safe handling

#### Advice on safe handling

Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse. Do not breathe vapor or mist. In case of insufficient ventilation, wear suitable respiratory equipment. Handle product only in closed system or provide appropriate exhaust ventilation.

#### General hygiene considerations

Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection. Do not eat, drink or smoke when using this product. Do not breathe vapor or mist. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Contaminated work clothing must not be allowed out of the workplace. Regular cleaning of equipment, work area and clothing is recommended. Wash hands before breaks and immediately after handling the product.

### Conditions for safe storage, including any incompatibilities

#### Storage Conditions

Keep containers tightly closed in a dry, cool and well-ventilated place. Store locked up. Keep out of the reach of children. Store according to product and label instructions.

## 8. Exposure controls/personal protection

### Working area parameters, subject to mandatory control (MAC or TSEL)

#### Exposure Limits

This product, as supplied, contains materials that do not have reportable occupational exposure limits or are not subject to the reporting requirements of the local jurisdiction.

#### Biological occupational exposure limits

This product, as supplied, contains materials that do not have reportable biological exposure limits or are not subject to the reporting requirements of the local jurisdiction.

### Appropriate engineering controls

#### Engineering controls

Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

#### Eye/face protection

Tight sealing safety goggles.

#### Hand protection

Wear suitable gloves. Impervious gloves.

#### Skin and body protection

Wear suitable protective clothing. Long sleeved clothing.

#### Respiratory protection

Appropriate respiratory protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

#### Appearance

aqueous solution

#### Physical state

Liquid

#### Color

colorless

Odor (includes odor threshold) Odorless

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      | 7                 |                         |
| Melting point / freezing point          | No data available | None known              |
| Initial boiling point and boiling range | No data available | None known              |
| Flash point                             | No data available | None known              |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapor pressure                          | No data available | None known              |
| Relative vapor density                  | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Miscible in water |                         |
| Solubility(ies)                         | No data available | None known              |
| Partition coefficient                   | No data available | None known              |
| Autoignition temperature                | No data available | None known              |
| Decomposition temperature               | No data available | None known              |
| Kinematic viscosity                     | No data available | None known              |
| Dynamic viscosity                       | No data available | None known              |

Other information

## 10. Stability and reactivity

|                                    |                                                      |
|------------------------------------|------------------------------------------------------|
| Reactivity                         | No information available.                            |
| Chemical stability                 | Stable under normal conditions.                      |
| Possibility of hazardous reactions | None under normal processing.                        |
| Conditions to avoid                | Excessive heat.                                      |
| Incompatible materials             | Strong acids. Strong bases. Strong oxidizing agents. |
| Hazardous decomposition products   | None known based on information supplied.            |

Information on likely routes of exposure

**Product Information**

|              |                                                                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. May cause irritation of respiratory tract. Toxic by inhalation (based on components). |
| Eye contact  | Specific test data for the substance or mixture is not available. Causes serious eye damage. May cause irreversible damage to eyes.                     |
| Skin contact | Specific test data for the substance or mixture is not available. Causes skin irritation (based on components).                                         |
| Ingestion    | Specific test data for the substance or mixture is not available. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea.       |

Symptoms related to the physical, chemical and toxicological characteristics

|          |                                                                                   |
|----------|-----------------------------------------------------------------------------------|
| Symptoms | Redness. Burning. May cause blindness. May cause redness and tearing of the eyes. |
|----------|-----------------------------------------------------------------------------------|

Coughing and/ or wheezing. Difficulty in breathing.

**Acute toxicity** Harmful by inhalation.**Numerical measures of toxicity**

The following ATE values have been calculated for the mixture

|                               |                 |
|-------------------------------|-----------------|
| ATEmix (oral)                 | 13,128.60 mg/kg |
| ATEmix (inhalation-dust/mist) | 0.5014 mg/l     |

**Unknown acute toxicity**

18.49 % of the mixture consists of ingredient(s) of unknown acute inhalation toxicity (dust/mist)

**Component Information**

| Chemical name                                                                                               | Oral LD50            | Dermal LD50 | Inhalation LC50             |
|-------------------------------------------------------------------------------------------------------------|----------------------|-------------|-----------------------------|
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)ph<br>enyl]-.omega.-hydroxy-<br>9002-93-1 | = 1800 mg/kg ( Rat ) | -           | -                           |
| Glycine, N-methyl-N-(1-oxododecyl)-,<br>sodium salt<br>137-16-6                                             | > 5000 mg/kg ( Rat ) | -           | 0.05 - 0.5 mg/L ( Rat ) 4 h |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure****Skin corrosion/irritation** Classification based on data available for ingredients. Causes skin irritation.**Serious eye damage/eye irritation** Classification based on data available for ingredients. Causes burns. Causes serious eye damage.**Respiratory or skin sensitization** No information available.**Germ cell mutagenicity** No information available.**Carcinogenicity** No information available.**Reproductive toxicity** No information available.**STOT - single exposure** No information available.**STOT - repeated exposure** No information available.**Aspiration hazard** No information available.**Other adverse effects** No information available.**Interactive effects** No information available.**12. Ecological information****Ecotoxicity** Toxic to aquatic life with long lasting effects. Harmful to aquatic life.

| Chemical name | Algae/aquatic plants | Fish | Toxicity to | Crustacea |
|---------------|----------------------|------|-------------|-----------|
|---------------|----------------------|------|-------------|-----------|

|                                                                    |   |                                      | microorganisms |   |
|--------------------------------------------------------------------|---|--------------------------------------|----------------|---|
| Glycine,<br>N-methyl-N-(1-oxododecyl)-,<br>sodium salt<br>137-16-6 | - | LC50: =107mg/L (96h,<br>Danio rerio) | -              | - |

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

**Other adverse effects** No information available.

| Chemical name                                                                                  | EU - REACH (1907/2006) - Article 59(1)<br>- Candidate List of Substances of Very<br>High Concern (SVHC) for Authorisation | EU - REACH (1907/2006) - Endocrine<br>Disruptor Assessment List of<br>Substances |
|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Poly(oxy-1,2-ethanediyl),<br>.alpha.-[4-(1,1,3,3-tetramethylbutyl)phenyl]-.ome<br>ga.-hydroxy- | Endocrine disrupting properties                                                                                           | -                                                                                |

### 13. Disposal considerations

#### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

### 14. Transport information

DOT Not regulated

TDG Not regulated

MEX Not regulated

ICAO (air) Not regulated

IATA Not regulated

IMDG Not regulated

### 15. Regulatory information

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories** Contact supplier for inventory compliance status

**US Federal Regulations****TSCA**

All chemical components are listed on the TSCA Inventory.

**SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

**CAA (Clean Air Act)**

This product does not contain any substances regulated as pollutants pursuant to Clean Air Act (CAA).

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**US State Regulations****California Proposition 65**

This product does not contain any Proposition 65 chemicals

**U.S. State Right-to-Know Regulations**

| Chemical name      | New Jersey | Massachusetts | Pennsylvania |
|--------------------|------------|---------------|--------------|
| Water<br>7732-18-5 | -          | -             | X            |

**U.S. EPA Label Information**

**EPA Pesticide Registration Number** Not applicable

**16. Other information**

|             |                         |                       |                           |                              |
|-------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> - |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend**

|                |                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------|
| <b>ADN</b>     | Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways (Europe) |
| <b>ADR</b>     | Agreement concerning the International Carriage of Dangerous Goods by Road (Europe)             |
| <b>ATE</b>     | Acute Toxicity Estimate                                                                         |
| <b>ASTM</b>    | American Society for the Testing of Materials                                                   |
| <b>BEL</b>     | Biological exposure limits                                                                      |
| <b>bw</b>      | Body weight                                                                                     |
| <b>Ceiling</b> | Maximum limit value                                                                             |

|                |                                                                                     |
|----------------|-------------------------------------------------------------------------------------|
| <b>CMR</b>     | Carcinogen, Mutagen or Reproductive Toxicant                                        |
| <b>DOT</b>     | Department of Transportation (United States)                                        |
| <b>EmS</b>     | Emergency Schedule                                                                  |
| <b>EPA</b>     | Environmental Protection Agency                                                     |
| <b>GHS</b>     | Globally Harmonized System                                                          |
| <b>HMIS</b>    | Hazardous Materials Identification System                                           |
| <b>IATA</b>    | International Air Transport Association                                             |
| <b>ICAO</b>    | International Civil Aviation Organization                                           |
| <b>IMDG</b>    | International Maritime Dangerous Goods                                              |
| <b>IMO</b>     | International Maritime Organization                                                 |
| <b>ISO</b>     | International Organization for Standardization                                      |
| <b>LC50</b>    | Lethal Concentration to 50% of a test population                                    |
| <b>LD50</b>    | Lethal Dose to 50% of a test population (Median Lethal Dose)                        |
| <b>MARPOL</b>  | International Convention for the Prevention of Pollution from Ships                 |
| <b>NFPA</b>    | National Fire Protection Association                                                |
| <b>NIOSH</b>   | National Institute for Occupational Safety and Health                               |
| <b>n.o.s.</b>  | Not Otherwise Specified                                                             |
| <b>NOAEL</b>   | No Observed Adverse Effect Level                                                    |
| <b>OECD</b>    | Organization for Economic Cooperation and Development                               |
| <b>OEL</b>     | Occupational exposure limits                                                        |
| <b>OSHA</b>    | Occupational Safety and Health Administration of the US Department of Labor         |
| <b>PBT</b>     | Persistent, Bioaccumulative and Toxic substance                                     |
| <b>PMT</b>     | Persistent, Mobile and Toxic                                                        |
| <b>PPE</b>     | Personal protective equipment                                                       |
| <b>RID</b>     | Agreement concerning the International Carriage of Dangerous Goods by Rail (Europe) |
| <b>SARA</b>    | Superfund Amendments and Reauthorization Act                                        |
| <b>SDS</b>     | Safety Data Sheet                                                                   |
| <b>STEL</b>    | Short Term Exposure Limit                                                           |
| <b>STOT RE</b> | Specific target organ toxicity - Repeated exposure                                  |
| <b>STOT SE</b> | Specific target organ toxicity - Single exposure                                    |
| <b>TDG</b>     | Transport of Dangerous Goods (Canada)                                               |
| <b>TSCA</b>    | Toxic Substances Control Act (United States)                                        |
| <b>TWA</b>     | Time-Weighted Average                                                               |
| <b>UN</b>      | United Nations                                                                      |
| <b>VOC</b>     | Volatile organic compounds                                                          |
| <b>vPvB</b>    | Very Persistent and Very Bioaccumulative                                            |
| <b>vPvM</b>    | Very Persistent and Very Mobile                                                     |
| <b>Sk</b>      | Skin notation                                                                       |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Prepared By**

Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date**

25-Jan-2023

**Revision Note**

This product has been discontinued and the Safety Data Sheet will not be updated.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage,

transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
US OSHA Hazard Communication Standard (29 CFR 1910.1200)

Revision date 14-May-2025

Revision Number 1.1

## 1. Identification

### Product identifier

**Product Name** Reagent B - Clarifying Solution

### Other means of identification

**Catalog Number(s)** 1641  
**UN number or ID number** UN1120

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals  
**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Life Science  
2000 Alfred Nobel Drive  
Hercules, California 94547

**Technical Service** 1-800-424-6723  
support@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification of the substance or mixture

|                                                                            |            |
|----------------------------------------------------------------------------|------------|
| Flammable liquids                                                          | Category 3 |
| Acute toxicity - Oral                                                      | Category 4 |
| Skin corrosion/irritation                                                  | Category 2 |
| Serious eye damage/eye irritation                                          | Category 1 |
| Specific target organ toxicity (single exposure)                           | Category 3 |
| Category 3 Target organ effects: Respiratory irritation, Narcotic effects. |            |

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements



Danger

**Hazard statements**

Flammable liquid and vapor.  
Harmful if swallowed.  
Causes skin irritation.  
Causes serious eye damage.  
May cause respiratory irritation.  
May cause drowsiness or dizziness.

**Precautionary Statements - Prevention**

Wash face, hands and any exposed skin thoroughly after handling.  
Do not eat, drink or smoke when using this product.  
Use only outdoors or in a well-ventilated area.  
Use only non-sparking tools.  
Take action to prevent static discharges.  
Wear protective gloves, protective clothing, eye protection and face protection.  
Avoid release to the environment.

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.  
Immediately call a POISON CENTER or doctor.  
If skin irritation occurs: Get medical advice/attention.  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower.  
IF INHALED: Remove person to fresh air and keep comfortable for breathing.  
Call a POISON CENTER or doctor if you feel unwell.  
IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell.  
Rinse mouth.  
In case of fire: Use CO<sub>2</sub>, dry chemical, or foam to extinguish.

**Precautionary Statements - Storage**

Store in a well-ventilated place. Keep container tightly closed.

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable.

**Hazards classified under paragraph (d)(1)(ii) of 1910.1200**

No information available.

**Other information**

Harmful to aquatic life with long lasting effects.

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

| Chemical name | CAS No. | Weight-% | Trade secret |
|---------------|---------|----------|--------------|
| 1-Butanol     | 71-36-3 | 50 - 100 | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

**4. First-aid measures****Description of first aid measures****General advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is

required.

|                                           |                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                         | Remove to fresh air. Get medical attention immediately if symptoms occur. IF exposed or concerned: Get medical advice/attention.                                                                                                                                                                               |
| <b>Eye contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Keep eye wide open while rinsing. Do not rub affected area. Get immediate medical attention. Remove contact lenses, if present and easy to do. Continue rinsing.                                                      |
| <b>Skin contact</b>                       | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention if irritation develops and persists.                                                                                                                                               |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Rinse mouth. Never give anything by mouth to an unconscious person. Call a physician.                                                                                                                                                                                                  |
| <b>Self-protection of the first aider</b> | Remove all sources of ignition. Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. Use personal protective equipment as required. See section 8 for more information. Avoid contact with skin, eyes or clothing. |

#### **Most important symptoms and effects, both acute and delayed**

|                            |                                                                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Symptoms</b>            | Burning sensation. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. |
| <b>Effects of Exposure</b> | No information available.                                                                                                               |

#### **Indication of any immediate medical attention and special treatment needed**

|                           |                        |
|---------------------------|------------------------|
| <b>Note to physicians</b> | Treat symptomatically. |
|---------------------------|------------------------|

### **5. Fire-fighting measures**

|                                                                       |                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>                                   | Dry chemical. Carbon dioxide (CO <sub>2</sub> ). Water spray. Alcohol resistant foam.                                                                                                                                                                           |
| <b>Unsuitable extinguishing media</b>                                 | Do not scatter spilled material with high pressure water streams.                                                                                                                                                                                               |
| <b>Specific hazards arising from the chemical</b>                     | Risk of ignition. Keep product and empty container away from heat and sources of ignition. In the event of fire, cool tanks with water spray. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. |
| <b>Explosion data</b>                                                 |                                                                                                                                                                                                                                                                 |
| <b>Sensitivity to mechanical impact</b>                               | None.                                                                                                                                                                                                                                                           |
| <b>Sensitivity to static discharge</b>                                | Yes.                                                                                                                                                                                                                                                            |
| <b>Special protective equipment and precautions for fire-fighters</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.                                                                                                                              |

### **6. Accidental release measures**

#### **Personal precautions, protective equipment and emergency procedures**

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal precautions</b> | Evacuate personnel to safe areas. Use personal protective equipment as required. See section 8 for more information. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. ELIMINATE all ignition sources (no smoking, flares, sparks or flames in immediate area). Pay attention to flashback. Take precautionary measures against static discharges. All equipment used when handling the product must be grounded. Do not touch or walk through spilled material. |
| <b>Other information</b>    | Ventilate the area. Refer to protective measures listed in Sections 7 and 8.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

**Methods and material for containment and cleaning up**

|                                        |                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Methods for containment</b>         | Stop leak if you can do it without risk. Do not touch or walk through spilled material. A vapor suppressing foam may be used to reduce vapors. Dike far ahead of spill to collect runoff water. Keep out of drains, sewers, ditches and waterways. Absorb with earth, sand or other non-combustible material and transfer to containers for later disposal. |
| <b>Methods for cleaning up</b>         | Take precautionary measures against static discharges. Dam up. Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.                                                                                                                                                                                                  |
| <b>Prevention of secondary hazards</b> | Clean contaminated objects and areas thoroughly observing environmental regulations.                                                                                                                                                                                                                                                                        |

**7. Handling and storage****Precautions for safe handling**

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Advice on safe handling</b>        | Use personal protection equipment. Avoid breathing vapors or mists. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Use grounding and bonding connection when transferring this material to prevent static discharge, fire or explosion. Use with local exhaust ventilation. Use spark-proof tools and explosion-proof equipment. Keep in an area equipped with sprinklers. Use according to package label instructions. Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse. In case of insufficient ventilation, wear suitable respiratory equipment. |
| <b>General hygiene considerations</b> | Do not eat, drink or smoke when using this product. Contaminated work clothing must not be allowed out of the workplace. Regular cleaning of equipment, work area and clothing is recommended. Wash hands before breaks and immediately after handling the product. Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection.                                                                                                                                                                                                                                                                                                                                                                                                                  |

**Conditions for safe storage, including any incompatibilities**

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Storage Conditions</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks, flame and other sources of ignition (i.e., pilot lights, electric motors and static electricity). Keep in properly labeled containers. Do not store near combustible materials. Keep in an area equipped with sprinklers. Store in accordance with the particular national regulations. Store in accordance with local regulations. Keep out of the reach of children. Store locked up. Store according to product and label instructions. |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**8. Exposure controls/personal protection****Working area parameters, subject to mandatory control (MAC or TSEL)****Exposure Limits**

| Chemical name        | ACGIH TLV   | OSHA PEL                                                                                                                             | NIOSH                                                               |
|----------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 1-Butanol<br>71-36-3 | TWA: 20 ppm | TWA: 100 ppm<br>TWA: 300 mg/m <sup>3</sup><br>(vacated) Sk*<br>(vacated) Ceiling: 50 ppm<br>(vacated) Ceiling: 150 mg/m <sup>3</sup> | IDLH: 1400 ppm<br>Ceiling: 50 ppm<br>Ceiling: 150 mg/m <sup>3</sup> |

|                                                |                                                                                                                                                                                  |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Biological occupational exposure limits</b> | This product, as supplied, contains materials that do not have reportable biological exposure limits or are not subject to the reporting requirements of the local jurisdiction. |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Appropriate engineering controls**

|                             |                             |
|-----------------------------|-----------------------------|
| <b>Engineering controls</b> | Showers<br>Eyewash stations |
|-----------------------------|-----------------------------|

Ventilation systems.

#### **Individual protection measures, such as personal protective equipment**

|                                 |                                                                                                                                                                                                                                                                                               |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye/face protection</b>      | Tight sealing safety goggles.                                                                                                                                                                                                                                                                 |
| <b>Hand protection</b>          | Wear suitable gloves. Impervious gloves.                                                                                                                                                                                                                                                      |
| <b>Skin and body protection</b> | Wear suitable protective clothing. Long sleeved clothing. Chemical resistant apron. Antistatic boots.                                                                                                                                                                                         |
| <b>Respiratory protection</b>   | Appropriate respiratory protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |

## **9. Physical and chemical properties**

#### **Information on basic physical and chemical properties**

|                                       |           |
|---------------------------------------|-----------|
| <b>Appearance</b>                     | Liquid    |
| <b>Physical state</b>                 | Liquid    |
| <b>Color</b>                          | colorless |
| <b>Odor (includes odor threshold)</b> | Alcohol   |

| <b>Property</b>                                | <b>Values</b>      | <b>Remarks • Method</b> |
|------------------------------------------------|--------------------|-------------------------|
| <b>pH</b>                                      |                    | None known              |
| <b>Melting point / freezing point</b>          | -89 °C / -128.2 °F |                         |
| <b>Initial boiling point and boiling range</b> | 116 °C / 240.8 °F  |                         |
| <b>Flash point</b>                             | 35 °C / 95 °F      |                         |
| <b>Evaporation rate</b>                        | No data available  | None known              |
| <b>Flammability</b>                            | No data available  | None known              |
| <b>Flammability Limit in Air</b>               |                    | None known              |
| <b>Upper flammability or explosive limits</b>  | No data available  |                         |
| <b>Lower flammability or explosive limits</b>  | No data available  |                         |
| <b>Vapor pressure</b>                          | No data available  | None known              |
| <b>Relative vapor density</b>                  | No data available  | None known              |
| <b>Relative density</b>                        | 0.81               | None known              |
| <b>Water solubility</b>                        | Miscible in water  |                         |
| <b>Solubility(ies)</b>                         | No data available  | None known              |
| <b>Partition coefficient</b>                   | No data available  | None known              |
| <b>Autoignition temperature</b>                | No data available  | None known              |
| <b>Decomposition temperature</b>               | No data available  | None known              |
| <b>Kinematic viscosity</b>                     | No data available  | None known              |
| <b>Dynamic viscosity</b>                       | No data available  | None known              |

#### **Other information**

## **10. Stability and reactivity**

|                                           |                                 |
|-------------------------------------------|---------------------------------|
| <b>Reactivity</b>                         | No information available.       |
| <b>Chemical stability</b>                 | Stable under normal conditions. |
| <b>Possibility of hazardous reactions</b> | None under normal processing.   |
| <b>Conditions to avoid</b>                | Heat, flames and sparks.        |

**Incompatible materials** Strong acids. Strong bases. Strong oxidizing agents.

**Hazardous decomposition products** None known based on information supplied.

#### Information on likely routes of exposure

##### Product Information

|                     |                                                                                                                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available. May cause irritation of respiratory tract. May cause drowsiness or dizziness.                                               |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available. Causes serious eye damage. May cause irreversible damage to eyes.                                                           |
| <b>Skin contact</b> | Specific test data for the substance or mixture is not available. Causes skin irritation (based on components).                                                                               |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea. Harmful if swallowed (based on components). |

#### Symptoms related to the physical, chemical and toxicological characteristics

**Symptoms** Redness. Burning. May cause blindness. May cause redness and tearing of the eyes. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

**Acute toxicity** Harmful if swallowed.

##### Numerical measures of toxicity

The following ATE values have been calculated for the mixture

|                               |              |
|-------------------------------|--------------|
| ATEmix (oral)                 | 790.10 mg/kg |
| ATEmix (inhalation-dust/mist) | 17.80 mg/l   |

##### Component Information

| Chemical name        | Oral LD50           | Dermal LD50             | Inhalation LC50          |
|----------------------|---------------------|-------------------------|--------------------------|
| 1-Butanol<br>71-36-3 | = 700 mg/kg ( Rat ) | = 3402 mg/kg ( Rabbit ) | > 17.76 mg/L ( Rat ) 4 h |

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Skin corrosion/irritation** Classification based on data available for ingredients. Causes skin irritation.

**Serious eye damage/eye irritation** Classification based on data available for ingredients. Causes burns. Causes serious eye damage.

**Respiratory or skin sensitization** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

**Reproductive toxicity** No information available.

|                                 |                                                                      |
|---------------------------------|----------------------------------------------------------------------|
| <b>STOT - single exposure</b>   | May cause respiratory irritation. May cause drowsiness or dizziness. |
| <b>STOT - repeated exposure</b> | No information available.                                            |
| <b>Target organ effects</b>     | Respiratory system, Eyes, Skin, Central nervous system.              |
| <b>Aspiration hazard</b>        | No information available.                                            |
| <b>Other adverse effects</b>    | No information available.                                            |
| <b>Interactive effects</b>      | No information available.                                            |

## 12. Ecological information

|                    |                                                    |
|--------------------|----------------------------------------------------|
| <b>Ecotoxicity</b> | Harmful to aquatic life with long lasting effects. |
|--------------------|----------------------------------------------------|

| Chemical name        | Algae/aquatic plants                                                                                           | Fish                                                                                                                                                                                                                                    | Toxicity to microorganisms | Crustacea                                                                                          |
|----------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------|
| 1-Butanol<br>71-36-3 | EC50: >500mg/L (96h, <i>Desmodesmus subspicatus</i> )<br>EC50: >500mg/L (72h, <i>Desmodesmus subspicatus</i> ) | LC50: 1730 - 1910mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: =1740mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 100000 - 500000µg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =1910000µg/L (96h, <i>Pimephales promelas</i> ) | -                          | EC50: =1983mg/L (48h, <i>Daphnia magna</i> )<br>EC50: 1897 - 2072mg/L (48h, <i>Daphnia magna</i> ) |

|                                      |                           |
|--------------------------------------|---------------------------|
| <b>Persistence and degradability</b> | No information available. |
|--------------------------------------|---------------------------|

### Bioaccumulation

#### Component Information

| Chemical name        | Partition coefficient |
|----------------------|-----------------------|
| 1-Butanol<br>71-36-3 | 1                     |

|                              |                           |
|------------------------------|---------------------------|
| <b>Other adverse effects</b> | No information available. |
|------------------------------|---------------------------|

## 13. Disposal considerations

### Disposal methods

|                                            |                                                                                                                                                                                                                                                                              |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Should not be released into the environment. Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| <b>Contaminated packaging</b>              | Empty containers pose a potential fire and explosion hazard. Do not cut, puncture or weld containers.                                                                                                                                                                        |

**California Hazardous Waste Status** This product contains one or more substances that are listed with the State of California as a hazardous waste.

## 14. Transport information

### DOT

|                                        |                                     |
|----------------------------------------|-------------------------------------|
| UN number or ID number                 | UN1120                              |
| Proper shipping name                   | BUTANOLS                            |
| Transport hazard class(es)             | 3                                   |
| Packing group                          | II                                  |
| Reportable quantity (lbs) (calculated) | 1-Butanol: RQ (lb)= 5001.00         |
| Special Provisions                     | IB2, T4, TP1, TP29                  |
| DOT Marine Pollutant                   | NP                                  |
| Description                            | UN1120, BUTANOLS (1-Butanol), 3, II |

### TDG

|                            |                                     |
|----------------------------|-------------------------------------|
| UN number or ID number     | UN1120                              |
| UN proper shipping name    | BUTANOLS                            |
| Transport hazard class(es) | 3                                   |
| Packing group              | II                                  |
| Description                | UN1120, BUTANOLS (1-Butanol), 3, II |

### MEX

|                            |                         |
|----------------------------|-------------------------|
| UN number or ID number     | UN1120                  |
| UN proper shipping name    | BUTANOLS                |
| Transport hazard class(es) | 3                       |
| Packing group              | II                      |
| Technical Name             | 1-Butanol               |
| Description                | UN1120, BUTANOLS, 3, II |

### ICAO (air)

|                            |                         |
|----------------------------|-------------------------|
| UN number or ID number     | UN1120                  |
| UN proper shipping name    | BUTANOLS                |
| Transport hazard class(es) | 3                       |
| Packing group              | II                      |
| Description                | UN1120, BUTANOLS, 3, II |
| Special Provisions         | A3                      |

### IATA

|                            |                         |
|----------------------------|-------------------------|
| UN number or ID number     | UN1120                  |
| UN proper shipping name    | Butanols                |
| Technical Name             | 1-Butanol               |
| Transport hazard class(es) | 3                       |
| Packing group              | II                      |
| Special Provisions         | A3                      |
| Description                | UN1120, Butanols, 3, II |

### IMDG

|                            |                                                  |
|----------------------------|--------------------------------------------------|
| UN number or ID number     | UN1120                                           |
| UN proper shipping name    | BUTANOLS                                         |
| Transport hazard class(es) | 3                                                |
| Packing group              | II                                               |
| Marine pollutant indicator | NP                                               |
| EmS-No.                    | F-E S-D                                          |
| Description                | UN1120, BUTANOLS (1-BUTANOL), 3, II, (35°C C.C.) |

## 15. Regulatory information

### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories** Contact supplier for inventory compliance status

### US Federal Regulations

#### **TSCA**

All chemical components are listed on the TSCA Inventory.

#### **SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

#### **SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

#### **CWA (Clean Water Act)**

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

#### **CAA (Clean Air Act)**

This product does not contain any substances regulated as pollutants pursuant to Clean Air Act (CAA).

#### **CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302).

| Chemical name        | Hazardous Substances RQs | Extremely Hazardous Substances RQs | Reportable Quantity (RQ)                   |
|----------------------|--------------------------|------------------------------------|--------------------------------------------|
| 1-Butanol<br>71-36-3 | 5000 lb                  | -                                  | RQ 5000 lb final RQ<br>RQ 2270 kg final RQ |

### US State Regulations

#### **California Proposition 65**

This product does not contain any Proposition 65 chemicals

#### **U.S. State Right-to-Know Regulations**

| Chemical name        | New Jersey | Massachusetts | Pennsylvania |
|----------------------|------------|---------------|--------------|
| 1-Butanol<br>71-36-3 | X          | X             | X            |

### U.S. EPA Label Information

**EPA Pesticide Registration Number** Not applicable

## **16. Other information**

**NFPA**  
**HMIS**

**Health hazards** 0  
**Health hazards** 0

**Flammability** 0  
**Flammability** 0

**Instability** 0  
**Physical hazards** 0

**Special hazards** -  
**Personal protection** -

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend**

|                |                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------|
| <b>ADN</b>     | Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways (Europe) |
| <b>ADR</b>     | Agreement concerning the International Carriage of Dangerous Goods by Road (Europe)             |
| <b>ATE</b>     | Acute Toxicity Estimate                                                                         |
| <b>ASTM</b>    | American Society for the Testing of Materials                                                   |
| <b>BEL</b>     | Biological exposure limits                                                                      |
| <b>bw</b>      | Body weight                                                                                     |
| <b>Ceiling</b> | Maximum limit value                                                                             |
| <b>CMR</b>     | Carcinogen, Mutagen or Reproductive Toxicant                                                    |
| <b>DOT</b>     | Department of Transportation (United States)                                                    |
| <b>EmS</b>     | Emergency Schedule                                                                              |
| <b>EPA</b>     | Environmental Protection Agency                                                                 |
| <b>GHS</b>     | Globally Harmonized System                                                                      |
| <b>HMIS</b>    | Hazardous Materials Identification System                                                       |
| <b>IATA</b>    | International Air Transport Association                                                         |
| <b>ICAO</b>    | International Civil Aviation Organization                                                       |
| <b>IMDG</b>    | International Maritime Dangerous Goods                                                          |
| <b>IMO</b>     | International Maritime Organization                                                             |
| <b>ISO</b>     | International Organization for Standardization                                                  |
| <b>LC50</b>    | Lethal Concentration to 50% of a test population                                                |
| <b>LD50</b>    | Lethal Dose to 50% of a test population (Median Lethal Dose)                                    |
| <b>MARPOL</b>  | International Convention for the Prevention of Pollution from Ships                             |
| <b>NFPA</b>    | National Fire Protection Association                                                            |
| <b>NIOSH</b>   | National Institute for Occupational Safety and Health                                           |
| <b>n.o.s.</b>  | Not Otherwise Specified                                                                         |
| <b>NOAEL</b>   | No Observed Adverse Effect Level                                                                |
| <b>OECD</b>    | Organization for Economic Cooperation and Development                                           |
| <b>OEL</b>     | Occupational exposure limits                                                                    |
| <b>OSHA</b>    | Occupational Safety and Health Administration of the US Department of Labor                     |
| <b>PBT</b>     | Persistent, Bioaccumulative and Toxic substance                                                 |
| <b>PMT</b>     | Persistent, Mobile and Toxic                                                                    |
| <b>PPE</b>     | Personal protective equipment                                                                   |
| <b>RID</b>     | Agreement concerning the International Carriage of Dangerous Goods by Rail (Europe)             |
| <b>SARA</b>    | Superfund Amendments and Reauthorization Act                                                    |
| <b>SDS</b>     | Safety Data Sheet                                                                               |
| <b>STEL</b>    | Short Term Exposure Limit                                                                       |
| <b>STOT RE</b> | Specific target organ toxicity - Repeated exposure                                              |
| <b>STOT SE</b> | Specific target organ toxicity - Single exposure                                                |
| <b>TDG</b>     | Transport of Dangerous Goods (Canada)                                                           |
| <b>TSCA</b>    | Toxic Substances Control Act (United States)                                                    |
| <b>TWA</b>     | Time-Weighted Average                                                                           |
| <b>UN</b>      | United Nations                                                                                  |
| <b>VOC</b>     | Volatile organic compounds                                                                      |
| <b>vPvB</b>    | Very Persistent and Very Bioaccumulative                                                        |
| <b>vPvM</b>    | Very Persistent and Very Mobile                                                                 |
| <b>Sk</b>      | Skin notation                                                                                   |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)

Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 14-May-2025

**Revision Note** Reviewed existing information and made minor updates.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
US OSHA Hazard Communication Standard (29 CFR 1910.1200)

Revision date 25-Jan-2023

Revision Number 2

## 1. Identification

### Product identifier

**Product Name** Discontinued - Reagent 2 - Clarifying Solution

### Other means of identification

**Catalog Number(s)** 1770  
**UN number or ID number** UN1120

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals  
**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Life Science  
2000 Alfred Nobel Drive  
Hercules, California 94547

**Technical Service** 1-800-424-6723  
support@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification of the substance or mixture

|                                                                            |            |
|----------------------------------------------------------------------------|------------|
| Flammable liquids                                                          | Category 3 |
| Acute toxicity - Oral                                                      | Category 4 |
| Skin corrosion/irritation                                                  | Category 2 |
| Serious eye damage/eye irritation                                          | Category 1 |
| Specific target organ toxicity (single exposure)                           | Category 3 |
| Category 3 Target organ effects: Respiratory irritation, Narcotic effects. |            |

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements



Danger

**Hazard statements**

Flammable liquid and vapor.  
Harmful if swallowed.  
Causes skin irritation.  
Causes serious eye damage.  
May cause respiratory irritation.  
May cause drowsiness or dizziness.

**Precautionary Statements - Prevention**

Wash face, hands and any exposed skin thoroughly after handling.  
Do not eat, drink or smoke when using this product.  
Use only outdoors or in a well-ventilated area.  
Use only non-sparking tools.  
Take action to prevent static discharges.  
Wear protective gloves, protective clothing, eye protection and face protection.  
Avoid release to the environment.

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.  
Immediately call a POISON CENTER or doctor.  
If skin irritation occurs: Get medical advice/attention.  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower.  
IF INHALED: Remove person to fresh air and keep comfortable for breathing.  
Call a POISON CENTER or doctor if you feel unwell.  
IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell.  
Rinse mouth.  
In case of fire: Use CO<sub>2</sub>, dry chemical, or foam to extinguish.

**Precautionary Statements - Storage**

Store in a well-ventilated place. Keep container tightly closed.

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable.

**Hazards classified under paragraph (d)(1)(ii) of 1910.1200**

No information available.

**Other information**

Harmful to aquatic life with long lasting effects.

**3. Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

| Chemical name | CAS No. | Weight-% | Trade secret |
|---------------|---------|----------|--------------|
| 1-Butanol     | 71-36-3 | 50 - 100 | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

**4. First-aid measures****Description of first aid measures****General advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is

required.

|                                           |                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                         | Remove to fresh air. Get medical attention immediately if symptoms occur. IF exposed or concerned: Get medical advice/attention.                                                                                                                                                                               |
| <b>Eye contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Keep eye wide open while rinsing. Do not rub affected area. Get immediate medical attention. Remove contact lenses, if present and easy to do. Continue rinsing.                                                      |
| <b>Skin contact</b>                       | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention if irritation develops and persists.                                                                                                                                               |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Rinse mouth. Never give anything by mouth to an unconscious person. Call a physician.                                                                                                                                                                                                  |
| <b>Self-protection of the first aider</b> | Remove all sources of ignition. Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. Use personal protective equipment as required. See section 8 for more information. Avoid contact with skin, eyes or clothing. |

#### **Most important symptoms and effects, both acute and delayed**

|                            |                                                                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Symptoms</b>            | Burning sensation. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. |
| <b>Effects of Exposure</b> | No information available.                                                                                                               |

#### **Indication of any immediate medical attention and special treatment needed**

|                           |                        |
|---------------------------|------------------------|
| <b>Note to physicians</b> | Treat symptomatically. |
|---------------------------|------------------------|

### **5. Fire-fighting measures**

|                                                                       |                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>                                   | Dry chemical. Carbon dioxide (CO <sub>2</sub> ). Water spray. Alcohol resistant foam.                                                                                                                                                                           |
| <b>Unsuitable extinguishing media</b>                                 | Do not scatter spilled material with high pressure water streams.                                                                                                                                                                                               |
| <b>Specific hazards arising from the chemical</b>                     | Risk of ignition. Keep product and empty container away from heat and sources of ignition. In the event of fire, cool tanks with water spray. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. |
| <b>Explosion data</b>                                                 |                                                                                                                                                                                                                                                                 |
| <b>Sensitivity to mechanical impact</b>                               | None.                                                                                                                                                                                                                                                           |
| <b>Sensitivity to static discharge</b>                                | Yes.                                                                                                                                                                                                                                                            |
| <b>Special protective equipment and precautions for fire-fighters</b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment.                                                                                                                              |

### **6. Accidental release measures**

#### **Personal precautions, protective equipment and emergency procedures**

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal precautions</b> | Evacuate personnel to safe areas. Use personal protective equipment as required. See section 8 for more information. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. ELIMINATE all ignition sources (no smoking, flares, sparks or flames in immediate area). Pay attention to flashback. Take precautionary measures against static discharges. All equipment used when handling the product must be grounded. Do not touch or walk through spilled material. |
| <b>Other information</b>    | Ventilate the area. Refer to protective measures listed in Sections 7 and 8.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

**Methods and material for containment and cleaning up**

|                                        |                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Methods for containment</b>         | Stop leak if you can do it without risk. Do not touch or walk through spilled material. A vapor suppressing foam may be used to reduce vapors. Dike far ahead of spill to collect runoff water. Keep out of drains, sewers, ditches and waterways. Absorb with earth, sand or other non-combustible material and transfer to containers for later disposal. |
| <b>Methods for cleaning up</b>         | Take precautionary measures against static discharges. Dam up. Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.                                                                                                                                                                                                  |
| <b>Prevention of secondary hazards</b> | Clean contaminated objects and areas thoroughly observing environmental regulations.                                                                                                                                                                                                                                                                        |

**7. Handling and storage****Precautions for safe handling**

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Advice on safe handling</b>        | Use personal protection equipment. Avoid breathing vapors or mists. Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Use grounding and bonding connection when transferring this material to prevent static discharge, fire or explosion. Use with local exhaust ventilation. Use spark-proof tools and explosion-proof equipment. Keep in an area equipped with sprinklers. Use according to package label instructions. Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash before reuse. In case of insufficient ventilation, wear suitable respiratory equipment. |
| <b>General hygiene considerations</b> | Do not eat, drink or smoke when using this product. Contaminated work clothing must not be allowed out of the workplace. Regular cleaning of equipment, work area and clothing is recommended. Wash hands before breaks and immediately after handling the product. Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection.                                                                                                                                                                                                                                                                                                                                                                                                                  |

**Conditions for safe storage, including any incompatibilities**

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Storage Conditions</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks, flame and other sources of ignition (i.e., pilot lights, electric motors and static electricity). Keep in properly labeled containers. Do not store near combustible materials. Keep in an area equipped with sprinklers. Store in accordance with the particular national regulations. Store in accordance with local regulations. Keep out of the reach of children. Store locked up. Store according to product and label instructions. |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**8. Exposure controls/personal protection****Working area parameters, subject to mandatory control (MAC or TSEL)****Exposure Limits**

| Chemical name        | ACGIH TLV   | OSHA PEL                                                                                                                             | NIOSH                                                               |
|----------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 1-Butanol<br>71-36-3 | TWA: 20 ppm | TWA: 100 ppm<br>TWA: 300 mg/m <sup>3</sup><br>(vacated) Sk*<br>(vacated) Ceiling: 50 ppm<br>(vacated) Ceiling: 150 mg/m <sup>3</sup> | IDLH: 1400 ppm<br>Ceiling: 50 ppm<br>Ceiling: 150 mg/m <sup>3</sup> |

|                                                |                                                                                                                                                                                  |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Biological occupational exposure limits</b> | This product, as supplied, contains materials that do not have reportable biological exposure limits or are not subject to the reporting requirements of the local jurisdiction. |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Appropriate engineering controls**

|                             |                             |
|-----------------------------|-----------------------------|
| <b>Engineering controls</b> | Showers<br>Eyewash stations |
|-----------------------------|-----------------------------|

Ventilation systems.

#### **Individual protection measures, such as personal protective equipment**

|                                 |                                                                                                                                                                                                                                                                                               |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye/face protection</b>      | Tight sealing safety goggles.                                                                                                                                                                                                                                                                 |
| <b>Hand protection</b>          | Wear suitable gloves. Impervious gloves.                                                                                                                                                                                                                                                      |
| <b>Skin and body protection</b> | Wear suitable protective clothing. Long sleeved clothing. Chemical resistant apron. Antistatic boots.                                                                                                                                                                                         |
| <b>Respiratory protection</b>   | Appropriate respiratory protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |

### **9. Physical and chemical properties**

#### **Information on basic physical and chemical properties**

|                                       |           |
|---------------------------------------|-----------|
| <b>Appearance</b>                     | Liquid    |
| <b>Physical state</b>                 | Liquid    |
| <b>Color</b>                          | colorless |
| <b>Odor (includes odor threshold)</b> | Alcohol   |

| <b><u>Property</u></b>                         | <b><u>Values</u></b> | <b><u>Remarks • Method</u></b> |
|------------------------------------------------|----------------------|--------------------------------|
| <b>pH</b>                                      |                      | None known                     |
| <b>Melting point / freezing point</b>          | -89 °C / -128.2 °F   |                                |
| <b>Initial boiling point and boiling range</b> | 116 °C / 240.8 °F    |                                |
| <b>Flash point</b>                             | 35 °C / 95 °F        |                                |
| <b>Evaporation rate</b>                        | No data available    | None known                     |
| <b>Flammability</b>                            | No data available    | None known                     |
| <b>Flammability Limit in Air</b>               |                      | None known                     |
| <b>Upper flammability or explosive limits</b>  | No data available    |                                |
| <b>Lower flammability or explosive limits</b>  | No data available    |                                |
| <b>Vapor pressure</b>                          | No data available    | None known                     |
| <b>Relative vapor density</b>                  | No data available    | None known                     |
| <b>Relative density</b>                        | 0.81                 | None known                     |
| <b>Water solubility</b>                        | Miscible in water    |                                |
| <b>Solubility(ies)</b>                         | No data available    | None known                     |
| <b>Partition coefficient</b>                   | No data available    | None known                     |
| <b>Autoignition temperature</b>                | No data available    | None known                     |
| <b>Decomposition temperature</b>               | No data available    | None known                     |
| <b>Kinematic viscosity</b>                     | No data available    | None known                     |
| <b>Dynamic viscosity</b>                       | No data available    | None known                     |

#### **Other information**

### **10. Stability and reactivity**

|                                           |                                 |
|-------------------------------------------|---------------------------------|
| <b>Reactivity</b>                         | No information available.       |
| <b>Chemical stability</b>                 | Stable under normal conditions. |
| <b>Possibility of hazardous reactions</b> | None under normal processing.   |
| <b>Conditions to avoid</b>                | Heat, flames and sparks.        |

**Incompatible materials** Strong acids. Strong bases. Strong oxidizing agents.

**Hazardous decomposition products** None known based on information supplied.

#### Information on likely routes of exposure

##### Product Information

|                     |                                                                                                                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available. May cause irritation of respiratory tract. May cause drowsiness or dizziness.                                               |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available. Causes serious eye damage. May cause irreversible damage to eyes.                                                           |
| <b>Skin contact</b> | Specific test data for the substance or mixture is not available. Causes skin irritation (based on components).                                                                               |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea. Harmful if swallowed (based on components). |

#### Symptoms related to the physical, chemical and toxicological characteristics

**Symptoms** Redness. Burning. May cause blindness. May cause redness and tearing of the eyes. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

**Acute toxicity** Harmful if swallowed.

##### Numerical measures of toxicity

The following ATE values have been calculated for the mixture

|                               |              |
|-------------------------------|--------------|
| ATEmix (oral)                 | 790.10 mg/kg |
| ATEmix (inhalation-dust/mist) | 17.80 mg/l   |

##### Component Information

| Chemical name        | Oral LD50           | Dermal LD50             | Inhalation LC50          |
|----------------------|---------------------|-------------------------|--------------------------|
| 1-Butanol<br>71-36-3 | = 700 mg/kg ( Rat ) | = 3402 mg/kg ( Rabbit ) | > 17.76 mg/L ( Rat ) 4 h |

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Skin corrosion/irritation** Classification based on data available for ingredients. Causes skin irritation.

**Serious eye damage/eye irritation** Classification based on data available for ingredients. Causes burns. Causes serious eye damage.

**Respiratory or skin sensitization** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

**Reproductive toxicity** No information available.

|                          |                                                                      |
|--------------------------|----------------------------------------------------------------------|
| STOT - single exposure   | May cause respiratory irritation. May cause drowsiness or dizziness. |
| STOT - repeated exposure | No information available.                                            |
| Target organ effects     | Respiratory system, Eyes, Skin, Central nervous system.              |
| Aspiration hazard        | No information available.                                            |
| Other adverse effects    | No information available.                                            |
| Interactive effects      | No information available.                                            |

## 12. Ecological information

|             |                                                    |
|-------------|----------------------------------------------------|
| Ecotoxicity | Harmful to aquatic life with long lasting effects. |
|-------------|----------------------------------------------------|

| Chemical name        | Algae/aquatic plants                                                                                           | Fish                                                                                                                                                                                                                                    | Toxicity to microorganisms | Crustacea                                                                                          |
|----------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------|
| 1-Butanol<br>71-36-3 | EC50: >500mg/L (96h, <i>Desmodesmus subspicatus</i> )<br>EC50: >500mg/L (72h, <i>Desmodesmus subspicatus</i> ) | LC50: 1730 - 1910mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: =1740mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 100000 - 500000µg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =1910000µg/L (96h, <i>Pimephales promelas</i> ) | -                          | EC50: =1983mg/L (48h, <i>Daphnia magna</i> )<br>EC50: 1897 - 2072mg/L (48h, <i>Daphnia magna</i> ) |

|                               |                           |
|-------------------------------|---------------------------|
| Persistence and degradability | No information available. |
|-------------------------------|---------------------------|

### Bioaccumulation

#### Component Information

| Chemical name        | Partition coefficient |
|----------------------|-----------------------|
| 1-Butanol<br>71-36-3 | 1                     |

|                       |                           |
|-----------------------|---------------------------|
| Other adverse effects | No information available. |
|-----------------------|---------------------------|

## 13. Disposal considerations

### Disposal methods

|                                     |                                                                                                                                                                                                                                                                              |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waste from residues/unused products | Should not be released into the environment. Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| Contaminated packaging              | Empty containers pose a potential fire and explosion hazard. Do not cut, puncture or weld containers.                                                                                                                                                                        |

**California Hazardous Waste Status** This product contains one or more substances that are listed with the State of California as a hazardous waste.

## 14. Transport information

### DOT

|                                        |                                     |
|----------------------------------------|-------------------------------------|
| UN number or ID number                 | UN1120                              |
| Proper shipping name                   | BUTANOLS                            |
| Transport hazard class(es)             | 3                                   |
| Packing group                          | II                                  |
| Reportable quantity (lbs) (calculated) | 1-Butanol: RQ (lb)= 5001.00         |
| Special Provisions                     | IB2, T4, TP1, TP29                  |
| DOT Marine Pollutant                   | NP                                  |
| Description                            | UN1120, BUTANOLS (1-Butanol), 3, II |

### TDG

|                            |                                     |
|----------------------------|-------------------------------------|
| UN number or ID number     | UN1120                              |
| UN proper shipping name    | BUTANOLS                            |
| Transport hazard class(es) | 3                                   |
| Packing group              | II                                  |
| Description                | UN1120, BUTANOLS (1-Butanol), 3, II |

### MEX

|                            |                         |
|----------------------------|-------------------------|
| UN number or ID number     | UN1120                  |
| UN proper shipping name    | BUTANOLS                |
| Transport hazard class(es) | 3                       |
| Packing group              | II                      |
| Technical Name             | 1-Butanol               |
| Description                | UN1120, BUTANOLS, 3, II |

### ICAO (air)

|                            |                         |
|----------------------------|-------------------------|
| UN number or ID number     | UN1120                  |
| UN proper shipping name    | BUTANOLS                |
| Transport hazard class(es) | 3                       |
| Packing group              | II                      |
| Description                | UN1120, BUTANOLS, 3, II |
| Special Provisions         | A3                      |

### IATA

|                            |                         |
|----------------------------|-------------------------|
| UN number or ID number     | UN1120                  |
| UN proper shipping name    | Butanols                |
| Technical Name             | 1-Butanol               |
| Transport hazard class(es) | 3                       |
| Packing group              | II                      |
| Special Provisions         | A3                      |
| Description                | UN1120, Butanols, 3, II |

### IMDG

|                            |                                                  |
|----------------------------|--------------------------------------------------|
| UN number or ID number     | UN1120                                           |
| UN proper shipping name    | BUTANOLS                                         |
| Transport hazard class(es) | 3                                                |
| Packing group              | II                                               |
| Marine pollutant indicator | NP                                               |
| EmS-No.                    | F-E S-D                                          |
| Description                | UN1120, BUTANOLS (1-BUTANOL), 3, II, (35°C C.C.) |

## 15. Regulatory information

### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories** Contact supplier for inventory compliance status

### US Federal Regulations

#### **TSCA**

All chemical components are listed on the TSCA Inventory.

#### **SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

#### **SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

#### **CWA (Clean Water Act)**

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

#### **CAA (Clean Air Act)**

This product does not contain any substances regulated as pollutants pursuant to Clean Air Act (CAA).

#### **CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302).

| Chemical name        | Hazardous Substances RQs | Extremely Hazardous Substances RQs | Reportable Quantity (RQ)                   |
|----------------------|--------------------------|------------------------------------|--------------------------------------------|
| 1-Butanol<br>71-36-3 | 5000 lb                  | -                                  | RQ 5000 lb final RQ<br>RQ 2270 kg final RQ |

### US State Regulations

#### **California Proposition 65**

This product does not contain any Proposition 65 chemicals

#### **U.S. State Right-to-Know Regulations**

| Chemical name        | New Jersey | Massachusetts | Pennsylvania |
|----------------------|------------|---------------|--------------|
| 1-Butanol<br>71-36-3 | X          | X             | X            |

### U.S. EPA Label Information

**EPA Pesticide Registration Number** Not applicable

## **16. Other information**

**NFPA**  
**HMIS**

**Health hazards** 0  
**Health hazards** 0

**Flammability** 0  
**Flammability** 0

**Instability** 0  
**Physical hazards** 0

**Special hazards** -  
**Personal protection** -

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend**

|                |                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------|
| <b>ADN</b>     | Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways (Europe) |
| <b>ADR</b>     | Agreement concerning the International Carriage of Dangerous Goods by Road (Europe)             |
| <b>ATE</b>     | Acute Toxicity Estimate                                                                         |
| <b>ASTM</b>    | American Society for the Testing of Materials                                                   |
| <b>BEL</b>     | Biological exposure limits                                                                      |
| <b>bw</b>      | Body weight                                                                                     |
| <b>Ceiling</b> | Maximum limit value                                                                             |
| <b>CMR</b>     | Carcinogen, Mutagen or Reproductive Toxicant                                                    |
| <b>DOT</b>     | Department of Transportation (United States)                                                    |
| <b>EmS</b>     | Emergency Schedule                                                                              |
| <b>EPA</b>     | Environmental Protection Agency                                                                 |
| <b>GHS</b>     | Globally Harmonized System                                                                      |
| <b>HMIS</b>    | Hazardous Materials Identification System                                                       |
| <b>IATA</b>    | International Air Transport Association                                                         |
| <b>ICAO</b>    | International Civil Aviation Organization                                                       |
| <b>IMDG</b>    | International Maritime Dangerous Goods                                                          |
| <b>IMO</b>     | International Maritime Organization                                                             |
| <b>ISO</b>     | International Organization for Standardization                                                  |
| <b>LC50</b>    | Lethal Concentration to 50% of a test population                                                |
| <b>LD50</b>    | Lethal Dose to 50% of a test population (Median Lethal Dose)                                    |
| <b>MARPOL</b>  | International Convention for the Prevention of Pollution from Ships                             |
| <b>NFPA</b>    | National Fire Protection Association                                                            |
| <b>NIOSH</b>   | National Institute for Occupational Safety and Health                                           |
| <b>n.o.s.</b>  | Not Otherwise Specified                                                                         |
| <b>NOAEL</b>   | No Observed Adverse Effect Level                                                                |
| <b>OECD</b>    | Organization for Economic Cooperation and Development                                           |
| <b>OEL</b>     | Occupational exposure limits                                                                    |
| <b>OSHA</b>    | Occupational Safety and Health Administration of the US Department of Labor                     |
| <b>PBT</b>     | Persistent, Bioaccumulative and Toxic substance                                                 |
| <b>PMT</b>     | Persistent, Mobile and Toxic                                                                    |
| <b>PPE</b>     | Personal protective equipment                                                                   |
| <b>RID</b>     | Agreement concerning the International Carriage of Dangerous Goods by Rail (Europe)             |
| <b>SARA</b>    | Superfund Amendments and Reauthorization Act                                                    |
| <b>SDS</b>     | Safety Data Sheet                                                                               |
| <b>STEL</b>    | Short Term Exposure Limit                                                                       |
| <b>STOT RE</b> | Specific target organ toxicity - Repeated exposure                                              |
| <b>STOT SE</b> | Specific target organ toxicity - Single exposure                                                |
| <b>TDG</b>     | Transport of Dangerous Goods (Canada)                                                           |
| <b>TSCA</b>    | Toxic Substances Control Act (United States)                                                    |
| <b>TWA</b>     | Time-Weighted Average                                                                           |
| <b>UN</b>      | United Nations                                                                                  |
| <b>VOC</b>     | Volatile organic compounds                                                                      |
| <b>vPvB</b>    | Very Persistent and Very Bioaccumulative                                                        |
| <b>vPvM</b>    | Very Persistent and Very Mobile                                                                 |
| <b>Sk</b>      | Skin notation                                                                                   |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)

Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date** 25-Jan-2023

**Revision Note** This product has been discontinued and the Safety Data Sheet will not be updated.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
US OSHA Hazard Communication Standard (29 CFR 1910.1200)

Revision date 14-May-2025

Revision Number 1.1

## 1. Identification

### Product identifier

**Product Name** Reagent C1 - Resolving Solution

### Other means of identification

**Catalog Number(s)** 5144D

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### **Corporate Headquarters**

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### **Manufacturer Address**

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### **Legal Entity / Contact Address**

Bio-Rad Laboratories  
Life Science  
2000 Alfred Nobel Drive  
Hercules, California 94547

#### **Technical Service**

1-800-424-6723  
support@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification of the substance or mixture

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200)

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

#### **Hazard statements**

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200)

#### **Hazards classified under paragraph (d)(1)(ii) of 1910.1200**

No information available.

### Other information

No information available.

### 3. Composition/information on ingredients

#### Substance

Not applicable.

#### Mixture

| Chemical name                                    | CAS No.  | Weight-%   | Trade secret |
|--------------------------------------------------|----------|------------|--------------|
| Glycine, N-methyl-N-(1-oxododecyl)-, sodium salt | 137-16-6 | 0.3 - 0.99 | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

### 4. First-aid measures

#### Description of first aid measures

|              |                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                    |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a physician. |
| Skin contact | Wash skin with soap and water.                                                                                          |
| Ingestion    | Rinse mouth.                                                                                                            |

#### Most important symptoms and effects, both acute and delayed

|                     |                           |
|---------------------|---------------------------|
| Symptoms            | No information available. |
| Effects of Exposure | No information available. |

#### Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

### 5. Fire-fighting measures

|                                                                |                                                                                                                                       |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                               |
| Unsuitable extinguishing media                                 | Do not scatter spilled material with high pressure water streams.                                                                     |
| Specific hazards arising from the chemical                     | No information available.                                                                                                             |
| Explosion data                                                 |                                                                                                                                       |
| Sensitivity to mechanical impact                               | None.                                                                                                                                 |
| Sensitivity to static discharge                                | None.                                                                                                                                 |
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.<br>Use personal protection equipment. |

### 6. Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Ensure adequate ventilation.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

**7. Handling and storage****Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store according to product and label instructions.

**8. Exposure controls/personal protection****Working area parameters, subject to mandatory control (MAC or TSEL)**

**Exposure Limits** This product, as supplied, contains materials that do not have reportable occupational exposure limits or are not subject to the reporting requirements of the local jurisdiction.

**Biological occupational exposure limits** This product, as supplied, contains materials that do not have reportable biological exposure limits or are not subject to the reporting requirements of the local jurisdiction.

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** Appropriate eye/face protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Hand protection** Appropriate hand protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Skin and body protection** Appropriate skin and body protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Respiratory protection** Appropriate respiratory protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                                |                  |
|--------------------------------|------------------|
| Appearance                     | aqueous solution |
| Physical state                 | Liquid           |
| Color                          | green            |
| Odor (includes odor threshold) | Odorless         |

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      | 7                 |                         |
| Melting point / freezing point          | No data available | None known              |
| Initial boiling point and boiling range | No data available | None known              |
| Flash point                             | No data available | None known              |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapor pressure                          | No data available | None known              |
| Relative vapor density                  | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Miscible in water |                         |
| Solubility(ies)                         | No data available | None known              |
| Partition coefficient                   | No data available | None known              |
| Autoignition temperature                | No data available | None known              |
| Decomposition temperature               | No data available | None known              |
| Kinematic viscosity                     | No data available | None known              |
| Dynamic viscosity                       | No data available | None known              |

### Other information

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

### Information on likely routes of exposure

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

**Symptoms related to the physical, chemical and toxicological characteristics**

**Symptoms** No information available.

**Acute toxicity** No information available.

**Numerical measures of toxicity**

The following ATE values have been calculated for the mixture

ATEmix (oral) 32,095.60 mg/kg

**Component Information**

| Chemical name                                                | Oral LD50            | Dermal LD50 | Inhalation LC50             |
|--------------------------------------------------------------|----------------------|-------------|-----------------------------|
| Glycine, N-methyl-N-(1-oxododecyl)-, sodium salt<br>137-16-6 | > 5000 mg/kg ( Rat ) | -           | 0.05 - 0.5 mg/L ( Rat ) 4 h |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** No information available.

**Serious eye damage/eye irritation** No information available.

**Respiratory or skin sensitization** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

**STOT - repeated exposure** No information available.

**Aspiration hazard** No information available.

**Other adverse effects** No information available.

**Interactive effects** No information available.

**12. Ecological information****Ecotoxicity**

| Chemical name                                                | Algae/aquatic plants | Fish                              | Toxicity to microorganisms | Crustacea |
|--------------------------------------------------------------|----------------------|-----------------------------------|----------------------------|-----------|
| Glycine, N-methyl-N-(1-oxododecyl)-, sodium salt<br>137-16-6 | -                    | LC50: =107mg/L (96h, Danio rerio) | -                          | -         |

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

**Other adverse effects** No information available.

### 13. Disposal considerations

#### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

### 14. Transport information

**DOT** Not regulated

**TDG** Not regulated

**MEX** Not regulated

**ICAO (air)** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

### 15. Regulatory information

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories** Contact supplier for inventory compliance status

#### US Federal Regulations

##### **TSCA**

All chemical components are listed on the TSCA Inventory.

##### **SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

##### **SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

**CAA (Clean Air Act)**

This product does not contain any substances regulated as pollutants pursuant to Clean Air Act (CAA).

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**US State Regulations****California Proposition 65**

This product does not contain any Proposition 65 chemicals

**U.S. State Right-to-Know Regulations**

| Chemical name                  | New Jersey | Massachusetts | Pennsylvania |
|--------------------------------|------------|---------------|--------------|
| Water<br>7732-18-5             | -          | -             | X            |
| C.I. Basic Green 4<br>569-64-2 | X          | X             | X            |

**U.S. EPA Label Information**

**EPA Pesticide Registration Number** Not applicable

**16. Other information**

|             |                         |                       |                           |                              |
|-------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> - |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend**

|                |                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------|
| <b>ADN</b>     | Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways (Europe) |
| <b>ADR</b>     | Agreement concerning the International Carriage of Dangerous Goods by Road (Europe)             |
| <b>ATE</b>     | Acute Toxicity Estimate                                                                         |
| <b>ASTM</b>    | American Society for the Testing of Materials                                                   |
| <b>BEL</b>     | Biological exposure limits                                                                      |
| <b>bw</b>      | Body weight                                                                                     |
| <b>Ceiling</b> | Maximum limit value                                                                             |
| <b>CMR</b>     | Carcinogen, Mutagen or Reproductive Toxicant                                                    |
| <b>DOT</b>     | Department of Transportation (United States)                                                    |
| <b>EmS</b>     | Emergency Schedule                                                                              |
| <b>EPA</b>     | Environmental Protection Agency                                                                 |
| <b>GHS</b>     | Globally Harmonized System                                                                      |
| <b>HMIS</b>    | Hazardous Materials Identification System                                                       |
| <b>IATA</b>    | International Air Transport Association                                                         |
| <b>ICAO</b>    | International Civil Aviation Organization                                                       |
| <b>IMDG</b>    | International Maritime Dangerous Goods                                                          |
| <b>IMO</b>     | International Maritime Organization                                                             |

|                                                                                                |
|------------------------------------------------------------------------------------------------|
| <b>ISO</b> International Organization for Standardization                                      |
| <b>LC50</b> Lethal Concentration to 50% of a test population                                   |
| <b>LD50</b> Lethal Dose to 50% of a test population (Median Lethal Dose)                       |
| <b>MARPOL</b> International Convention for the Prevention of Pollution from Ships              |
| <b>NFPA</b> National Fire Protection Association                                               |
| <b>NIOSH</b> National Institute for Occupational Safety and Health                             |
| <b>n.o.s.</b> Not Otherwise Specified                                                          |
| <b>NOAEL</b> No Observed Adverse Effect Level                                                  |
| <b>OECD</b> Organization for Economic Cooperation and Development                              |
| <b>OEL</b> Occupational exposure limits                                                        |
| <b>OSHA</b> Occupational Safety and Health Administration of the US Department of Labor        |
| <b>PBT</b> Persistent, Bioaccumulative and Toxic substance                                     |
| <b>PMT</b> Persistent, Mobile and Toxic                                                        |
| <b>PPE</b> Personal protective equipment                                                       |
| <b>RID</b> Agreement concerning the International Carriage of Dangerous Goods by Rail (Europe) |
| <b>SARA</b> Superfund Amendments and Reauthorization Act                                       |
| <b>SDS</b> Safety Data Sheet                                                                   |
| <b>STEL</b> Short Term Exposure Limit                                                          |
| <b>STOT RE</b> Specific target organ toxicity - Repeated exposure                              |
| <b>STOT SE</b> Specific target organ toxicity - Single exposure                                |
| <b>TDG</b> Transport of Dangerous Goods (Canada)                                               |
| <b>TSCA</b> Toxic Substances Control Act (United States)                                       |
| <b>TWA</b> Time-Weighted Average                                                               |
| <b>UN</b> United Nations                                                                       |
| <b>VOC</b> Volatile organic compounds                                                          |
| <b>vPvB</b> Very Persistent and Very Bioaccumulative                                           |
| <b>vPvM</b> Very Persistent and Very Mobile                                                    |
| <b>Sk</b> Skin notation                                                                        |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGL(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.  
**Revision date** 14-May-2025  
**Revision Note** Reviewed existing information and made minor updates.  
**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
US OSHA Hazard Communication Standard (29 CFR 1910.1200)

Revision date 25-Jan-2023

Revision Number 2

## 1. Identification

### Product identifier

**Product Name** Discontinued - Reagent 3 - Resolving Buffer

### Other means of identification

**Catalog Number(s)** 1768

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Life Science  
2000 Alfred Nobel Drive  
Hercules, California 94547

**Technical Service** 1-800-424-6723  
support@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification of the substance or mixture

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200)

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

#### Hazard statements

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200)

### Hazards classified under paragraph (d)(1)(ii) of 1910.1200

No information available.

### Other information

No information available.

### 3. Composition/information on ingredients

#### Substance

Not applicable.

#### Mixture

| Chemical name                                    | CAS No.  | Weight-%   | Trade secret |
|--------------------------------------------------|----------|------------|--------------|
| Glycine, N-methyl-N-(1-oxododecyl)-, sodium salt | 137-16-6 | 0.3 - 0.99 | *            |

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

### 4. First-aid measures

#### Description of first aid measures

|              |                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                    |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a physician. |
| Skin contact | Wash skin with soap and water.                                                                                          |
| Ingestion    | Rinse mouth.                                                                                                            |

#### Most important symptoms and effects, both acute and delayed

|                     |                           |
|---------------------|---------------------------|
| Symptoms            | No information available. |
| Effects of Exposure | No information available. |

#### Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

### 5. Fire-fighting measures

|                                                                |                                                                                                                                       |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                               |
| Unsuitable extinguishing media                                 | Do not scatter spilled material with high pressure water streams.                                                                     |
| Specific hazards arising from the chemical                     | No information available.                                                                                                             |
| Explosion data                                                 |                                                                                                                                       |
| Sensitivity to mechanical impact                               | None.                                                                                                                                 |
| Sensitivity to static discharge                                | None.                                                                                                                                 |
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.<br>Use personal protection equipment. |

### 6. Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Ensure adequate ventilation.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labeled containers.

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

**7. Handling and storage****Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**General hygiene considerations** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store according to product and label instructions.

**8. Exposure controls/personal protection****Working area parameters, subject to mandatory control (MAC or TSEL)**

**Exposure Limits** This product, as supplied, contains materials that do not have reportable occupational exposure limits or are not subject to the reporting requirements of the local jurisdiction.

**Biological occupational exposure limits** This product, as supplied, contains materials that do not have reportable biological exposure limits or are not subject to the reporting requirements of the local jurisdiction.

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** Appropriate eye/face protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Hand protection** Appropriate hand protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Skin and body protection** Appropriate skin and body protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.

**Respiratory protection** Appropriate respiratory protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

## 9. Physical and chemical properties

### Information on basic physical and chemical properties

|                                |                  |
|--------------------------------|------------------|
| Appearance                     | aqueous solution |
| Physical state                 | Liquid           |
| Color                          | green            |
| Odor (includes odor threshold) | Odorless         |

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      | 7                 |                         |
| Melting point / freezing point          | No data available | None known              |
| Initial boiling point and boiling range | No data available | None known              |
| Flash point                             | No data available | None known              |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapor pressure                          | No data available | None known              |
| Relative vapor density                  | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Miscible in water |                         |
| Solubility(ies)                         | No data available | None known              |
| Partition coefficient                   | No data available | None known              |
| Autoignition temperature                | No data available | None known              |
| Decomposition temperature               | No data available | None known              |
| Kinematic viscosity                     | No data available | None known              |
| Dynamic viscosity                       | No data available | None known              |

### Other information

## 10. Stability and reactivity

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

### Information on likely routes of exposure

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

**Symptoms related to the physical, chemical and toxicological characteristics**

**Symptoms** No information available.

**Acute toxicity** No information available.

**Numerical measures of toxicity**

The following ATE values have been calculated for the mixture  
ATEmix (oral) 32,095.60 mg/kg

**Component Information**

| Chemical name                                                | Oral LD50            | Dermal LD50 | Inhalation LC50             |
|--------------------------------------------------------------|----------------------|-------------|-----------------------------|
| Glycine, N-methyl-N-(1-oxododecyl)-, sodium salt<br>137-16-6 | > 5000 mg/kg ( Rat ) | -           | 0.05 - 0.5 mg/L ( Rat ) 4 h |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** No information available.

**Serious eye damage/eye irritation** No information available.

**Respiratory or skin sensitization** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

**STOT - repeated exposure** No information available.

**Aspiration hazard** No information available.

**Other adverse effects** No information available.

**Interactive effects** No information available.

**12. Ecological information****Ecotoxicity**

| Chemical name                                                | Algae/aquatic plants | Fish                              | Toxicity to microorganisms | Crustacea |
|--------------------------------------------------------------|----------------------|-----------------------------------|----------------------------|-----------|
| Glycine, N-methyl-N-(1-oxododecyl)-, sodium salt<br>137-16-6 | -                    | LC50: =107mg/L (96h, Danio rerio) | -                          | -         |

**Persistence and degradability** No information available.

**Bioaccumulation** There is no data for this product.

**Other adverse effects** No information available.

### 13. Disposal considerations

#### Disposal methods

**Waste from residues/unused products** Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

**Contaminated packaging** Do not reuse empty containers.

### 14. Transport information

**DOT** Not regulated

**TDG** Not regulated

**MEX** Not regulated

**ICAO (air)** Not regulated

**IATA** Not regulated

**IMDG** Not regulated

### 15. Regulatory information

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories** Contact supplier for inventory compliance status

#### US Federal Regulations

##### **TSCA**

All chemical components are listed on the TSCA Inventory.

##### **SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

##### **SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

**CAA (Clean Air Act)**

This product does not contain any substances regulated as pollutants pursuant to Clean Air Act (CAA).

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**US State Regulations****California Proposition 65**

This product does not contain any Proposition 65 chemicals

**U.S. State Right-to-Know Regulations**

| Chemical name                  | New Jersey | Massachusetts | Pennsylvania |
|--------------------------------|------------|---------------|--------------|
| Water<br>7732-18-5             | -          | -             | X            |
| C.I. Basic Green 4<br>569-64-2 | X          | X             | X            |

**U.S. EPA Label Information**

**EPA Pesticide Registration Number** Not applicable

**16. Other information**

|             |                         |                       |                           |                              |
|-------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> - |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend**

|                |                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------|
| <b>ADN</b>     | Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways (Europe) |
| <b>ADR</b>     | Agreement concerning the International Carriage of Dangerous Goods by Road (Europe)             |
| <b>ATE</b>     | Acute Toxicity Estimate                                                                         |
| <b>ASTM</b>    | American Society for the Testing of Materials                                                   |
| <b>BEL</b>     | Biological exposure limits                                                                      |
| <b>bw</b>      | Body weight                                                                                     |
| <b>Ceiling</b> | Maximum limit value                                                                             |
| <b>CMR</b>     | Carcinogen, Mutagen or Reproductive Toxicant                                                    |
| <b>DOT</b>     | Department of Transportation (United States)                                                    |
| <b>EmS</b>     | Emergency Schedule                                                                              |
| <b>EPA</b>     | Environmental Protection Agency                                                                 |
| <b>GHS</b>     | Globally Harmonized System                                                                      |
| <b>HMIS</b>    | Hazardous Materials Identification System                                                       |
| <b>IATA</b>    | International Air Transport Association                                                         |
| <b>ICAO</b>    | International Civil Aviation Organization                                                       |
| <b>IMDG</b>    | International Maritime Dangerous Goods                                                          |
| <b>IMO</b>     | International Maritime Organization                                                             |

|                                                                                                |
|------------------------------------------------------------------------------------------------|
| <b>ISO</b> International Organization for Standardization                                      |
| <b>LC50</b> Lethal Concentration to 50% of a test population                                   |
| <b>LD50</b> Lethal Dose to 50% of a test population (Median Lethal Dose)                       |
| <b>MARPOL</b> International Convention for the Prevention of Pollution from Ships              |
| <b>NFPA</b> National Fire Protection Association                                               |
| <b>NIOSH</b> National Institute for Occupational Safety and Health                             |
| <b>n.o.s.</b> Not Otherwise Specified                                                          |
| <b>NOAEL</b> No Observed Adverse Effect Level                                                  |
| <b>OECD</b> Organization for Economic Cooperation and Development                              |
| <b>OEL</b> Occupational exposure limits                                                        |
| <b>OSHA</b> Occupational Safety and Health Administration of the US Department of Labor        |
| <b>PBT</b> Persistent, Bioaccumulative and Toxic substance                                     |
| <b>PMT</b> Persistent, Mobile and Toxic                                                        |
| <b>PPE</b> Personal protective equipment                                                       |
| <b>RID</b> Agreement concerning the International Carriage of Dangerous Goods by Rail (Europe) |
| <b>SARA</b> Superfund Amendments and Reauthorization Act                                       |
| <b>SDS</b> Safety Data Sheet                                                                   |
| <b>STEL</b> Short Term Exposure Limit                                                          |
| <b>STOT RE</b> Specific target organ toxicity - Repeated exposure                              |
| <b>STOT SE</b> Specific target organ toxicity - Single exposure                                |
| <b>TDG</b> Transport of Dangerous Goods (Canada)                                               |
| <b>TSCA</b> Toxic Substances Control Act (United States)                                       |
| <b>TWA</b> Time-Weighted Average                                                               |
| <b>UN</b> United Nations                                                                       |
| <b>VOC</b> Volatile organic compounds                                                          |
| <b>vPvB</b> Very Persistent and Very Bioaccumulative                                           |
| <b>vPvM</b> Very Persistent and Very Mobile                                                    |
| <b>Sk</b> Skin notation                                                                        |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AEGl(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
 Organization for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Prepared By**

Bio-Rad Laboratories, Environmental Health and Safety.

**Revision date**

25-Jan-2023

**Revision Note**

This product has been discontinued and the Safety Data Sheet will not be updated.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
US OSHA Hazard Communication Standard (29 CFR 1910.1200)

Revision date 14-May-2025

Revision Number 1.1

## 1. Identification

### Product identifier

**Product Name** R1 - Solid Phase Antibody

### Other means of identification

**Catalog Number(s)** 3551103C1

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Restrictions on use** No information available

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer Address

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories  
Life Science  
2000 Alfred Nobel Drive  
Hercules, California 94547

#### **Technical Service**

1-800-424-6723  
support@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC USA: 1 (800) 424-9300

## 2. Hazard(s) identification

### Classification of the substance or mixture

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200).

### Hazards not otherwise classified (HNOC)

Not applicable

### Label elements

#### Hazard statements

This product is not considered hazardous by the US OSHA Hazard Communication Standard 2024 (29 CFR 1910.1200).

### **Hazards classified under paragraph (d)(1)(ii) of 1910.1200**

No information available.

### Other information

No information available.

### 3. Composition/information on ingredients

#### Substance

\*The exact percentage (concentration) of composition has been withheld as a trade secret.

### 4. First-aid measures

#### Description of first aid measures

|              |                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                    |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a physician. |
| Skin contact | Wash skin with soap and water.                                                                                          |
| Ingestion    | Rinse mouth.                                                                                                            |

#### Most important symptoms and effects, both acute and delayed

|                     |                           |
|---------------------|---------------------------|
| Symptoms            | No information available. |
| Effects of Exposure | No information available. |

#### Indication of any immediate medical attention and special treatment needed

|                    |                        |
|--------------------|------------------------|
| Note to physicians | Treat symptomatically. |
|--------------------|------------------------|

### 5. Fire-fighting measures

|                                                                |                                                                                                                                       |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media                                   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                               |
| Unsuitable extinguishing media                                 | Do not scatter spilled material with high pressure water streams.                                                                     |
| Specific hazards arising from the chemical                     | No information available.                                                                                                             |
| Explosion data                                                 |                                                                                                                                       |
| Sensitivity to mechanical impact                               | None.                                                                                                                                 |
| Sensitivity to static discharge                                | None.                                                                                                                                 |
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.<br>Use personal protection equipment. |

### 6. Accidental release measures

#### Personal precautions, protective equipment and emergency procedures

|                      |                              |
|----------------------|------------------------------|
| Personal precautions | Ensure adequate ventilation. |
|----------------------|------------------------------|

**Methods and material for containment and cleaning up**

|                                        |                                                                                      |
|----------------------------------------|--------------------------------------------------------------------------------------|
| <b>Methods for containment</b>         | Prevent further leakage or spillage if safe to do so.                                |
| <b>Methods for cleaning up</b>         | Pick up and transfer to properly labeled containers.                                 |
| <b>Prevention of secondary hazards</b> | Clean contaminated objects and areas thoroughly observing environmental regulations. |

**7. Handling and storage****Precautions for safe handling**

|                                       |                                                                        |
|---------------------------------------|------------------------------------------------------------------------|
| <b>Advice on safe handling</b>        | Handle in accordance with good industrial hygiene and safety practice. |
| <b>General hygiene considerations</b> | Handle in accordance with good industrial hygiene and safety practice. |

**Conditions for safe storage, including any incompatibilities**

|                           |                                                    |
|---------------------------|----------------------------------------------------|
| <b>Storage Conditions</b> | Store according to product and label instructions. |
|---------------------------|----------------------------------------------------|

**8. Exposure controls/personal protection****Working area parameters, subject to mandatory control (MAC or TSEL)**

|                                                |                                                                                                                                                                                    |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Limits</b>                         | This product, as supplied, contains materials that do not have reportable occupational exposure limits or are not subject to the reporting requirements of the local jurisdiction. |
| <b>Biological occupational exposure limits</b> | This product, as supplied, contains materials that do not have reportable biological exposure limits or are not subject to the reporting requirements of the local jurisdiction.   |

**Appropriate engineering controls**

|                             |                                                     |
|-----------------------------|-----------------------------------------------------|
| <b>Engineering controls</b> | Showers<br>Eyewash stations<br>Ventilation systems. |
|-----------------------------|-----------------------------------------------------|

**Individual protection measures, such as personal protective equipment**

|                                 |                                                                                                                                                                                                                                                                                               |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye/face protection</b>      | Appropriate eye/face protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.                                                                                                              |
| <b>Hand protection</b>          | Appropriate hand protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.                                                                                                                  |
| <b>Skin and body protection</b> | Appropriate skin and body protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction.                                                                                                         |
| <b>Respiratory protection</b>   | Appropriate respiratory protection should be selected and used according to the chemical nature, hazards and use of this product and safety requirements of the local jurisdiction. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |

**9. Physical and chemical properties**

**Information on basic physical and chemical properties**

|                                |          |
|--------------------------------|----------|
| Appearance                     | solid    |
| Physical state                 | Solid    |
| Color                          | white    |
| Odor (includes odor threshold) | Odorless |

| <b><u>Property</u></b>                  | <b><u>Values</u></b> | <b><u>Remarks • Method</u></b> |
|-----------------------------------------|----------------------|--------------------------------|
| pH                                      |                      | None known                     |
| Melting point / freezing point          | No data available    | None known                     |
| Initial boiling point and boiling range | No data available    | None known                     |
| Flash point                             | No data available    | None known                     |
| Evaporation rate                        | No data available    | None known                     |
| Flammability                            | No data available    | None known                     |
| Flammability Limit in Air               |                      | None known                     |
| Upper flammability or explosive limits  | No data available    |                                |
| Lower flammability or explosive limits  | No data available    |                                |
| Vapor pressure                          | No data available    | None known                     |
| Relative vapor density                  | No data available    | None known                     |
| Relative density                        | No data available    | None known                     |
| Water solubility                        | Soluble in water     | None known                     |
| Solubility(ies)                         | No data available    | None known                     |
| Partition coefficient                   | No data available    | None known                     |
| Autoignition temperature                | No data available    | None known                     |
| Decomposition temperature               | No data available    | None known                     |
| Kinematic viscosity                     | No data available    | None known                     |
| Dynamic viscosity                       | No data available    | None known                     |

**Other information****10. Stability and reactivity**

|                                    |                                           |
|------------------------------------|-------------------------------------------|
| Reactivity                         | No information available.                 |
| Chemical stability                 | Stable under normal conditions.           |
| Possibility of hazardous reactions | None under normal processing.             |
| Conditions to avoid                | None known based on information supplied. |
| Incompatible materials             | None known based on information supplied. |
| Hazardous decomposition products   | None known based on information supplied. |

**Information on likely routes of exposure**

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

**Symptoms related to the physical, chemical and toxicological characteristics**

|          |                           |
|----------|---------------------------|
| Symptoms | No information available. |
|----------|---------------------------|

Acute toxicity No information available.

Numerical measures of toxicity No information available.

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

Skin corrosion/irritation No information available.

Serious eye damage/eye irritation No information available.

Respiratory or skin sensitization No information available.

Germ cell mutagenicity No information available.

Carcinogenicity No information available.

Reproductive toxicity No information available.

STOT - single exposure No information available.

STOT - repeated exposure No information available.

Aspiration hazard No information available.

Other adverse effects No information available.

Interactive effects No information available.

### **12. Ecological information**

Ecotoxicity The environmental impact of this product has not been fully investigated.

Persistence and degradability No information available.

Bioaccumulation There is no data for this product.

Other adverse effects No information available.

### **13. Disposal considerations**

#### Disposal methods

|                                            |                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation. |
| <b>Contaminated packaging</b>              | Do not reuse empty containers.                                                                                  |

## 14. Transport information

|                          |               |
|--------------------------|---------------|
| <b><u>DOT</u></b>        | Not regulated |
| <b><u>TDG</u></b>        | Not regulated |
| <b><u>MEX</u></b>        | Not regulated |
| <b><u>ICAO (air)</u></b> | Not regulated |
| <b><u>IATA</u></b>       | Not regulated |
| <b><u>IMDG</u></b>       | Not regulated |

## 15. Regulatory information

### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories** Contact supplier for inventory compliance status

### US Federal Regulations

#### **TSCA**

All chemical components are listed on the TSCA Inventory.

#### **SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

#### **SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

#### **CWA (Clean Water Act)**

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42).

#### **CAA (Clean Air Act)**

This product does not contain any substances regulated as pollutants pursuant to Clean Air Act (CAA).

#### **CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**US State Regulations****California Proposition 65**

This product does not contain any Proposition 65 chemicals

**U.S. State Right-to-Know Regulations**

This product does not contain any substances regulated under applicable state right-to-know regulations

**U.S. EPA Label Information**

**EPA Pesticide Registration Number** Not applicable

**16. Other information**

|             |                         |                       |                           |                              |
|-------------|-------------------------|-----------------------|---------------------------|------------------------------|
| <b>NFPA</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Instability</b> 0      | <b>Special hazards</b> -     |
| <b>HMIS</b> | <b>Health hazards</b> 0 | <b>Flammability</b> 0 | <b>Physical hazards</b> 0 | <b>Personal protection</b> - |

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend**

|                |                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------|
| <b>ADN</b>     | Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways (Europe) |
| <b>ADR</b>     | Agreement concerning the International Carriage of Dangerous Goods by Road (Europe)             |
| <b>ATE</b>     | Acute Toxicity Estimate                                                                         |
| <b>ASTM</b>    | American Society for the Testing of Materials                                                   |
| <b>BEL</b>     | Biological exposure limits                                                                      |
| <b>bw</b>      | Body weight                                                                                     |
| <b>Ceiling</b> | Maximum limit value                                                                             |
| <b>CMR</b>     | Carcinogen, Mutagen or Reproductive Toxicant                                                    |
| <b>DOT</b>     | Department of Transportation (United States)                                                    |
| <b>EmS</b>     | Emergency Schedule                                                                              |
| <b>EPA</b>     | Environmental Protection Agency                                                                 |
| <b>GHS</b>     | Globally Harmonized System                                                                      |
| <b>HMIS</b>    | Hazardous Materials Identification System                                                       |
| <b>IATA</b>    | International Air Transport Association                                                         |
| <b>ICAO</b>    | International Civil Aviation Organization                                                       |
| <b>IMDG</b>    | International Maritime Dangerous Goods                                                          |
| <b>IMO</b>     | International Maritime Organization                                                             |
| <b>ISO</b>     | International Organization for Standardization                                                  |
| <b>LC50</b>    | Lethal Concentration to 50% of a test population                                                |
| <b>LD50</b>    | Lethal Dose to 50% of a test population (Median Lethal Dose)                                    |
| <b>MARPOL</b>  | International Convention for the Prevention of Pollution from Ships                             |
| <b>NFPA</b>    | National Fire Protection Association                                                            |
| <b>NIOSH</b>   | National Institute for Occupational Safety and Health                                           |
| <b>n.o.s.</b>  | Not Otherwise Specified                                                                         |
| <b>NOAEL</b>   | No Observed Adverse Effect Level                                                                |
| <b>OECD</b>    | Organization for Economic Cooperation and Development                                           |
| <b>OEL</b>     | Occupational exposure limits                                                                    |
| <b>OSHA</b>    | Occupational Safety and Health Administration of the US Department of Labor                     |
| <b>PBT</b>     | Persistent, Bioaccumulative and Toxic substance                                                 |
| <b>PMT</b>     | Persistent, Mobile and Toxic                                                                    |
| <b>PPE</b>     | Personal protective equipment                                                                   |
| <b>RID</b>     | Agreement concerning the International Carriage of Dangerous Goods by Rail (Europe)             |
| <b>SARA</b>    | Superfund Amendments and Reauthorization Act                                                    |
| <b>SDS</b>     | Safety Data Sheet                                                                               |
| <b>STEL</b>    | Short Term Exposure Limit                                                                       |
| <b>STOT RE</b> | Specific target organ toxicity - Repeated exposure                                              |
| <b>STOT SE</b> | Specific target organ toxicity - Single exposure                                                |
| <b>TDG</b>     | Transport of Dangerous Goods (Canada)                                                           |
| <b>TSCA</b>    | Toxic Substances Control Act (United States)                                                    |
| <b>TWA</b>     | Time-Weighted Average                                                                           |
| <b>UN</b>      | United Nations                                                                                  |

|             |                                          |
|-------------|------------------------------------------|
| <b>VOC</b>  | Volatile organic compounds               |
| <b>vPvB</b> | Very Persistent and Very Bioaccumulative |
| <b>vPvM</b> | Very Persistent and Very Mobile          |
| <b>Sk</b>   | Skin notation                            |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
Environmental Protection Agency  
Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
National Institute of Technology and Evaluation (NITE)  
Australia National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organization for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organization for Economic Co-operation and Development High Production Volume Chemicals Program  
Organization for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

**Prepared By** Bio-Rad Laboratories, Environmental Health and Safety.  
**Revision date** 14-May-2025  
**Revision Note** Reformatted and updated existing information.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

**End of Safety Data Sheet**